
CMS Analysis Note

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2026/05/29

Search for disappearing tracks in 2022 and 2023 data at 13.6 TeV

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Abstract

Presented here is a search for long-lived charged particles that decay within the inner CMS tracker and produce the signature of a disappearing track. Candidate disappearing tracks are identified as isolated tracks with no associated hits in the muon detectors, little associated calorimeter energy deposits, and missing hits in the outer layers of the silicon tracker. The search uses pp collision data recorded at $\sqrt{s} = 13.6$ TeV, during 2022 and 2023, corresponding to a total integrated luminosity of 62.5 fb^{-1} . The results are interpreted in the context of anomaly-mediated supersymmetry breaking (AMSB), in terms of upper limits on the cross section of direct chargino production in terms of the chargino mass and lifetime for both a purely wino-like and higgsino-like neutralino LSP.

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Change log

v2

Adapted the note from Run 2 analysis. Target plots with Run2022F only (some plots w/ more).

v3

Included chapters for signal systematics and results, with preliminary estimations.

v4

Answering Christina and Lisa's comments. Bottom line of this update is adding all remaining data samples (2022 CD, EFG, 2023 C and D), lepton background estimates from all lepton flavours, additional checks in the fake track background estimates, background systematics and full expected limit plots.

- Moved to the recommended tau lepton selection path from the Tau POG.
- Added new jet veto maps and ECAL maps in Figures 8 and 9.
- We have all the fiducial maps for all eras, but we still choose to display only 2022 F in Figure 10.
- Added 2023 data to angular requirements in Figure 12.
- Table 24 now has P_{veto} for all lepton flavours and data taking periods.
- Changed the plots in Figures 20 and 21 ($p_{\text{T}}^{\text{miss, no } \mu}$ vs $|\Delta\Phi(\text{leading jet}, p_{\text{T}}^{\text{miss, no } \mu})|$) to use 2022 postEE data, and have EGamma, Muon and Tau data, for P_{offline} calculation.
- Added the $p_{\text{T}}^{\text{miss, no } \mu}$ turn-on curves for EGamma 2023 and Muon 2022 and 2023.
- Figure 25 now has plots for electrons, muons and tau leptons, for P_{trigger} calculation.
- Corrected N_{ctrl}^1 for input for the lepton background estimates by the trigger efficiency.
- Table 28 now has electron background estimates for 2023 data. Tables 29 and 30 have muon and tau background estimates for 2022 and 2023 data.
- Figure 26 and Table 34 now have the fits for the transfer factor ζ and the fake track background estimates for all data taking periods. Also added a parallel estimation from $Z \rightarrow \mu\mu$ data.
- Added a check on the independence of the fake track estimate dependence on the d_{xy} sideband definition, in Figure 27.
- We now have missing hits corrections for all periods. Uncorrected data and MC are shown in Figures 30 and 31, corrected data and MC in Figures 32 and 33, and Table 36 shows the hit removal probability for all periods.
- Added ratio plots to the trigger turn-on curves in Figure 35.
- Added 2023 data to the $Z\text{to}\mu\mu$ p_{T} distributions in Figure 37, for the ISR reweighting.
- Added a new subsection (7.1) on background estimates systematic uncertainties. Subsection on signal systematic uncertainties is now 7.2.
- Tables 42–46 now have signal systematic uncertainties and scale factors for all periods.
- Added ratio plots to the trigger turn-on curve in Figure 42.

- 40 • Added track reconstruction efficiency for all data taking periods in Figure 44 (straight
41 from Tracking POG).
- 42 • Table 46 now has complete signal systematic uncertainties for 2022 wino-like sam-
43 ples. Added Tables 47–49 for 2023 wino-like, and 2022–2023 higgsino-like samples.
- 44 • **Table 50 has the summary of numbers of events for the estimated backgrounds.**
45 Data are still blinded, of course.
- 46 • Added Figures 46, 48 and 50 for expected limits in the higgsino-like case.
- 47 • Added Figure 51 for the expected combination with Run 2 results in the wino-like
48 case.

Executive Summary

This note documents the search for disappearing tracks in 2022 and 2023 data at $\sqrt{s} = 13.6$ TeV pp collisions. This analysis continues the searches done with the Run 1 [1] and Run 2 [2, 3] datasets. The goal of the analysis is to search for beyond the standard model processes in which a heavy, long-lived charged particle decays in the volume of the silicon tracker, but its decay products are not detected / not reconstructed. The relevant particle masses are in the 100–1000 GeV range, and with lifetimes in the range of 10^{-2} – 10^2 ns.

The chosen theory benchmark is the anomaly-mediated SUSY breaking model [4, 5], in which the neutralino $\tilde{\chi}_1^0$ is the lightest supersymmetric particle (dark matter candidate) and the chargino $\tilde{\chi}_1^\pm$ is just slightly heavier than it. In this scenario the dominant decay is $\tilde{\chi}_1^\pm \rightarrow \tilde{\chi}_1^0 \pi^\pm$, but the pion has very low p_T , hence going undetected. The process signature is then a high- p_T , isolated, charged particle track that has missing hits in the outer layers of the tracker; since the particle does not reach the calorimeters, it has an absence of associated calorimeter deposits.

Since CMS does not yet have Level-1 Track Triggering, collection of such events is challenging, and we rely on the presence of initial state radiation, in the form of an hadronic jet, together with missing transverse momentum to trigger them. The collision datasets used are discussed in Section 2.1, with a discussion on the triggers in Section 3. Conversely, the simulation of signal samples is discussed in Section 2.2, with additional details in Section 6.

There are very few standard model processes that can mimic the disappearing tracks signature. Therefore, the background to the search is dominated by instrumental sources, essentially hardware and track reconstruction failures. A large part of the analysis consists of identifying and preventing those sources, going beyond the standard selection of good physics objects. The definition of *fiducial track selections* is described in Section 4.2; it includes tight selection of detector regions, as well as a dedicated study of track inefficiency (the *fiducial maps*) that aims at uncovering additional noisy channel prior to any unblinding. The event selection is described in Section 4.3 and is divided in four steps:

- a *basic selection* that targets the jet + p_T^{miss} signature (Section 4.3.1),
- an *isolated track selection* that targets fiducial, high- p_T , isolated tracks (Section 4.3.2),
- a *candidate track selection* that aims at removing the charged leptons from weak boson decays (Section 4.3.3),
- a *disappearing track selection* that targets the signature in question (Section 4.3.4).

A final categorization in the number of tracker layers with measurement associated to the selected track is done to increase sensitivity. The category in which the track has six layers has the highest sensitivity, with the expected background being essentially zero.

The instrumental backgrounds are of two kinds: charged leptons from weak boson decays that survive candidate track selection, and fake tracks that comprise random hits in the tracker, i.e. errors in the track reconstruction. All the background estimations are data-driven, with closure tests done in simulated samples. The analysis is still blinded, and the estimated backgrounds for the 2022 and 2023 datasets are discussed in Section 5. The charged lepton background is estimated via the calculation of the relevant probabilities: that of lepton failure reconstruction while still leaving a track in the silicon tracker, and that of appearance of associated p_T^{miss} of a large enough magnitude as to be selected both at the trigger level and offline. The details are described in Section 5.1. Conversely, the fake track background is estimated from a sample of tracks with high $|d_{xy}|$ in dilepton control regions, assumed to be composed exclusively of fake tracks. The details are described in Section 5.2.

1 Introduction

This note presents a search for long-lived, charged, beyond-the-standard-model (BSM) particles that decay within the volume of the silicon tracker and produce the signature of a “disappearing track”. A disappearing track is produced by a BSM charged particle whose decay products are undetected because they are either too soft to be reconstructed or interact only weakly such that they do not interact with the tracker material nor deposit significant energy in the calorimeters.

Anomaly-mediated supersymmetry breaking (AMSB) [4, 5] is one of the most often cited BSM scenarios that can produce a disappearing track. AMSB predicts a particle mass spectrum in which there is a small mass splitting between the lightest chargino ($\tilde{\chi}_1^\pm$) and neutralino ($\tilde{\chi}_1^0$), with the latter being the lightest supersymmetric particle (LSP). The chargino decays to a neutralino and a pion, $\tilde{\chi}_1^\pm \rightarrow \tilde{\chi}_1^0 \pi^\pm$. The phase space for this decay is limited by the small chargino-neutralino mass splitting. As a consequence, the chargino has a lifetime on the order of 1 ns, and the daughter pion has low momentum (~ 100 MeV), typically too small for the pion to be reconstructed as a track. If the chargino decays inside the tracker volume, it thus produces a disappearing track. We benchmark our search in terms of the chargino mass and lifetime in AMSB for several production modes, although we note there are other BSM scenarios that produce the disappearing track signature [6–11].

Because no tracking is currently done in the Level 1 (L1) trigger of CMS, events are collected with triggers based on L1 seeds requiring missing transverse momentum. The missing transverse momentum vector $\vec{p}_T^{\text{miss}}$ is defined as the negative vector sum of the momenta of all reconstructed particle-flow (PF) [12] objects in the event, projected onto the plane perpendicular to the beam axis. The magnitude of $\vec{p}_T^{\text{miss}}$ is referred to as p_T^{miss} . In the L1 trigger, p_T^{miss} is produced by the recoil of the BSM particles against initial-state radiation (ISR).

To select candidate disappearing tracks, we first require events to contain an isolated, high- p_T track. Next, this track is required to not be consistent with a charged lepton. Finally, candidate tracks are required to have small associated energy deposits in the calorimeter, and to have several missing outer hits in the silicon tracker, the distinctive features of disappearing tracks from signal processes. By contrast, tracks from background standard model processes typically have few missing outer tracker hits or have large associated calorimeter energy deposits, as illustrated in Figure 1. Standard model processes only produce a disappearing track signature if some aspect of the reconstruction fails, for example if the electromagnetic calorimeter does not reconstruct all of the energy of an electron. These background contributions are estimated with data-driven techniques.

Three previous CMS analyses searched for disappearing tracks in 19.5 fb^{-1} of 8 TeV, 38.4 fb^{-1} of 13 TeV and 101 fb^{-1} of 13 TeV pp collisions recorded in 2012 [1], 2015-6 [2] and 2017-8 [3], respectively. These searches exclude at 95% confidence level the direct electroweak production of charginos with a mass less than 884 GeV for a mean proper lifetime of 3 ns. This search differs from previous searches due to several different variables used in the events’ analysis to improve the workflow of the analysis, by taking advantage of quantities available in the MINIAOD data format. Specifically the calculation of energy deposited in the ECAL around candidate disappearing tracks is changed to use a sum of the `isolatedTracks matchedCaloJetEMEnergy` and `matchedCaloJetHadEnergy` quantities which provide the energy of the closest jets (using dR matching). In the run 2 version of this analysis the ECAL energy deposits were calculated as a sum of all AOD rec hits meeting the requirement $dR < 0.5$ around a candidate track. It also uses a new method for events generation based on MadGraph in order to decrease one of the largest sources of systematic uncertainties of the analysis. In the latest Run 2 analysis [3]

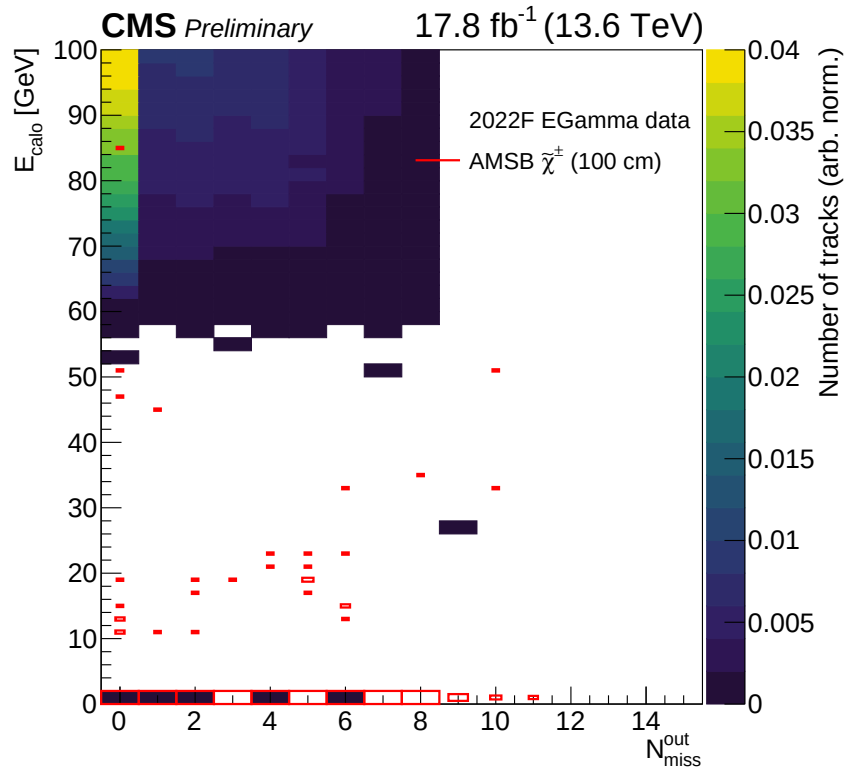


Figure 1: Associated calorimeter energy versus number of missing outer hits for selected tracks with $n_{\text{layers}} \geq 6$ from a control region dominated by electrons, the largest background in this category. The data collected by CMS in 2022F is shown as colored bins, while a simulation of electroweak AMSB chargino production is shown as black boxes, with the size of each box indicating the relative size of the contents in that bin. The particular signal model shown here has a chargino with a mass of 700 GeV and a lifetime of $c\tau = 100$ cm. Associated calorimeter energy and number of missing outer hits are the primary variables used to distinguish signal from background in the search, as an obvious separation between the two is seen in this plane.

this uncertainty source was related to scale factors applied over the signal samples to correct the Pythia simulation to be closer to a MadGraph simulation, which is known to make a better estimation of initial state radiation momentum. In the latest analysis, this systematic source was on the order of 15%, and in this analysis it was reduced to about 7%. This method was extensively validated in close contact with the CMS EXO MC&I and GEN groups. Lastly, as the luminosity of the dataset used in this analysis is smaller than that of Run 2, the plan is to combine our results with those analyses. Candidate disappearing tracks in this search are separated into three signal regions depending on how many layers of the tracker they traverse (n_{layers}): exactly 4, exactly 5, and 6 or more layers. Furthermore in the expression of this search's results in the context of AMSB, an additional production mode is included over the previous analyses, the direct electroweak production of charginos with a purely higgsino-like neutralino.

This note is organized as follows. Section 2 lists all data and simulated samples used for this search, and describes the generation of the benchmark AMSB models. The triggers used to collect events are described in Section 3, and Section 4 details the offline selection requirements used to identify candidate signal tracks. Section 5 describes the methods used to estimate the backgrounds and Section 6 describes the methods used to correct the simulated signal samples. Section 7 details the evaluation of systematic uncertainties, and Section 8 presents the results of the search and their interpretation.

2 Samples of experimental and simulated data

2.1 Experimental data

The search is performed in pp collision data at $\sqrt{s} = 13.6$ TeV collected in 2022 and 2023, corresponding respectively to an integrated luminosity of 34.7 fb^{-1} and 27.8 fb^{-1} . The 2022 data were reconstructed in various versions of CMSSW_13_0_X in the 22Sep2023-v1 (2) re-miniAOD of the 22Sep2023 re-reconstruction. The 2023 data were reconstructed in various versions of CMSSW_13_0_X in the 22Sep2023-v4 miniAOD re-reconstruction. The analysis of the data proceeded in CMSSW_13_0_13 with global tags 130X_dataRun3_v2 and 130X_dataRun3_PromptAnalysis_v1 for 2022, and in CMSSW_13_0_13 with global tag 130X_dataRun3_PromptAnalysis_v1 for 2023. The full list of data samples used in 2022 and 2023 are summarized in Tables 1 and 2 respectively. To select luminosity blocks certified for use in physics analyses, we apply the JSONs:

- Cert_Collisions2022_355100_362760_Golden.json
- Cert_Collisions2023_366442_370790_Golden.json

Table 1: Samples used for the search and for background estimation and their respective integrated luminosities.

Sample	Integrated Luminosity [fb^{-1}]
/MET/Run2022C-22Sep2023-v1/MINIAOD	0.64
/JetMET/Run2022C-22Sep2023-v1/MINIAOD	4.47
/JetMET/Run2022D-22Sep2023-v1/MINIAOD	2.97
/JetMET/Run2022E-22Sep2023-v1/MINIAOD	5.81
/JetMET/Run2022F-22Sep2023-v2/MINIAOD	17.8
/JetMET/Run2022G-22Sep2023-v2/MINIAOD	3.08
/EGamma/Run2022C-22Sep2023-v1/MINIAOD	5.01
/EGamma/Run2022D-22Sep2023-v1/MINIAOD	2.97
/EGamma/Run2022E-22Sep2023-v1/MINIAOD	5.81
/EGamma/Run2022F-22Sep2023-v1/MINIAOD	17.8
/EGamma/Run2022G-22Sep2023-v2/MINIAOD	3.08
/Muon/Run2022C-22Sep2023-v1/MINIAOD	5.01
/Muon/Run2022D-22Sep2023-v1/MINIAOD	2.97
/Muon/Run2022E-22Sep2023-v1/MINIAOD	5.81
/Muon/Run2022F-22Sep2023-v2/MINIAOD	17.8
/Muon/Run2022G-22Sep2023-v1/MINIAOD	3.08

2.2 Simulated signal events

AMSB is used as a benchmark for evaluating the sensitivity of the search, simulated in a variety of production modes. The mass spectrum in AMSB is determined by four parameters: the gravitino mass $m_{3/2}$, the universal scalar mass m_0 , the ratio of vacuum expectation values of the Higgs field $\tan\beta$, and the sign of the higgsino mass term $\text{sign}(\mu)$. Of these four only $m_{3/2}$ has a significant impact on the mass of the chargino, and by varying this parameter the mass spectrum is calculated using ISAJET 7.80 [13] for chargino masses between 100 and 1200 GeV (1000 GeV) in intervals of 100 GeV for the case of a purely wino-like (higgsino-like) LSP case. The SUSY parameters for each mass point are listed in Table 3.

Direct electroweak chargino production is simulated via $pp \rightarrow \tilde{\chi}_1^\pm \tilde{\chi}_1^\mp$ and $pp \rightarrow \tilde{\chi}_{1,2}^0 \tilde{\chi}_1^\pm$, with the addition of up to one extra parton. This simulation is different than what was performed in

Table 2: Samples used for the search and for background estimation and their respective integrated luminosities. The suffix N indicate that the datasets are further divided into two primary datasets: Dataset0 and Dataset1. Both datasets for every version/process were processed.

Sample	Integrated Luminosity [fb^{-1}]
/JetMETN/Run2023C-22Sep2023_v1-v1/MINIAOD	4.35
/JetMETN/Run2023C-22Sep2023_v2-v1/MINIAOD	1.27
/JetMETN/Run2023C-22Sep2023_v3-v1/MINIAOD	1.60
/JetMETN/Run2023C-22Sep2023_v4-v1/MINIAOD	10.9
/JetMETN/Run2023D-22Sep2023_v1-v1/MINIAOD	7.99
/JetMETN/Run2023D-22Sep2023_v2-v1/MINIAOD	1.70
/EGammaN/Run2023C-22Sep2023_v1-v1/MINIAOD	4.35
/EGammaN/Run2023C-22Sep2023_v2-v1/MINIAOD	1.27
/EGammaN/Run2023C-22Sep2023_v3-v1/MINIAOD	1.60
/EGammaN/Run2023C-22Sep2023_v4-v1/MINIAOD	10.9
/EGammaN/Run2023D-22Sep2023_v1-v1/MINIAOD	7.99
/EGammaN/Run2023D-22Sep2023_v2-v1/MINIAOD	1.70
/MuonN/Run2023C-22Sep2023_v1-v1/MINIAOD	4.35
/MuonN/Run2023C-22Sep2023_v2-v1/MINIAOD	1.27
/MuonN/Run2023C-22Sep2023_v3-v1/MINIAOD	1.60
/MuonN/Run2023C-22Sep2023_v4-v1/MINIAOD	10.9
/MuonN/Run2023D-22Sep2023_v1-v1/MINIAOD	7.99
/MuonN/Run2023D-22Sep2023_v2-v1/MINIAOD	1.71

Table 3: SUSY parameters used to generate simulated signal samples for each considered chargino mass. The large mass scale of the squarks is denoted by the mass of the down squark $m(\tilde{d})$. The other parameters are described in the text.

m_0 [GeV]	$m_{3/2}$ [TeV]	$\tan \beta$	$\text{sign}(\mu)$	$m(\tilde{\chi}_1^\pm)$ [GeV]	Δm [GeV]	$m(\tilde{d})$ [GeV]
1500	35.1	5	+1	100.0	0.189	1649
1500	68.9	5	+1	200.0	0.174	2019
1500	103.7	5	+1	300.0	0.172	2494
1500	139.0	5	+1	400.0	0.173	3018
1500	174.4	5	+1	500.0	0.173	3561
1500	210.3	5	+1	600.0	0.173	4121
1500	246.4	5	+1	700.0	0.173	4686
1500	282.9	5	+1	800.0	0.174	5261
1500	320.2	5	+1	900.0	0.174	5845
1500	320.2	5	+1	1000.0	0.174	5845
1500	320.2	5	+1	1100.0	0.174	5845
1500	320.2	5	+1	1200.0	0.174	5845

the last analyses, and the details on the methods used and checks performed are in appendix A. In the case of a purely wino-like LSP, only $\tilde{\chi}_1^0$ is produced as the $\tilde{\chi}_2^0$ is decoupled at large mass, and the chargino branching fraction for $\tilde{\chi}_1^\pm \rightarrow \tilde{\chi}_1^0 \pi^\pm$ is set to 100%. In the case of a purely higgsino-like LSP, the chargino branching fraction is set to 95.5% for $\tilde{\chi}_1^\pm \rightarrow \tilde{\chi}_{1,2}^0 \pi^\pm$, 3% for $\tilde{\chi}_1^\pm \rightarrow \tilde{\chi}_{1,2}^0 e \bar{\nu}_e$, and 1.5% for $\tilde{\chi}_1^\pm \rightarrow \tilde{\chi}_{1,2}^0 \mu \bar{\nu}_\mu$ [14], where $\tilde{\chi}_{1,2}^0$ are completely mass degenerate with equal branching fractions and cross sections.

The cross sections for electroweak production of charginos [15] are calculated as a function of chargino mass and are listed in Table 4 for the wino-like case and Table 5 for the higgsino-like case.

Table 4: Cross sections for electroweak production of wino-like charginos calculated at aNNLO+NNLL. The cross sections for $pp \rightarrow \tilde{\chi}_1^\pm \tilde{\chi}_1^\mp$ and $pp \rightarrow \tilde{\chi}_1^0 \tilde{\chi}_1^\pm$ are listed separately, along with the total cross section, which is the sum of these two contributions.

$m(\tilde{\chi}_1^\pm)$ [GeV]	$\sigma(pp \rightarrow \tilde{\chi}_1^\pm \tilde{\chi}_1^\mp)$ [pb]	$\sigma(pp \rightarrow \tilde{\chi}_1^0 \tilde{\chi}_1^\pm)$ [pb]	total σ [pb]
100	$12.84^{+0.44}_{-0.51}$	$24.92^{+0.64}_{-0.74}$	$37.76^{+0.78}_{-0.90}$
200	$1.012^{+0.035}_{-0.035}$	$2.016^{+0.058}_{-0.059}$	$3.028^{+0.068}_{-0.069}$
300	$0.2155^{+0.0087}_{-0.0088}$	$0.437^{+0.015}_{-0.015}$	$0.652^{+0.018}_{-0.018}$
400	$0.0667^{+0.0032}_{-0.0032}$	$0.1378^{+0.0058}_{-0.0058}$	$0.2048^{+0.0067}_{-0.0067}$
500	$0.0255^{+0.0014}_{-0.0014}$	$0.0532^{+0.0026}_{-0.0026}$	$0.0787^{+0.0030}_{-0.0030}$
600	$0.0110^{+0.00073}_{-0.00073}$	$0.0233^{+0.0013}_{-0.0013}$	$0.0343^{+0.0015}_{-0.0015}$
700	$0.00519^{+0.00040}_{-0.00040}$	$0.01111^{+0.00074}_{-0.00074}$	$0.01629^{+0.00084}_{-0.00084}$
800	$0.00260^{+0.00023}_{-0.00023}$	$0.00562^{+0.00044}_{-0.00044}$	$0.00822^{+0.00050}_{-0.00050}$
900	$0.00137^{+0.00015}_{-0.00015}$	$0.00298^{+0.00027}_{-0.00027}$	$0.00435^{+0.00031}_{-0.00031}$
1000	$0.000745^{+0.000095}_{-0.000095}$	$0.00163^{+0.00018}_{-0.00018}$	$0.00238^{+0.00020}_{-0.00020}$
1100	$0.000418^{+0.000064}_{-0.000064}$	$0.00092^{+0.00012}_{-0.00012}$	$0.00134^{+0.00014}_{-0.00014}$
1200	$0.000241^{+0.000045}_{-0.000045}$	$0.000531^{+0.000084}_{-0.000084}$	$0.000771^{+0.000096}_{-0.000096}$

Table 5: Cross sections for electroweak production of higgsino-like charginos calculated at aNNLO+NNLL. The $\tilde{\chi}_2^0$ is completely mass degenerate with the $\tilde{\chi}_1^0$ with the same production cross sections, so the total includes an additional contribution equal to that listed for $\sigma(pp \rightarrow \tilde{\chi}_1^0 \tilde{\chi}_1^\pm)$.

$m(\tilde{\chi}_1^\pm)$ [GeV]	$\sigma(pp \rightarrow \tilde{\chi}_1^\pm \tilde{\chi}_1^\mp)$ [fb]	$\sigma(pp \rightarrow \tilde{\chi}_1^0 \tilde{\chi}_1^\pm)$ [fb]	total σ [fb]
100	$3328.8^{+109.525}_{-129.701}$	$6229.35^{+160.588}_{-185.265}$	$15787.5^{+252.137}_{-292.35}$
200	$282.34^{+9.60048}_{-9.78659}$	$504.02^{+14.484}_{-14.683}$	$1290.38^{+22.62}_{-22.96}$
300	$61.2041^{+2.45229}_{-2.45521}$	$109.167^{+3.84543}_{-3.84871}$	$279.5^{+5.97}_{-5.97}$
400	$19.1958^{+0.913918}_{-0.914351}$	$34.4632^{+1.46005}_{-1.45856}$	$88.12^{+2.26}_{-2.26}$
500	$7.35264^{+0.412507}_{-0.412511}$	$13.3129^{+0.661957}_{-0.661395}$	$33.98^{+1.02}_{-1.02}$
600	$3.19172^{+0.209581}_{-0.209431}$	$5.82839^{+0.336897}_{-0.336617}$	$14.85^{+0.52}_{-0.52}$
700	$1.50946^{+0.116121}_{-0.116051}$	$2.77757^{+0.185965}_{-0.185817}$	$7.06^{+0.29}_{-0.29}$
800	$0.759528^{+0.0688722}_{-0.0688436}$	$1.4066^{+0.109758}_{-0.109697}$	$3.57^{+0.17}_{-0.17}$
900	$0.400664^{+0.0432108}_{-0.043196}$	$0.745364^{+0.0682122}_{-0.0681905}$	$1.89^{+0.11}_{-0.11}$
1000	$0.219273^{+0.0285098}_{-0.0285049}$	$0.40882^{+0.0444191}_{-0.0444085}$	$1.04^{+0.07}_{-0.07}$

Although in AMSB the mass spectrum (specifically the mass splitting between the chargino and neutralino) determines the chargino lifetime, by treating the lifetime as a free parameter we are able to evaluate the sensitivity of the search to a broader range of models. With just this selection of lifetimes, we are also able to study intermediate chargino lifetimes via lifetime

197 reweighting. To reweight a sample generated with a mean lifetime τ_{src} to a target mean lifetime
 198 of τ_{dst} , we define a weight:

$$w_{\tau} = \prod_i \frac{\left(\frac{1}{\tau_{\text{dst}}} e^{-\tau_i / \tau_{\text{dst}}} \right)}{\left(\frac{1}{\tau_{\text{src}}} e^{-\tau_i / \tau_{\text{src}}} \right)}, \quad (1)$$

199 where τ_i is the actual lifetime of the i -th chargino in a given event ($i = 1, 2$ for a $\text{pp} \rightarrow \tilde{\chi}_1^{\pm} \tilde{\chi}_1^{\mp}$
 200 event and $i = 1$ for a $\text{pp} \rightarrow \tilde{\chi}_{1,2}^0 \tilde{\chi}_1^{\pm}$ event). This weight is calculated and applied to each event¹,
 201 reshaping the distribution of actual chargino lifetimes with mean τ_{src} to that expected for a
 202 lifetime of τ_{dst} , as shown in Figure 2. The target lifetimes used and the original lifetimes of the
 203 samples from which they are reweighted are listed in Table 6.

Figure 2: Distributions of actual chargino lifetime at generator level. Both plots come from a sample generated with a mean chargino lifetime of 100 cm/c. The left plot is with no lifetime reweighting applied, while the right plot is reweighted to a target lifetime of 50 cm/c. The red curve in both cases is the result of fitting the distribution with an exponential function.

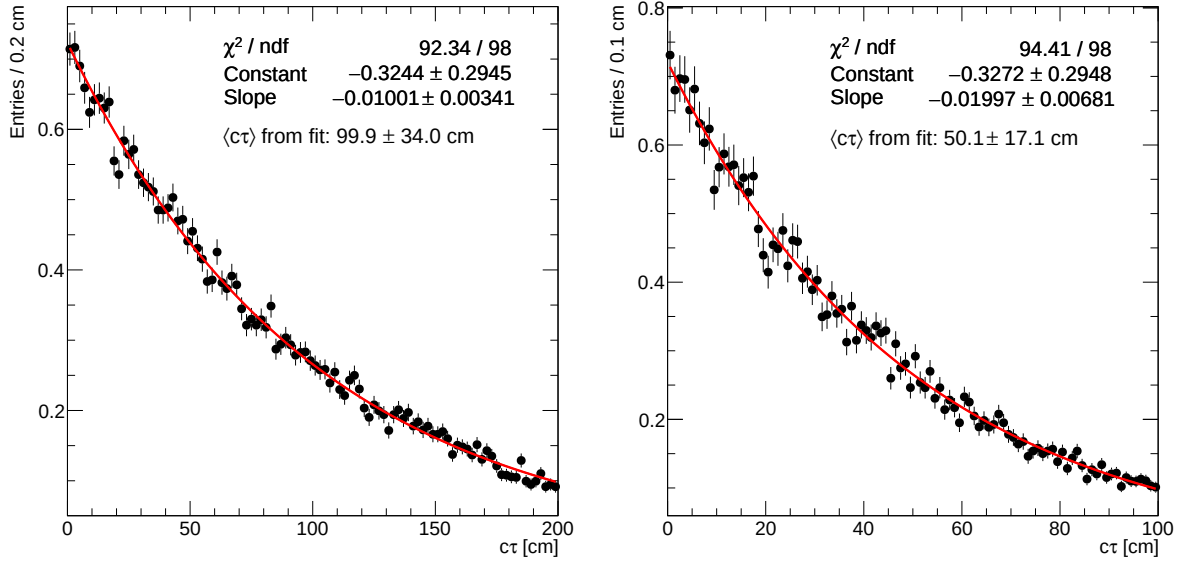


Table 6: Target chargino lifetimes achieved via lifetime reweighting and the original lifetime of the sample from which they are reweighted.

original lifetime [cm/c]	target lifetimes [cm/c]
10	0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10
100	20, 30, 40, 50, 60, 70, 80, 90, 100
1 000	200, 300, 400, 500, 600, 700, 800, 900, 1 000
10 000	2 000, 3 000, 4 000, 5 000, 6 000, 7 000, 8 000, 9 000, 10 000

204 The events are generated with MADGRAPH 2.9.9 [16, 17] at LO using the NNPDF3.1 NNLO [18]
 205 PDF sets, with PYTHIA 8.306 [19], and FASTJET version 3.4.0 [20] to simulate parton hadroni-
 206 zation and to execute the MLM jet matching algorithm [21, 22], respectively. Interactions with

¹The reweighting is always applied such that $\tau_{\text{dst}} < \tau_{\text{src}}$ so that events on the tail of the distribution of actual chargino lifetimes are never weighted up.

the detector and decays of long-lived particles, including the decays of charginos, are simulated with GEANT4 10.7.2 [23–25]. Events from samples of simulated minimum bias events are mixed in the digitization and reconstruction step to simulate the effects of pileup. The MC simulation conditions are different throughout 2022 and 2023 because of a couple of issues. In 2022, due to the ECAL endcap water leak described in section 4.2, which affected collected data from eras E, F and G, the MC campaigns are referred to 2022 and 2022EE for pre and post ECAL endcap issue, respectively. In 2023, an issue with two pixel barrel layers after era C of 2023, described in section 4.3.2, caused a smaller efficiency in HLT and offline tracks reconstruction and the MC campaigns are referred to 2023 and 2023BPix for pre and post pixel barrel issue, respectively. In 2022 conditions the event generation, digitization and reconstruction are performed in CMSSW_12_4_11_patch3, and the MiniAOD step is performed in CMSSW_13_0_13 following the recipe for the Run3Summer22MiniAODv4 campaign with global tags 124X_mcRun3_2022_realistic_v12 and 130X_mcRun3_2022_realistic_v5. In 2022-EE conditions the event generation, digitization and reconstruction are performed in CMSSW_12_4_11_patch3, and the MiniAOD step is performed in CMSSW_13_0_13 following the recipe for the Run3Summer22EEMiniAODv4 campaign with global tags 124X_mcRun3_2022_realistic_postEE_v1 and 130X_mcRun3_2022_realistic_postEE_v6. In 2023 conditions the event generation, digitization and reconstruction are performed in CMSSW_13_0_13 following the recipe for the Run3Summer23MiniAODv4 campaign with global tag 130X_mcRun3_2022_realistic_v14. In 2023BPix conditions the event generation, digitization and reconstruction are performed in CMSSW_13_0_13 following the recipe for the Run3Summer23MiniAODv4 campaign with global tags 130X_mcRun3_2023_realistic_postBPix_v2 and 130X_mcRun3_2023_realistic_postBPix_v6.

2.3 Simulated background events

The backgrounds in this search are estimated using experimental data only, however samples of simulated background events are used to study and validate the data-driven methods. These samples are also used to assign systematic uncertainties to the expected signal acceptances.

All such samples are from the central Run3Summer22MiniAODv4, Run3Summer22EEMiniAODv4, Run3Summer23MiniAODv4 and Run3Summer23BPixMiniAODv4 campaigns, reconstructed respectively in CMSSW_12_4_11_patch3 and CMSSW_13_0_13 with global tags 124X_mcRun3_2022_realistic_v12 and 130X_mcRun3_2022_realistic_v5, 124X_mcRun3_2022_realistic_postEE_v1 and 130X_mcRun3_2022_realistic_postEE_v6, 130X_mcRun3_2022_realistic_v14, and 130X_mcRun3_2023_realistic_postBPix_v2 and 130X_mcRun3_2023_realistic_postBPix_v6, respectively for 2022 and 2023 conditions. The names of these samples are summarized in Table 7, and their cross sections and integrated luminosities are given in Table 8.

Table 7: Names of the samples of simulated background processes used to study and validate the background estimation methods in 2022 and 2023. Samples without superscript have the same naming in both years. Samples with superscript $[a]$ were used in 2022 and samples with superscript $[b]$ were used in 2023.

Composite Sample	Constituent Sample	Sample Name
$Z \rightarrow \ell\ell$	$10 \text{ GeV} < M_{\ell\ell} < 50 \text{ GeV}$	DYto2L-4Jets_MLL-10to50_TuneCP5.13p6TeV_madgraphMLM-pythia8
	$M_{\ell\ell} > 50 \text{ GeV}^{[a]}$	DYJetsToLL_M-50_TuneCP5.13p6TeV-madgraphMLM-pythia8
	$M_{\ell\ell} > 50 \text{ GeV}^{[b]}$	DYto2L-4Jets_MLL-50_TuneCP5.13p6TeV-madgraphMLM-pythia8
single top	$\bar{t}b \rightarrow l\nu b$	TbarBtoLminusNuB-s-channel-4FS_TuneCP5.13p6TeV_amcatnlo-pythia8
	$\bar{t}b \rightarrow l\nu \bar{b}$	TBbartoLplusNuBbar-s-channel-4FS_TuneCP5.13p6TeV_amcatnlo-pythia8
	$\bar{t}q \rightarrow l\nu$	TbarQtoLnu-t-channel_TuneCP5.13p6TeV_powheg-pythia8
	$t\bar{q} \rightarrow l\nu$	TQbartoLnu-t-channel_TuneCP5.13p6TeV_powheg-pythia8
	$tW \rightarrow 2l2\nu$	TbarWplusto2L2Nu_TuneCP5.13p6TeV_powheg-pythia8
	$tW \rightarrow l\nu 2q$	TbarWplustoLNU2Q_TuneCP5.13p6TeV_powheg-pythia8
	$\bar{t}W \rightarrow 2l2\nu$	TWminusto2L2Nu_TuneCP5.13p6TeV_powheg-pythia8
	$\bar{t}W \rightarrow l\nu 2q$	TWminustoLNU2Q_TuneCP5.13p6TeV_powheg-pythia8
$t\bar{t}$	dilepton	TTto2L2Nu_TuneCP5.13p6TeV_powheg-pythia8
	semi-leptonic	TTtoLNU2Q_TuneCP5.13p6TeV_powheg-pythia8
	hadronic	TTto4Q_TuneCP5.13p6TeV_powheg-pythia8
diboson	WW	WW_TuneCP5.13p6TeV-pythia8
	WZ	WZ_TuneCP5.13p6TeV-pythia8
	ZZ	ZZ_TuneCP5.13p6TeV-pythia8
$W \rightarrow l\nu$	inclusive	WtoLNU-4Jets_TuneCP5.13p6TeV_madgraphMLM-pythia8
$Z \rightarrow \nu\bar{\nu}$	$100 \text{ GeV} < H_T < 200 \text{ GeV}$	Zto2Nu-4Jets_HT-100To200.13p6TeV-madgraphMLM-pythia8
	$200 \text{ GeV} < H_T < 400 \text{ GeV}$	Zto2Nu-4Jets_HT-200To400.13p6TeV-madgraphMLM-pythia8
	$400 \text{ GeV} < H_T < 800 \text{ GeV}$	Zto2Nu-4Jets_HT-400To800.13p6TeV-madgraphMLM-pythia8
	$800 \text{ GeV} < H_T < 1500 \text{ GeV}$	Zto2Nu-4Jets_HT-800To1500.13p6TeV-madgraphMLM-pythia8
	$1500 \text{ GeV} < H_T < 2500 \text{ GeV}$	Zto2Nu-4Jets_HT-1500To2500.13p6TeV-madgraphMLM-pythia8
	$H_T > 2500 \text{ GeV}$	Zto2Nu-4Jets_HT-2500.13p6TeV-madgraphMLM-pythia8
QCD	$15 \text{ GeV} < \hat{p}_T < 30 \text{ GeV}$	QCD_PT-15to30_TuneCP5.13p6TeV-pythia8
	$30 \text{ GeV} < \hat{p}_T < 50 \text{ GeV}$	QCD_PT-30to50_TuneCP5.13p6TeV-pythia8
	$50 \text{ GeV} < \hat{p}_T < 80 \text{ GeV}$	QCD_PT-50to80_TuneCP5.13p6TeV-pythia8
	$80 \text{ GeV} < \hat{p}_T < 120 \text{ GeV}$	QCD_PT-80to120_TuneCP5.13p6TeV-pythia8
	$120 \text{ GeV} < \hat{p}_T < 170 \text{ GeV}$	QCD_PT-120to170_TuneCP5.13p6TeV-pythia8
	$170 \text{ GeV} < \hat{p}_T < 300 \text{ GeV}$	QCD_PT-170to300_TuneCP5.13p6TeV-pythia8
	$300 \text{ GeV} < \hat{p}_T < 470 \text{ GeV}$	QCD_PT-300to470_TuneCP5.13p6TeV-pythia8
	$470 \text{ GeV} < \hat{p}_T < 600 \text{ GeV}$	QCD_PT-470to600_TuneCP5.13p6TeV-pythia8
	$600 \text{ GeV} < \hat{p}_T < 800 \text{ GeV}$	QCD_PT-600to800_TuneCP5.13p6TeV-pythia8
	$800 \text{ GeV} < \hat{p}_T < 1000 \text{ GeV}$	QCD_PT-800to1000_TuneCP5.13p6TeV-pythia8
	$1000 \text{ GeV} < \hat{p}_T < 1400 \text{ GeV}$	QCD_PT-1000to1400_TuneCP5.13p6TeV-pythia8
	$1400 \text{ GeV} < \hat{p}_T < 1800 \text{ GeV}$	QCD_PT-1400to1800_TuneCP5.13p6TeV-pythia8
	$1800 \text{ GeV} < \hat{p}_T < 2400 \text{ GeV}$	QCD_PT-1800to2400_TuneCP5.13p6TeV-pythia8
	$2400 \text{ GeV} < \hat{p}_T < 3200 \text{ GeV}$	QCD_PT-2400to3200_TuneCP5.13p6TeV-pythia8
	$\hat{p}_T > 3200 \text{ GeV}$	QCD_PT-3200_TuneCP5.13p6TeV-pythia8

Table 8: Cross sections and equivalent integrated luminosities of the samples of simulated background processes used to study and validate the background estimation methods.

composite sample	constituent sample	$\sigma \times \text{BR}$ [pb]	$\mathcal{L}_{\text{equiv.}}$ [pb^{-1}]	
			2022	2023
$Z \rightarrow \text{ll}$	$10 \text{ GeV} < M_{\text{ll}} < 50 \text{ GeV}$	20 950	32 668	21 832
	$M_{\text{ll}} > 50 \text{ GeV}$	6 331.5	19 317	31 791
single top	$\bar{t}b \rightarrow l\nu b$	1.35	2 628 287	1 765 926
	$t\bar{b} \rightarrow l\nu \bar{b}$	2.30	2 498 263	1 681 739
	$\bar{t}q \rightarrow l\nu$	26.18	964 416	661 306
	$t\bar{q} \rightarrow l\nu$	43.91	950 409	635 868
	$tW \rightarrow 2l2\nu$	14.38	767 809	519 054
	$tW \rightarrow l\nu 2q$	29.63	695 782	492 995
	$\bar{t}W \rightarrow 2l2\nu$	14.38	765 558	516 412
	$\bar{t}W \rightarrow l\nu 2q$	29.63	734 585	499 221
$t\bar{t}$	dilepton	98.88	1 108 676	736 377
	semi-leptonic	407.43	862 356	578 978
	hadronic	419.69	563 723	377 157
diboson	WW	80.23	863 196	626 212
	WZ	29.1	1 183 632	864 124
	ZZ	12.75	410 131	295 765
$W \rightarrow l\nu$	inclusive	64 451.4	6 770	4 435
$Z \rightarrow \nu\bar{\nu}$	$100 \text{ GeV} < H_T < 200 \text{ GeV}$	314.03	435 927	283 889
	$200 \text{ GeV} < H_T < 400 \text{ GeV}$	87.15	991 034	673 258
	$400 \text{ GeV} < H_T < 800 \text{ GeV}$	15.13	594 631	389 161
	$800 \text{ GeV} < H_T < 1500 \text{ GeV}$	1.56	1 402 519	881 855
	$1500 \text{ GeV} < H_T < 2500 \text{ GeV}$	0.113	20 050 938	13 061 841
	$H_T > 2500 \text{ GeV}$	0.0077	284 995 584	183 999 221
QCD	$15 \text{ GeV} < \hat{p}_T < 30 \text{ GeV}$	1.301×10^9	0.004	0.003
	$30 \text{ GeV} < \hat{p}_T < 50 \text{ GeV}$	1.133×10^8	0.046	0.031
	$50 \text{ GeV} < \hat{p}_T < 80 \text{ GeV}$	1.676×10^7	2.366	1.073
	$80 \text{ GeV} < \hat{p}_T < 120 \text{ GeV}$	2.534×10^6	23.56	10.63
	$120 \text{ GeV} < \hat{p}_T < 170 \text{ GeV}$	445 800	129.4	60.40
	$170 \text{ GeV} < \hat{p}_T < 300 \text{ GeV}$	113 700	509.9	235.9
	$300 \text{ GeV} < \hat{p}_T < 470 \text{ GeV}$	7 559	15 020	6 876
	$470 \text{ GeV} < \hat{p}_T < 600 \text{ GeV}$	626.4	87 164	40 150
	$600 \text{ GeV} < \hat{p}_T < 800 \text{ GeV}$	178.6	742 003	341 915
	$800 \text{ GeV} < \hat{p}_T < 1000 \text{ GeV}$	30.57	2 552 698	1 173 831
	$1000 \text{ GeV} < \hat{p}_T < 1400 \text{ GeV}$	8.92	4 395 612	2 010 762
	$1400 \text{ GeV} < \hat{p}_T < 1800 \text{ GeV}$	0.810 3	14 410 919	6 651 857
	$1800 \text{ GeV} < \hat{p}_T < 2400 \text{ GeV}$	0.114 8	51 425 217	23 449 477
	$2400 \text{ GeV} < \hat{p}_T < 3200 \text{ GeV}$	0.007 542	511 565 367	238 530 894
	$\hat{p}_T > 3200 \text{ GeV}$	2.331×10^{-4}	6.797×10^9	3.080×10^9

3 Triggers

3.1 Use of p_T^{miss}

As CMS currently has no tracking at Level 1 (L1), it is not possible to trigger on tracks alone. Therefore we need to use another object, and we employ the same strategy as do collider searches for dark matter by triggering on p_T^{miss} . Our signal can generate p_T^{miss} in one of three ways:

- a $\tilde{\chi}_1^\pm$ can be produced with a $\tilde{\chi}_1^0$, where the $\tilde{\chi}_1^\pm$ reaches the muon chambers (see Figure 3, top)
- two $\tilde{\chi}_1^\pm$ can be produced and only one reaches the muon chambers (see Figure 3, bottom)²
- the ewkino pair can be accompanied by initial-state radiation (ISR) (see Figure 4)

In the first two cases, the $\tilde{\chi}_1^\pm$ that reaches the muon chambers is reconstructed as a muon, and therefore contributes visible momentum to PF p_T^{miss} , but not to calorimeter-based p_T^{miss} . On the other hand, the $\tilde{\chi}_1^0$ contributes no visible momentum since it interacts only weakly. The $\tilde{\chi}_1^\pm$ that fails to reach the muon chambers also contributes no visible momentum because, being a track with no calorimeter energy nor muon chamber hits, it is considered a fake track by particle flow and ignored. Finally, in the third case, if the ISR is a gluon and has sufficient p_T , then it can contribute substantially to both PF p_T^{miss} and to calorimeter-based p_T^{miss} .

Again as there is no tracking available at L1, the only type of p_T^{miss} that can be used at L1 is calorimeter-based p_T^{miss} . As such a signal event with p_T^{miss} generated by a high- p_T ISR gluon jet is precisely the category of event we target.

²This type of event could be selected using a single-muon trigger. This was investigated, however, this muon + disappearing track topology, which would be relevant for longer lifetimes, is expected to be completely covered by the HSCP search.

Figure 3: Event displays of simulated signal events, one of $pp \rightarrow \tilde{\chi}_1^0 \tilde{\chi}_1^\pm$ (top) and one of $pp \rightarrow \tilde{\chi}_1^\pm \tilde{\chi}_1^\mp$ (bottom). The PF $\vec{p}_T^{\text{miss}}$ vector is shown as a red arrow and the calorimeter-based $\vec{p}_T^{\text{miss}}$ vector is shown as a gold arrow.

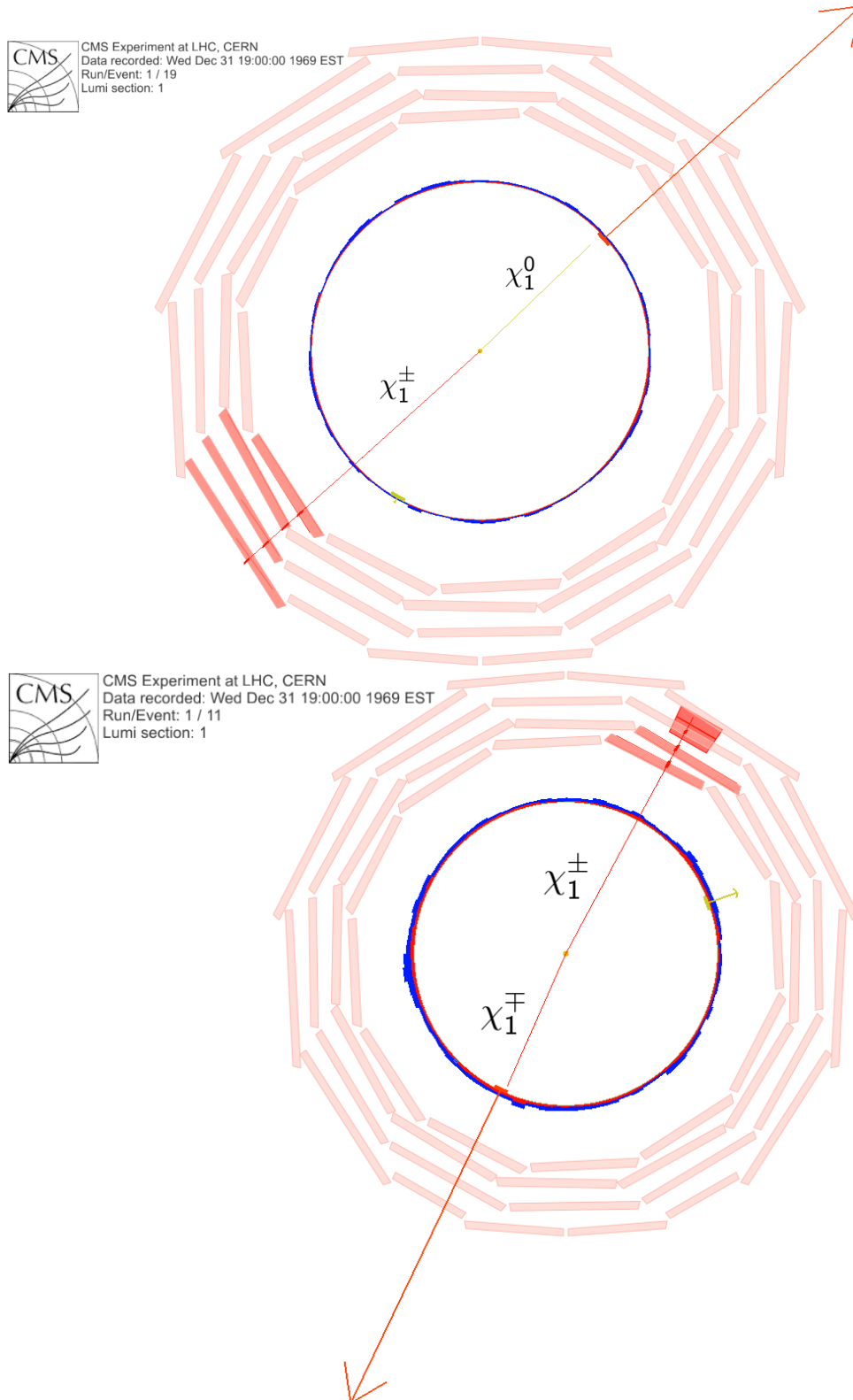
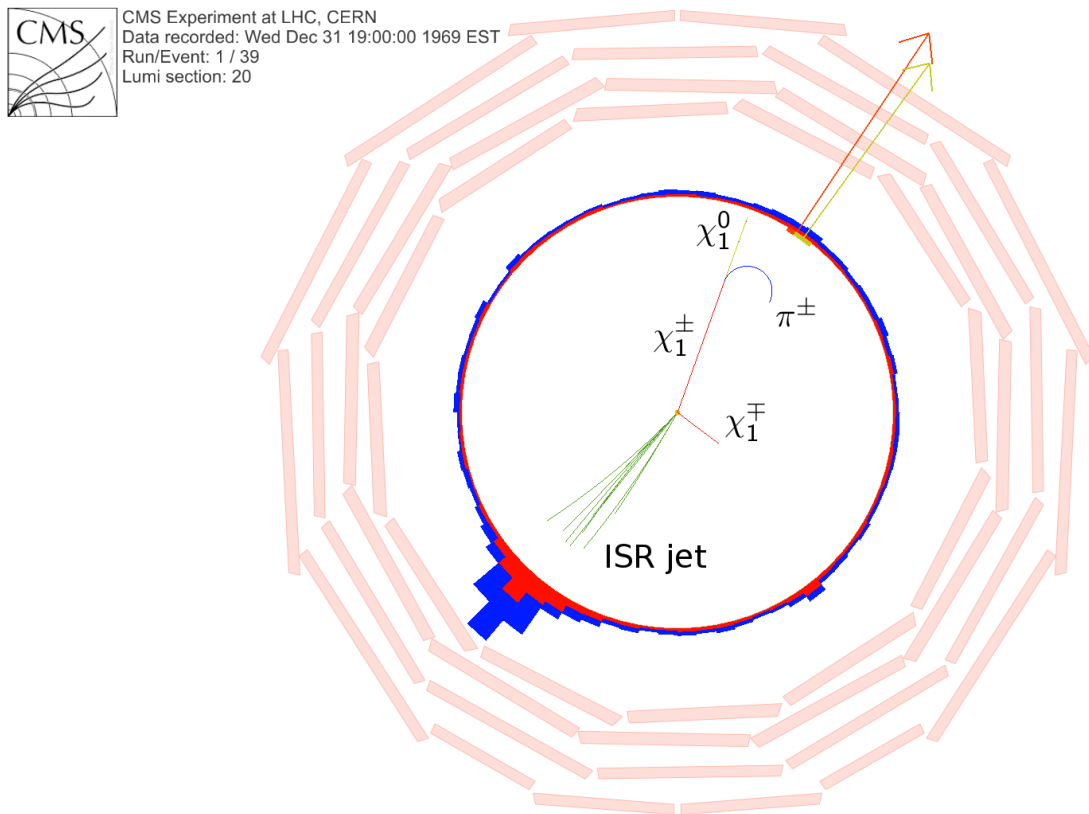


Figure 4: Event display of a simulated $pp \rightarrow \tilde{\chi}_1^0 \tilde{\chi}_1^\pm$ event with an ISR gluon jet. The PF $\vec{p}_T^{\text{miss}}$ vector is shown as a red arrow and the calorimeter-based $\vec{p}_T^{\text{miss}}$ vector is shown as a gold arrow.



3.2 Triggers used in the search

This search uses data collected with the logical OR of several p_T^{miss} triggers. Various single lepton triggers are also used for the estimation of backgrounds and for studies of systematic uncertainties. The complete list of triggers used is shown in Table 9.

Some issues affecting these paths arose throughout the 2022 and 2023 data-taking periods. The HLT_PFMETNoMu110_PFMHTNoMu110_IDTight_FilterHF was disabled from era G of 2022 onwards. In the middle of era C of 2022, an issue with the jet scale calibrations at the Level-1 trigger cause a decrease in efficiency of several inclusive p_T^{miss} paths³. Finally, in era D of 2023, the issue with the pixel barrel caused a decrease in efficiency of tracks reconstruction. However, due to the addition of a recovery module on the HLT tracks reconstruction, the effect on the efficiency of the used paths is found to be negligible.

Table 9: Triggers used in the search. The MET data set is used as the search data set and the evaluation of fake track background systematic uncertainties, and the single lepton data sets are used for background estimation and systematic uncertainty evaluation. The check marks indicate which data-taking periods in which a particular trigger was used. Triggers with notes have the following remarks: a) was disabled in the two highest-rate prescale columns during 2022 G and 2023.

Primary Data Set	Trigger Path	2022	2023	Notes
MET	HLT_MET105_IsoTrk50_v*	✓	✓	
	HLT_MET120_IsoTrk50_v*	✓	✓	
	HLT_PFMET105_IsoTrk50_v*	✓	✓	
	HLT_PFMET120_PFMHT120_IDTight_v*	✓	✓	
	HLT_PFMET130_PFMHT130_IDTight_v*	✓	✓	
	HLT_PFMET140_PFMHT140_IDTight_v*	✓	✓	
	HLT_PFMETNoMu120_PFMHTNoMu120_IDTight_v*	✓	✓	
	HLT_PFMETNoMu130_PFMHTNoMu130_IDTight_v*	✓	✓	
	HLT_PFMETNoMu140_PFMHTNoMu140_IDTight_v*	✓	✓	
	HLT_PFMETNoMu120_PFMHTNoMu120_IDTight_PFHT60_v*	✓	✓	
	HLT_PFMETNoMu110_PFMHTNoMu110_IDTight_FilterHF_v*	✓	✓	a
	HLT_PFMETNoMu120_PFMHTNoMu120_IDTight_FilterHF_v*	✓	✓	
	HLT_PFMETNoMu130_PFMHTNoMu130_IDTight_FilterHF_v*	✓	✓	
	HLT_PFMETNoMu140_PFMHTNoMu140_IDTight_FilterHF_v*	✓	✓	
	HLT_PFMET120_PFMHT120_IDTight_PFHT60_v*	✓	✓	
EGamma	HLT_Ele32_WPTight_Gsf_v*	✓	✓	
Muon	HLT_IsoMu24_v*	✓	✓	

The HLT_MET105_IsoTrk50_v* path was developed specifically for this search, was deployed and monitored throughout the Run 2 data-taking, and is still used for the Run 3 data analysis. The p_T^{miss} threshold was kept the same with respect to the Run 2 algorithm. For Run 3, there was also the deployment and monitoring of path HLT_PFMET105_IsoTrk50_v*, which makes use of lower threshold PF p_T^{miss} filter to increase the rates of relevant events to the analysis. The paths consist of three/four steps depending on the type of p_T^{miss} :

1. logical OR of several L1 p_T^{miss} seeds

³See the september 23rd 2022 Jets/MET HLT report on the impact of this issue in the efficiency of jet/MET paths for details.

2. high-level trigger (HLT) p_T^{miss} filter, `hltMET105` (or `hltMET75` in the PFMET path)
3. HLT track filter, `hltTrk50Filter`
4. high-level trigger (HLT) PF p_T^{miss} filter, `hltPFMET105` (only in the PFMET path)

At HLT the p_T^{miss} filter requires `hltMET` > 105 GeV, that is calorimeter-based p_T^{miss} calculated in the HLT, and the PF p_T^{miss} filter requires `hltPFMET` > 105 GeV, where the HLT uses information from all subsystems to calculate the p_T^{miss} . The track filter is processed only for events passing the p_T^{miss} filter and requires at least one HLT track to pass the following criteria:

- $p_T > 50$ GeV
- track-based isolation < $0.05p_T$
- ≥ 5 silicon hits
- no missing inner hits
- no missing middle hits

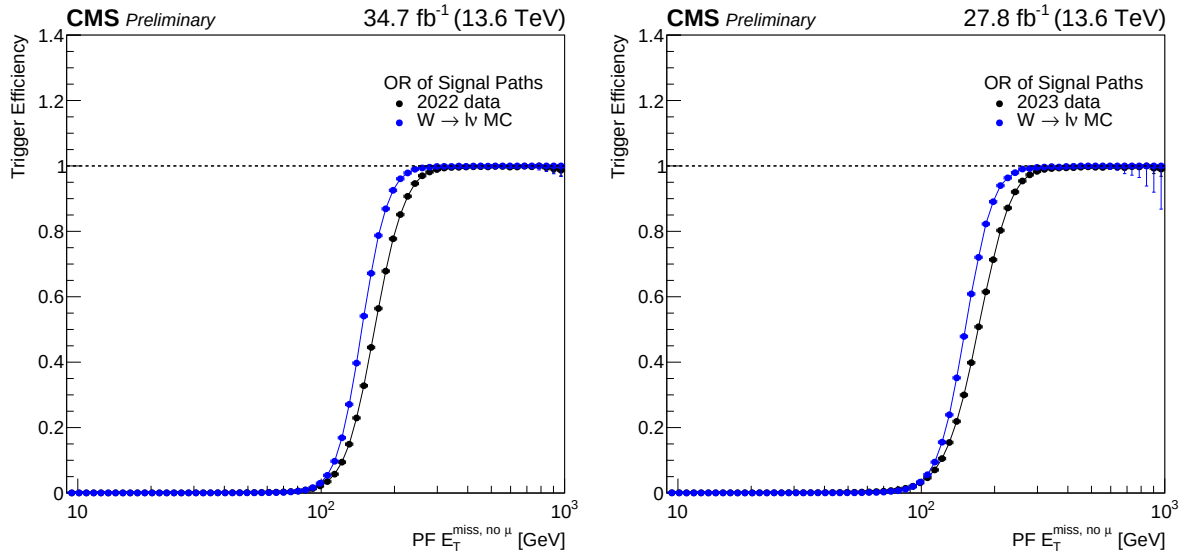
where missing inner hits are defined as how many hits should be there from the trajectory crossing with active layers, but are not there, counting them inwards from the first layer that has a hit, and missing middle hits are how many hits should be there in between the innermost and outermost hits associated to the track. In other words, the hits on the HLT track are required to be consecutive, beginning in the innermost layer of the tracker through which the track traversed. The track-based isolation is calculated simply as the scalar sum of the p_T of all tracks reconstructed in the HLT down to $p_T = 50$ MeV within $\Delta R < 0.3$ of the track in question⁴.

The calculation of efficiencies for the above triggers are detailed in Section 6.3, however for reference when discussing the event selections below, these efficiencies are plotted as a function of $p_T^{\text{miss, no } \mu}$, which is p_T^{miss} calculated offline with the p_T of all reconstructed muons counted as invisible, as in Figure 5.

The effect of requiring at least five silicon hits in the track leg on the signal acceptance for the $n_{\text{layers}} = 4, 5$ categories will be studied as a systematic source, which is in preparation.

⁴No accounting for pileup was present in the HLT track isolation calculation during 2022 and 2023.

Figure 5: Efficiency of the combination of p_T^{miss} triggers used in the search for simulated $W \rightarrow l\nu$ and experimental data for 2022 (left) and 2023 (right) data and simulations. The efficiency is plotted only with respect to $p_T^{\text{miss, no } \mu}$, with the dependence on the track p_T integrated over, as described in Section 6.3.



4 Event selections

4.1 Object definitions

Whenever we use p_T^{miss} , we use PF p_T^{miss} with the type 1 corrections applied and with the p_T of all reconstructed muons counted as invisible, which we denote as $p_T^{\text{miss, no } \mu}$. This is done so that the offline p_T^{miss} is consistent with the calorimeter-based p_T^{miss} used in the triggers, both at L1 and HLT. Whenever we use jets, they are PF jets with $p_T > 30 \text{ GeV}$ and $|\eta| < 4.5$, reconstructed with the anti- k_T clustering algorithm [20, 26] with a distance parameter of $R = 0.4$ and with charged hadron subtraction applied. When we select well-reconstructed muons, electrons, and hadronic tau candidates, they are reconstructed with the relevant PF algorithms [12].

Since missing transverse momentum is used in this search, we apply a set of standard p_T^{miss} filters designed to remove events with badly reconstructed p_T^{miss} .⁵ In particular, they are:

- primary vertex filter
- beam halo filter
- ECAL trigger primitive filter
- bad PF muon filter
- bad PF Muon Dz Filter
- HF noisy hits filter
- ee bad supercluster noise filter
- ECAL bad calibration filter (updated)

The effect of these filters on the signal acceptance has been found to be negligible.

The search, background estimation, and studies of systematic uncertainties use several physics objects to which standard sets of selections are applied. These sets of selections for primary vertices, jets, and leptons are listed in Tables 10, 11, 12, 13, and 14. Whenever a single primary vertex needs to be selected, the reconstructed vertex with the largest value of summed physics-object p_T^2 is taken to be the primary pp interaction vertex, where the physics objects are the objects returned by a jet finding algorithm [20, 26, 27] applied to all charged tracks associated with the vertex, plus the corresponding associated missing transverse momentum. The transverse and longitudinal impact parameters of a track with respect to the primary vertex are defined as

$$d_{xy} \equiv \frac{-(v_x - x)p_y + (v_y - y)p_x}{p_T} \quad \text{and} \quad d_z \equiv (v_z - z) - \frac{(v_x - x)p_x + (v_y - y)p_y}{p_T} \cdot \frac{p_z}{p_T}, \quad (2)$$

respectively, where (v_x, v_y, v_z) is the point of closest approach on the track to the CMS origin, and (x, y, z) are the coordinates of the primary vertex.

Finally, the key object for this analysis are the charged particle tracks. We use the “isolated-Tracks” available in MINIAOD, which are `pat::IsolatedTracks` drawn from the union of `packedPFCandidates`, `lostTracks`, and `generalTracks` collections, with a duplicate removal algorithm. All `isolatedTracks` with $p_T > 20 \text{ GeV}$ or with $p_T > 5 \text{ GeV}$ and a minimal isolation are saved in the event. As our track selection will include a $p_T > 50 \text{ GeV}$ selection

⁵See the MET Filter Recommendations for Run III from the JME POG.

Table 10: Standard criteria used to select primary vertices in both the 2022 and 2023 data sets.

selection name	variable	selection
good primary vertex	isValid	= true
	ndof	≥ 4
	$ z $	< 24 cm
	ρ	< 2 cm

Table 11: Tight jet ID with lepton veto defined by the Jet POG for 2022 and 2023.

variable	requirement
neutral hadron fraction	< 0.99
charged hadron fraction	> 0.01
neutral EM fraction	< 0.9
charged EM fraction	< 0.8
number of constituents	> 1
charged multiplicity	> 0
muon fraction	< 0.8

Table 12: Tight and loose electron IDs defined by the EGamma POG, including the recommended selections on impact parameter. The first values given are used when $|\eta_{\text{supercluster}}| < 1.479$, and the second values in parentheses are used when $|\eta_{\text{supercluster}}| > 1.479$, if different. These values correspond to the cutBasedElectronID-RunIIIWinter22-V1 VID, used for both 2022 and 2023 data sets.

variable	requirement	
	tight ID	loose ID
full $5 \times 5 \sigma_{i\eta i\eta}$	$< 0.0101(0.027)$	$< 0.0107(0.0275)$
$ \Delta\eta_{\text{seed}} $	$< 0.00411(0.00938)$	$< 0.00691(0.0121)$
$ \Delta\phi_{\text{in}} $	$< 0.116(0.164)$	$< 0.175(0.228)$
	$< 0.02(0.02)$	$< 0.05(0.05)$
H/E	$+1.16(0.5)/E_{\text{SC}} +$ $+0.0422(0.262) * \rho/E_{\text{SC}} +$	$+1.28(2.3)/E_{\text{SC}} +$ $+0.0422(0.262) * \rho/E_{\text{SC}} +$
rel. iso. with ρ -corr.	$< 0.0388(0.0544)$ $+0.535(0.519)/p_{\text{T}}$	$< 0.194(0.184)$ $+0.535(0.519)/p_{\text{T}}$
$ 1/E - 1/p $	$< 0.023(0.018)$	$< 0.138(0.127)$
$ d_{xy} $	$< 0.05(0.10)$ cm	$< 0.05(0.10)$ cm
$ d_z $	$< 0.10(0.20)$ cm	$< 0.10(0.20)$ cm
missing inner hits	≤ 1	≤ 1
has matched conversion	= true	= true

Table 13: Tight and loose muon IDs defined by the muon POG for the 2022 and 2023 data sets.

variable	requirement	
	tight ID	loose ID
isGlobalMuon	= true	= true
isPFMuon	= true	= true
χ^2/ndof	< 10	
valid muon hits	> 0	
matched muon stations	> 1	
$ d_{xy} $	< 0.2 cm	
$ d_z $	< 0.5 cm	
valid pixel hits	> 0	
tracker layers with hits	≥ 10	
rel. iso. with $\Delta\beta$ -corr.	< 0.15	

Table 14: Hadronic tau ID with light flavor rejection using tau ID sources defined by the tau POG for the 2022 and 2023 data sets.

selection name	variable	requirement
hadronic tau ID	decayModeFindingNewDMs	> 0.5
	byTightCombinedIsolationDeltaBetaCorr3Hits	> 0.5
	byVVVLooseDeepTau2018v2p5VSe	> 0.5
	byVLooseDeepTau2018v2p5VSmu	> 0.5

later on (Section 4.3.2), this collection is adequate for our purposes. Crucially, these tracks allow access to the `reco::HitPattern` and the track quality by name.

Finally, we also use calorimeter jets, albeit indirectly through the `isolatedTracks` collection. They are the regular “ak4CaloJets” available in AOD, which are `reco::CaloJets` with $\Delta R = 0.4$, made from calorimetric towers. Since these jets don’t have much usage in CMS nowadays, we report here some details on their construction. The `ak4CaloJets` inputs have an E_T threshold of 0.3 GeV, and the minimum jet p_T is set at 10 GeV. They don’t have any kind of jet energy corrections applied. They are used to construct the `matchedCaloJetEmEnergy` and `matchedCaloJetHadEnergy` variables that are used as proxies for calorimeter energy near the `isolatedTracks`.

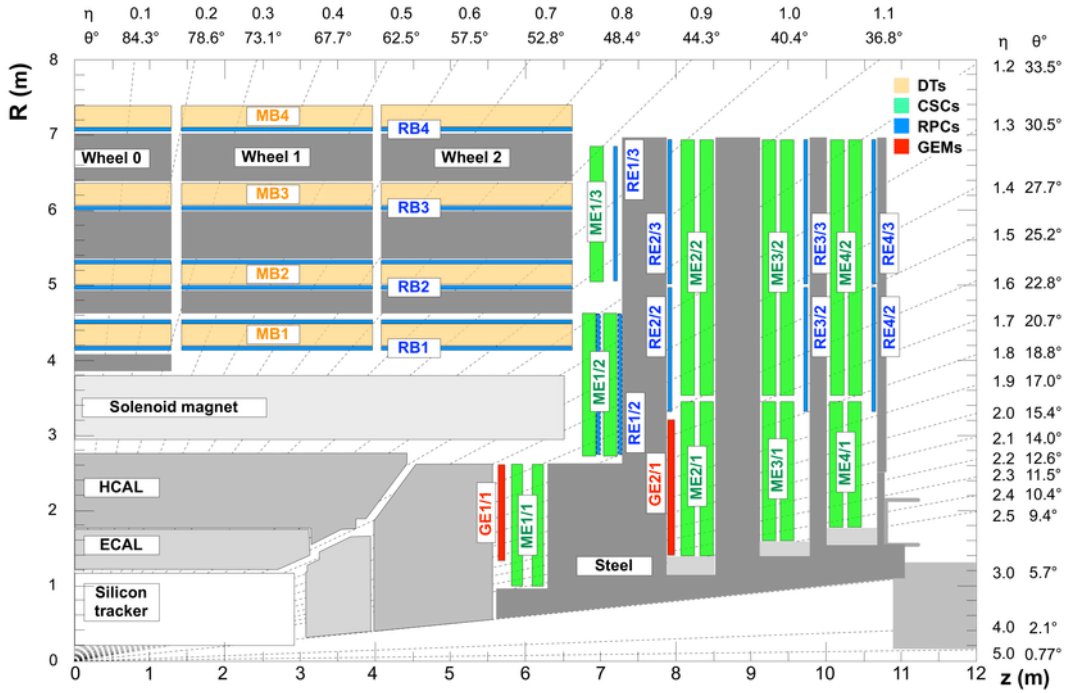
4.2 Fiducial track selections

As most signal tracks are produced centrally, especially for higher mass charginos, we only select tracks with $|\eta| < 2.1$. In order to avoid selecting tracks which are missing muon chamber hits and associated calorimeter energy due to detector inefficiencies, we veto tracks that fall into known regions of high inefficiency. First, there are three regions in $|\eta|$ which are vetoed:

- $0.15 < |\eta| < 0.35$: there is a gap here between wheels 0 and 1 of the DT stations
- $1.42 < |\eta| < 1.65$: this is the region where the barrel and endcap sections of the ECAL overlap
- $1.55 < |\eta| < 1.85$: this is the transition region between the outermost rings and innermost ring of the CSCs

A schematic layout of one quarter of the CMS detector, in which these three regions can be identified, is shown in Figure 6.

Figure 6: Layout of one quarter of the CMS detector, with the muon system detailed.

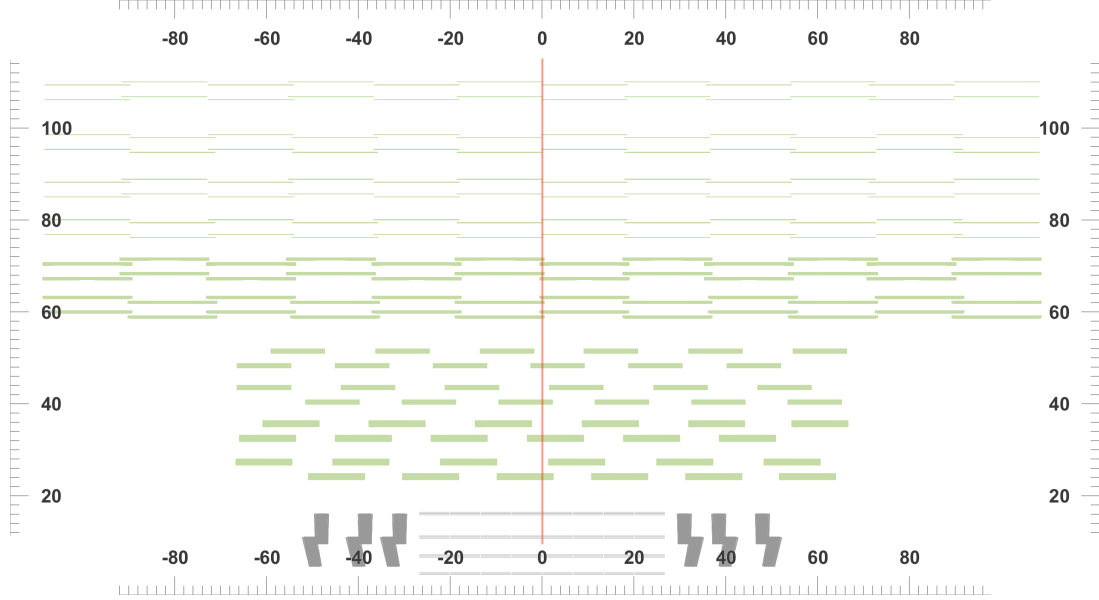


There is also a region of low hit efficiency in the middle of the TOB region of the strip tracker. In Figure 7, one can see that if a track originates from very close to $z = 0$ and has a pseudorapidity very close to $\eta = 0$, then it will pass through only the edges of all the modules in the TOB, and potentially have missing hits in all six layers. Two such tracks were found previously in 2015 data (see AN-15-213), and we veto tracks that have $|d_z^{\text{w.r.t. origin}}| < 0.5 \text{ cm}$ and $|\lambda| < 10^{-3}$, where $\lambda = \frac{\pi}{2} - \theta$ is the pitch angle of the track, to avoid this poorly instrumented region of the TOB.

There is an additional veto region in 2022 for eras E, F, and G when there was a water leak in the ECAL endcap. This issue can be seen in the recommended jet veto map for 2022EFG⁶ in figure 8.

⁶These come from JME POG recommendations for Run 3.

Figure 7: Schematic of the silicon tracker in the ρ - z plane, with a trajectory with $\eta \simeq 0$. That trajectory passes through only the edges of all the modules in the TOB, potentially having missing hits in all six layers. Note that the TID and TEC are not shown here.



In addition to these fixed fiducial selections, the ECAL channel statuses provided via the global tag are checked for each run, and tracks that are within $\Delta R < 0.05$ of any noisy or dead channels are vetoed. The map of the noisy or dead ECAL channels for one run in 2022 is shown in Figure 9.

Additional regions of detector inefficiency are identified in the η - ϕ plane using a tag-and-probe study. For the electron reconstruction inefficiency, events are selected using the requirements in Table 60⁷. The reconstructed electron serves as the tag and the track is the probe. For this study, if there are more than one possible tags or more than one possible probes in a given event, only one unique pair is chosen at random. Then the electron reconstruction inefficiency is defined as the fraction of selected tracks for which

$$\min \Delta R_{\text{track, loose electron}} > 0.15 \quad (3)$$

is true; i.e., the fraction of selected tracks that are not matched to an electron passing loose ID⁸. The muon reconstruction inefficiency is defined similarly, using the selections in Table 16. Again, only one unique tag-probe pair per event is selected.

The reconstruction inefficiency is calculated in bins of size 0.1×0.1 in the probe track η - ϕ plane (see Figure 10). We then use any bins that are at least $2 \sigma_{\text{mean}}$ above the mean inefficiency to veto tracks in the search, where σ_{mean} is the standard deviation of the mean inefficiency across all $\eta - \phi$. Any tracks within the radius⁹ of such a bin is rejected. This is done separately for each run era of 2022 (C-F) and 2023 (CD).

⁷Except for the p_T criterion, the selections on the track come from the *disappearing track selection* with the electron veto removed, which are defined below in Section 4.3.

⁸No fit to the tag-probe invariant mass spectrum in order to separate true $Z \rightarrow \ell\ell$ events from background is done as the purpose of this study is to identify regions of abnormally high reconstruction inefficiency, not to measure the actual value of the inefficiency.

⁹ $\sqrt{0.05^2 + 0.05^2} = \sqrt{0.005} \approx 0.071$

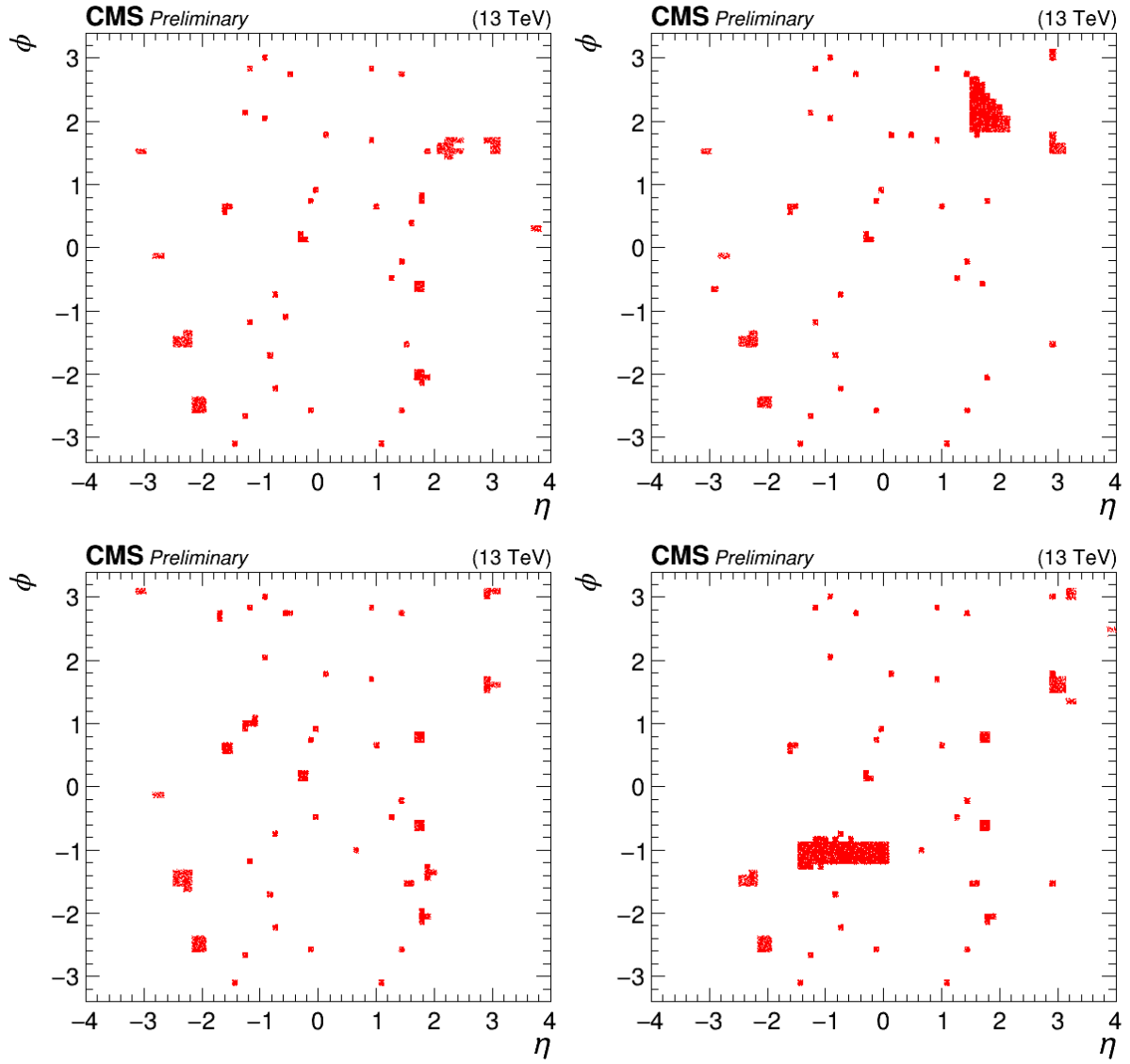


Figure 8: Recommended jet veto maps for 2022 CD (top left), 2022 EFG (top right), 2023 C (bottom left), and 2023 D (bottom right). Note the large veto region in the 2022 EFG map at $\eta \approx 2$ due to the water leak in the ECAL endcap.

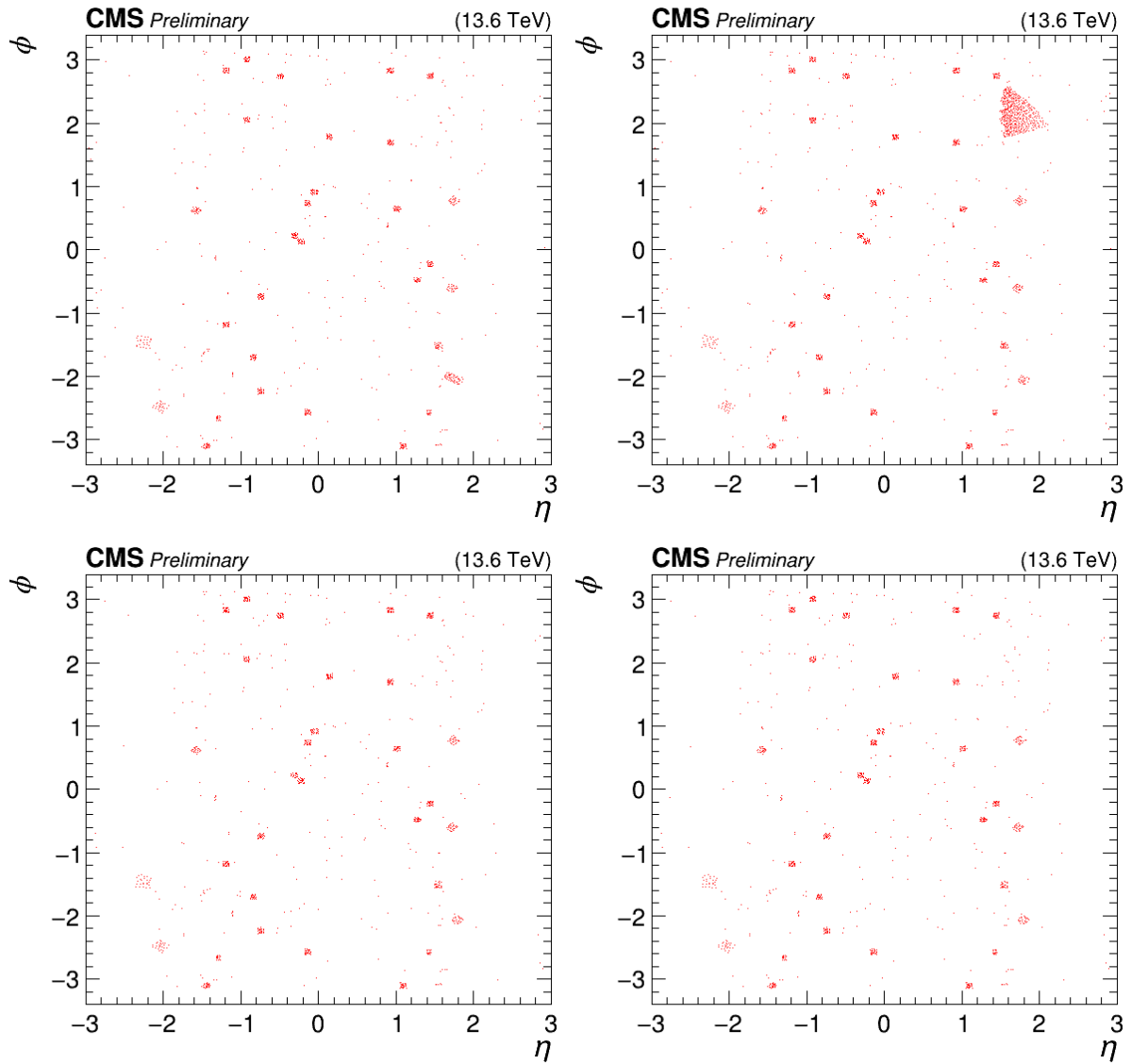


Figure 9: Channels identified as noisy or dead in the ECAL channel status record provided by the global tag for runs in 2022CD (top left), 2022 EFG (top right), 2023 C (bottom left), and 2023 D (bottom right). Tracks within $\Delta R < 0.05$ of any of these channels are vetoed.

Table 15: Selections used in the tag-and-probe study of electron reconstruction inefficiency. The requirements are cumulative, with only the objects passing a given criterion being considered in subsequent criteria. The selected electron is the tag, while the selected track is the probe. With the exception of the p_T and tag-and-probe pair requirements, the probe track selection is the disappearing track selection with the electron veto (defined in Table 21) removed.

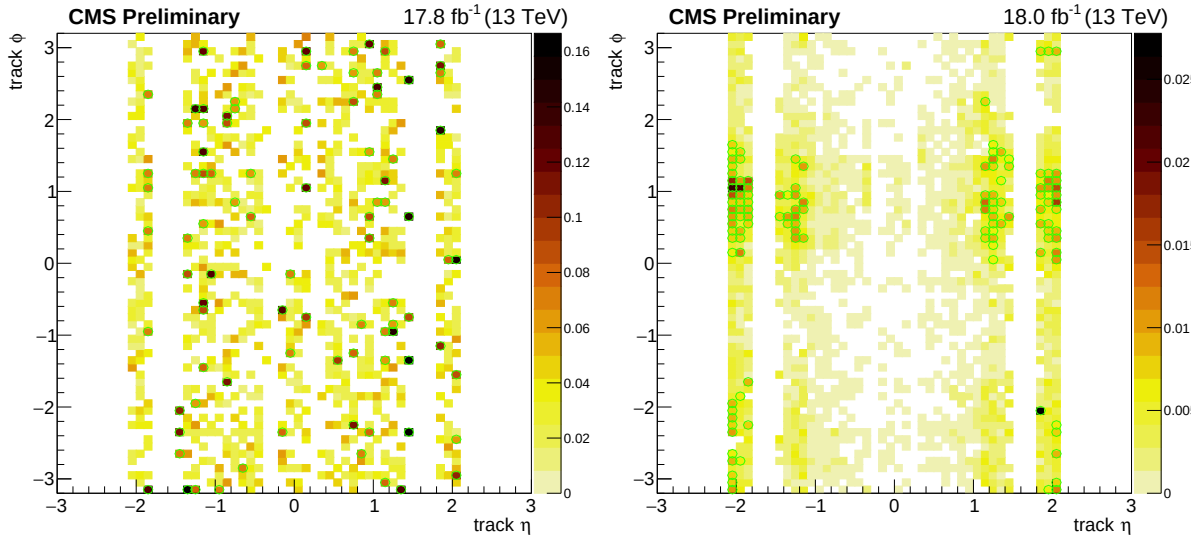
quantity	object	selection
event passes SingleElectron/EGamma triggers		
≥ 1	electrons	$p_T > 35 \text{ GeV}$ ($> 32 \text{ GeV}$ in 2018)
≥ 1	electrons	$ \eta < 2.1$
≥ 1	electrons	passing tight electron ID
	exactly one passing electron chosen randomly	
≥ 1	tracks	$p_T > 30 \text{ GeV}$
≥ 1	tracks	$ \eta < 2.1$
≥ 1	tracks	$ \eta < 0.15 \text{ OR } \eta > 0.35$
≥ 1	tracks	$ \eta < 1.42 \text{ OR } \eta > 1.65$
≥ 1	tracks	$ \eta < 1.55 \text{ OR } \eta > 1.85$
≥ 1	tracks	$\min \Delta R_{\text{track,noisy/dead ECAL ch.}} > 0.05$
≥ 1	tracks	$ d_z > 0.5 \text{ cm OR } \lambda > 10^{-3}$
≥ 1	tracks	$\eta < 0 \text{ OR } \eta > 1.42 \text{ OR } \phi < 2.7$
≥ 1	tracks	number of pixel hits ≥ 4
≥ 1	tracks	missing inner hits = 0
≥ 1	tracks	missing middle hits = 0
≥ 1	tracks	rel. PF-based iso. < 0.05
≥ 1	tracks	$ d_{xy} < 0.02 \text{ cm}$
≥ 1	tracks	$ d_z < 0.5 \text{ cm}$
≥ 1	track-jet pairs	$\Delta R_{\text{track,jet}} > 0.5$
≥ 1	track-electron pairs	$M_{\text{track,electron}} > 10 \text{ GeV}$
≥ 1	tracks	$\min \Delta R_{\text{track,muon}} > 0.15$
≥ 1	tracks	$\min \Delta R_{\text{track,had. tau}} > 0.15$
	exactly one passing track chosen randomly	
= 1	track-electron pairs	$ M_{\text{track,electron}} - M_Z < 10 \text{ GeV}$
= 1	track-electron pairs	$q_{\text{track}} \cdot q_{\text{electron}} < 0$
≥ 1	track	$n_{\text{layers}} \geq 4$ (three signal region bins)

Table 16: Selections used in the tag-and-probe study of muon reconstruction inefficiency. The requirements are cumulative, with only the objects passing a given criterion being considered in subsequent criteria. The selected muon is the tag, while the selected track is the probe. With the exception of the p_T and tag-and-probe pair requirements, the probe track selection is the disappearing track selection with the muon veto (defined in Table 21) removed.

quantity	object	selection
event passes SingleMuon triggers		
≥ 1	muons	$p_T > 26 \text{ GeV}$
≥ 1	muons	$ \eta < 2.1$
≥ 1	muons	passing tight muon ID
	exactly one passing muon chosen randomly	
≥ 1	tracks	$p_T > 30 \text{ GeV}$
≥ 1	tracks	$ \eta < 2.1$
≥ 1	tracks	$ \eta < 0.15 \text{ OR } \eta > 0.35$
≥ 1	tracks	$ \eta < 1.42 \text{ OR } \eta > 1.65$
≥ 1	tracks	$ \eta < 1.55 \text{ OR } \eta > 1.85$
≥ 1	tracks	$\min \Delta R_{\text{track,noisy/dead ECAL ch.}} > 0.05$
≥ 1	tracks	$ d_z > 0.5 \text{ cm OR } \lambda > 10^{-3}$
≥ 1	tracks	number of pixel hits ≥ 4
≥ 1	tracks	missing inner hits = 0
≥ 1	tracks	missing middle hits = 0
≥ 1	tracks	rel. PF-based iso. < 0.05
≥ 1	tracks	$ d_{xy} < 0.02 \text{ cm}$
≥ 1	tracks	$ d_z < 0.5 \text{ cm}$
≥ 1	track-jet pairs	$\Delta R_{\text{track,jet}} > 0.5$
≥ 1	track-muon pairs	$M_{\text{track,muon}} > 10 \text{ GeV}$
≥ 1	tracks	$\min \Delta R_{\text{track,electron}} > 0.15$
≥ 1	tracks	$\min \Delta R_{\text{track,had. tau}} > 0.15$
≥ 1	tracks	$E_{\text{calo}} < 10 \text{ GeV}$
	exactly one passing track chosen randomly	
= 1	track-muon pairs	$ M_{\text{track,muon}} - M_Z < 10 \text{ GeV}$
= 1	track-muon pairs	$q_{\text{track}} \cdot q_{\text{muon}} < 0$
≥ 1	track	$n_{\text{layers}} \geq 4$ (three signal region bins)

393 The application of this final veto (the “fiducial map” veto) is found to be necessary due to our
 394 experience with the early analysis of 2015 data in the previous search. In the initial unblinding,
 395 an excess in the search sample was found with many tracks originating from limited regions
 396 in the detector (“hot spots”) that corresponded to a high lepton veto inefficiency. After con-
 397 structing the fiducial map for that period, no excess was observed. Furthermore soon after that
 398 time, an updated global tag was provided for the 2015 data that mitigated this issue with better
 399 maps of noisy detector channels. The effect of this veto on background estimates in this search
 400 is seen to be a 3–4% reduction in the electron backgrounds, and a 27–29% reduction in the muon
 401 backgrounds. While this is not a considerably large background suppression, this process is ef-
 402 fective at preventing the possibility of needing to retroactively remove signal candidate tracks
 403 after unblinding due to noisy channels that may later be noted in an updated global tag (or not
 404 at all). This process improves our confidence in any observation.

Figure 10: Lepton reconstruction inefficiencies as measured using the tag-and-probe study de-
 scribed in the text. Electrons (left) and muons (right) are calculated separately for each era
 within the Run3 data sets. Shown is 2022 F. Bins having an inefficiency of at least $2\sigma_{\text{mean}}$ above
 the mean inefficiency are used to veto tracks in the search for the given flavor and era, and are
 marked with green circles in each plot.



4.3 Search region

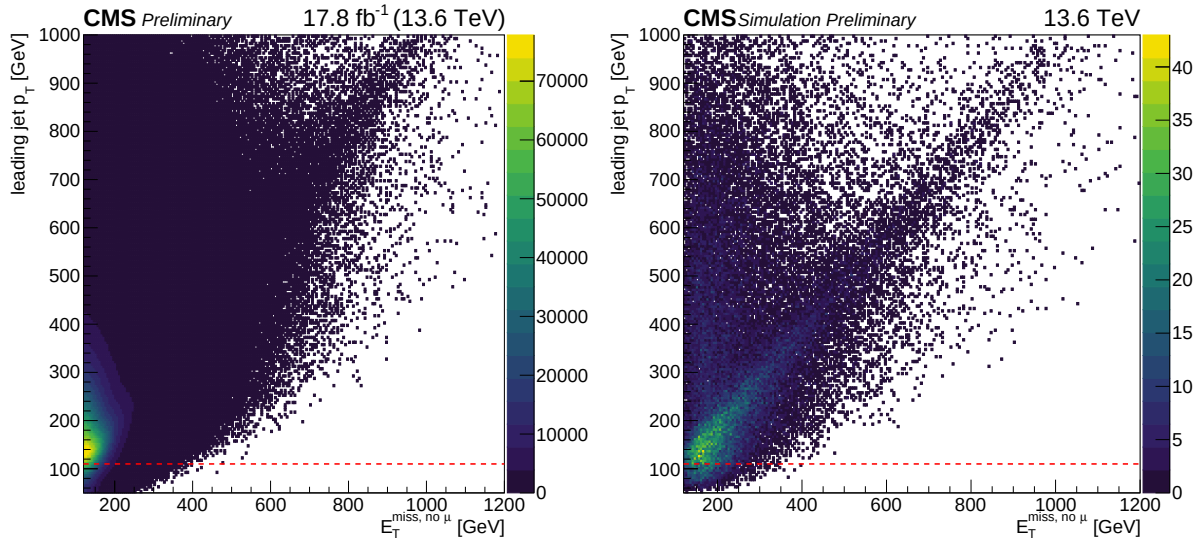
4.3.1 Basic selection

The selections for the search begin with the *basic selection*, which defines an event topology consistent with having a high- p_T jet from ISR and a subsequently large $p_T^{\text{miss, no } \mu}$. The criteria of the *basic selection* are listed in Table 17. As seen from the trigger efficiencies in Section 3.2, the criteria of $p_T^{\text{miss, no } \mu} > 120 \text{ GeV}$ is well below the point at which the triggers become fully efficient. This is chosen to maximize the number of potential signal events that are selected. Although the leading jet p_T is largely correlated with $p_T^{\text{miss, no } \mu}$ for signal events, events are required to be consistent with this trigger strategy and contain at least one jet with $p_T > 110 \text{ GeV}$. This also serves to remove the sample of events in data where the leading jet p_T is considerably lower than the $p_T^{\text{miss, no } \mu}$, as seen in Figure 11.

Table 17: The criteria defining the *basic selection*. The requirements are cumulative, with only the objects passing a given criterion being considered in subsequent criteria.

quantity	object	selection
event passes MET triggers		
≥ 1	primary vertices	passing good primary vertex requirements
	$p_T^{\text{miss, no } \mu}$	$p_T > 120 \text{ GeV}$
≥ 1	jets	$p_T > 110 \text{ GeV}$
≥ 1	jets	$ \eta < 2.4$
≥ 1	jets	passing tight ID with lepton veto requirements
	jet pairs (if any)	$\max \Delta\phi_{\text{jet, jet}} < 2.5$
	$p_T^{\text{miss, no } \mu}$	$ \Delta\Phi(\text{leading jet}, p_T^{\text{miss, no } \mu}) > 0.5$

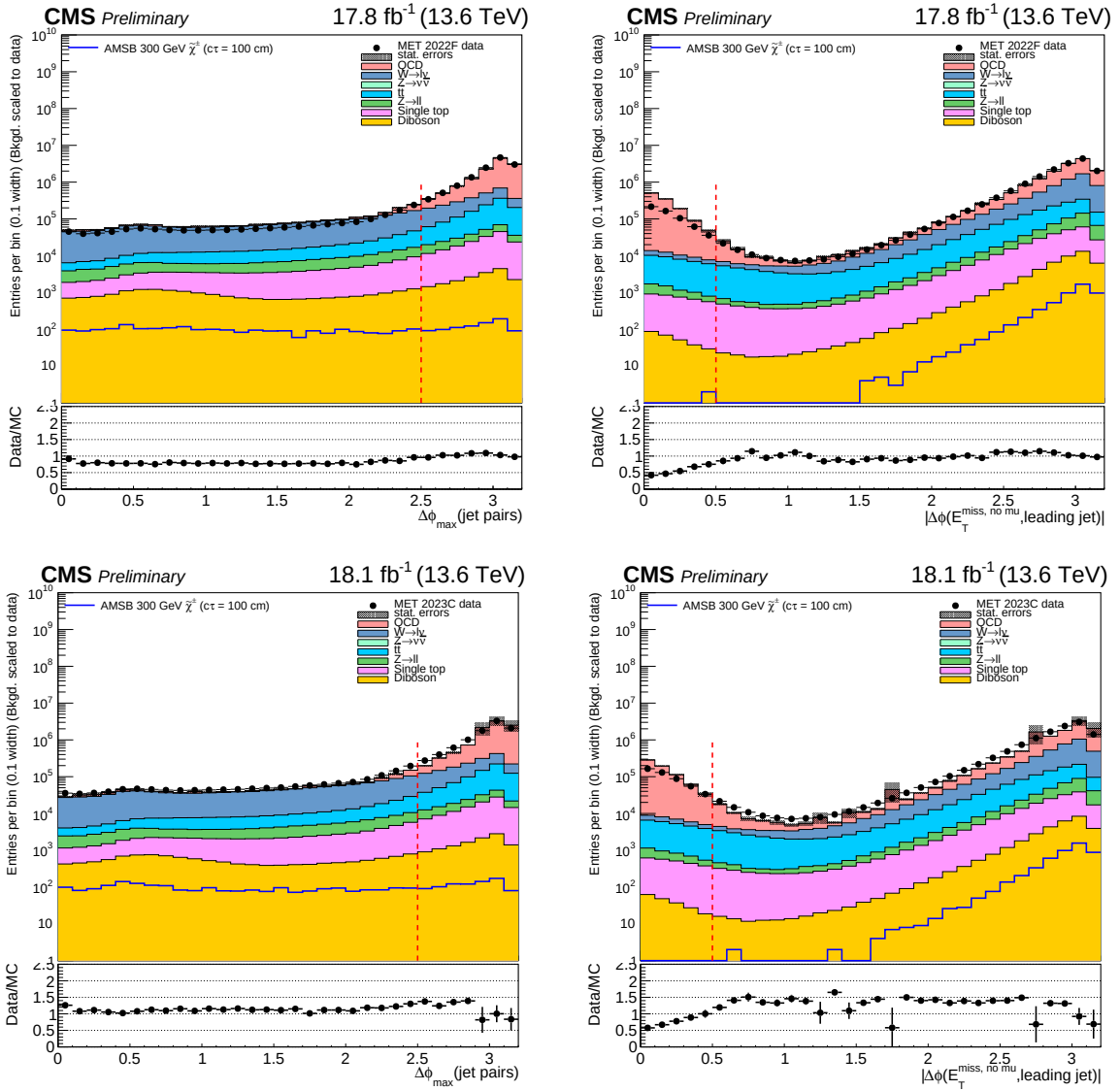
Figure 11: Distributions of the leading jet p_T vs. $p_T^{\text{miss, no } \mu}$ for events in the *basic selection* before any jet requirements are imposed. The distribution is shown for the 2022 F experimental data (left) and for simulated signal events (right) with a 700 GeV chargino with a lifetime of 10 000 cm. The red dashed line indicates the chosen value for the jet p_T requirement.



The angular requirements of the *basic selection* are designed to remove the large, very reducible

background from QCD multijet events. These enter the selection largely in the form of back-to-back dijet events where a mismeasurement of the jet energy conspires to produce a large value for $p_T^{\text{miss, no } \mu}$. The requirement of $\max |\Delta\phi_{\text{jet, jet}}| < 2.5$ targets the dijet topology, while the requirement of $|\Delta\phi(\text{leading jet}, p_T^{\text{miss, no } \mu})| > 0.5$ targets events with badly mismeasured jet energy. The requirement on $\max |\Delta\phi_{\text{jet, jet}}|$ is only applied if multiple jets with $p_T > 30$ GeV, $|\eta| < 4.5$, and passing the jet ID exist in the event. Signal events are essentially flat in the former variable, while being strongly peaked at π in the latter, as seen in Figure 12.

Figure 12: Distributions of $\max |\Delta\phi_{\text{jet, jet}}|$ (left) and $|\Delta\phi(\text{leading jet}, p_T^{\text{miss, no } \mu})|$ (right) for events in the *basic selection* before the angular requirements on the jets/ $p_T^{\text{miss, no } \mu}$ are imposed, for the 2022 (top) and 2023 (bottom) data-taking periods. The red dashed lines indicates the chosen value for the requirement on each variable.



4.3.2 Isolated track selection

Events passing the *basic selection* are then required to pass the *isolated track selection*. As the name suggests, this selection defines an event topology consistent with having a high- p_T isolated

track. The requirements of the *isolated track selection* are listed in Table 18. This is where the fiducial selections described in the previous section are applied. The 2023 pixel barrel issue, in which layers 3 and 4 (BPix3 and BPix4) of the pixel barrel detector had readout issues which implied in the loss of 27 modules, decreased the efficiency of tracks reconstruction in the region of $-1.5 < \eta < -0.2$ and $-1.1 < \phi < -0.9$. Given that recovering those modules means that the electronics of the service cylinder would need to be replaced, a recovery was not considered. There are no special instructions to handle tracks in era D of 2023, but we consider the two 2023 campaigns, regular and BPix, in the analysis.

Table 18: The criteria defining the *isolated track selection*. The requirements are cumulative, with only the objects passing a given criterion being considered in subsequent criteria.

quantity	object	selection
event passes <i>basic selection</i>		
≥ 1	tracks	$p_T > 55 \text{ GeV}$
≥ 1	tracks	passing fiducial selections (see Section 4.2)
≥ 1	tracks	number of pixel hits ≥ 4
≥ 1	tracks	missing inner hits = 0
≥ 1	tracks	missing middle hits = 0
≥ 1	tracks	rel. PF-based iso. < 0.05
≥ 1	tracks	$ d_{xy} < 0.02 \text{ cm}$ (w.r.t. primary vertex)
≥ 1	tracks	$ d_z < 0.5 \text{ cm}$ (w.r.t. primary vertex)
≥ 1	track-jet pairs	$\Delta R_{\text{track,jet}} > 0.5$

The track p_T requirement is chosen to be approximately at the point at which the track leg of `HLT_MET105_IsoTrk50_v*` becomes fully efficient. The fiducial track selections, as well as the requirements on the number of hits and number of missing hits, are designed so that we are selecting the highest quality tracks possible.

The track is also required to have no missing inner hits and no missing middle hits. The missing inner hits variable, accessed through the `IsolatedTrack::lostInnerLayers()` method, counts how many hits should be there from the trajectory crossing with active layers, but are not there, counting them *inwards* from the first layer that has a hit. In other words, it counts how many missing hits you have from where the production vertex is to the first actual measurement. The missing outer hits variable, accessed through the `IsolatedTrack::lostOuterLayers()` method, does the same counting *outwards* from the last layer that has a hit; in this search, we expect that the disappearing tracks has a non-zero number of missing outer hits. Missing middle hits do not have a dedicated method, but can be calculated from the total number of missing hits (`IsolatedTrack::lostLayers()`) minus the inner and outer hits. Figure 13 shows graphically all three kinds of missing hits.

The PF-based isolation used in the *isolated track selection* is calculated as the scalar sum of the p_T of the charged PF candidates within a cone of $\Delta R < 0.3$ relative to the candidate track in question. The value of the isolation requirement is chosen to match the requirement made in `HLT_MET105_IsoTrk50_v*`. This calculation only takes into account charged PF objects that come from the PV as well, i.e., their longitudinal impact parameter $|d_z|$ have to be smaller than 0.1 cm with respect to the PV. This procedure removes the pileup dependence from the calculation.

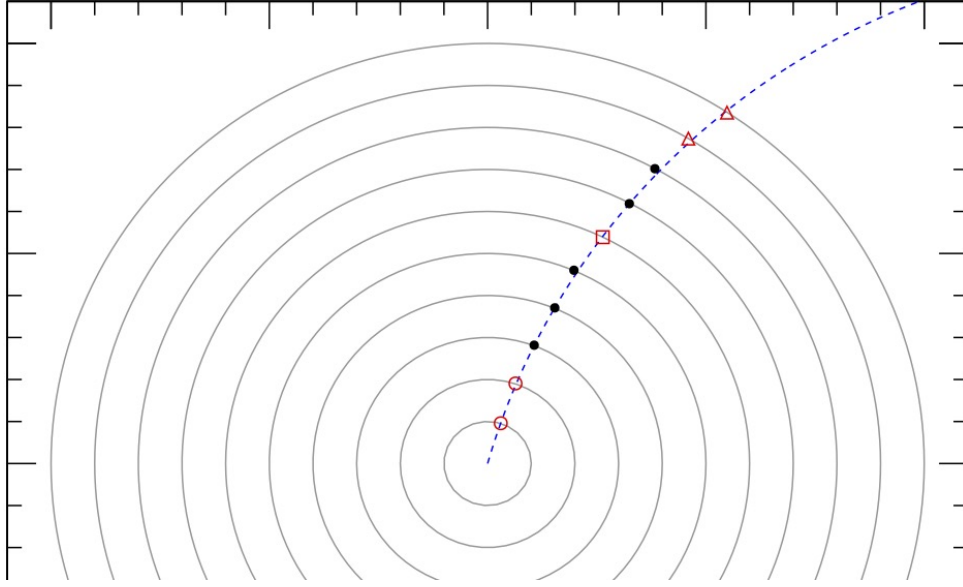


Figure 13: Cartoon of a track with measured hits (black points), missing inner hits (red open circles), missing middle hits (red open squares), and missing outer hits (red open triangles). This track would not pass the *isolated track selection*, since it has two missing inner hits and one missing middle hit.

4.3.3 Candidate track selection

The vast majority of events selected by the *isolated track selection* are events with leptons from the decays of gauge bosons. Most of these are removed in the next phase of the selection, called the *candidate track selection*, where events with tracks matched to leptons are rejected. The requirements of the *candidate track selection* are listed in Table 19. All reconstructed electrons and muons available in MINIAOD are used for this matching to reject as many leptons as possible. This includes ¹⁰:

- Muons:

- $p_T > 5 \text{ GeV}$ (no ID requirement)
- $p_T > 3 \text{ GeV}$ if passing any ID of: PF, global, arbitrated tracker, standalone, RPCMuLoose
- any p_T if passing the PF ID

- Electrons:

- all `gedGsfElectrons`

Hadronic taus for this matching are identified using the criteria listed above in Table 14, but with isolation criterion removed. The particular choice of $\Delta R > 0.15$ was made as this value removes the vast majority of events with $\min \Delta R_{\text{track, lepton}}$ near zero, which results from successfully reconstructed leptons.

4.3.4 Disappearing track selection

Finally, we require the tracks passing the *candidate track selection* to “disappear” in the *disappearing track selection*, which defines the search region. The *disappearing track selection* is defined in Table 20. In Run 2, the calorimeter energy sum, E_{calo} , was calculated as the sum of all calori-

¹⁰See WorkbookMiniAOD2017 for details.

Table 19: The criteria defining the *candidate track selection*. The requirements are cumulative, with only the objects passing a given criterion being considered in subsequent criteria. All reconstructed electrons and muons are used during this matching, in order to reject as many leptons as possible. Hadronic taus for this matching are identified using the criteria listed above in Table 14, but with the the isolation criterion removed.

quantity	object	selection
event passes <i>isolated track selection</i>		
≥ 1	tracks	$\min \Delta R_{\text{track,electron}} > 0.15$
≥ 1	tracks	$\min \Delta R_{\text{track,muon}} > 0.15$
≥ 1	tracks	$\min \Delta R_{\text{track,had. tau}} > 0.15$

meter towers (from both the ECAL and HCAL) within a cone of $\Delta R < 0.5$ of the selected track, and the sum was then corrected for contamination from pileup using ρ , the median calorimeter energy density of the event, which was simply multiplied by the cone size of 0.5. This procedure required that the calorimeter hits were extracted from the AOD data format, which is very costly in terms of processing time. In this iteration of the disappearing tracks analysis, some effort was put to try to use a variable that is available directly from the MINIAOD data format, which is the energy of the calorimeter jet that is closest to the candidate track, up to a radius of $\Delta R < 0.3$. This serves as a proxy to the actual energy associated to the track in the calorimeter. This modification has only a small impact in the background estimation method and signal selection efficiency. Finally, the events that pass the selection using this proxy variable can have their E_{calo} calculated as in Run 2, thereby providing the most accurate result, while reducing processing time.

Table 20: The criteria defining the *disappearing track selection*. The requirements are cumulative, with only the objects passing a given criterion being considered in subsequent criteria.

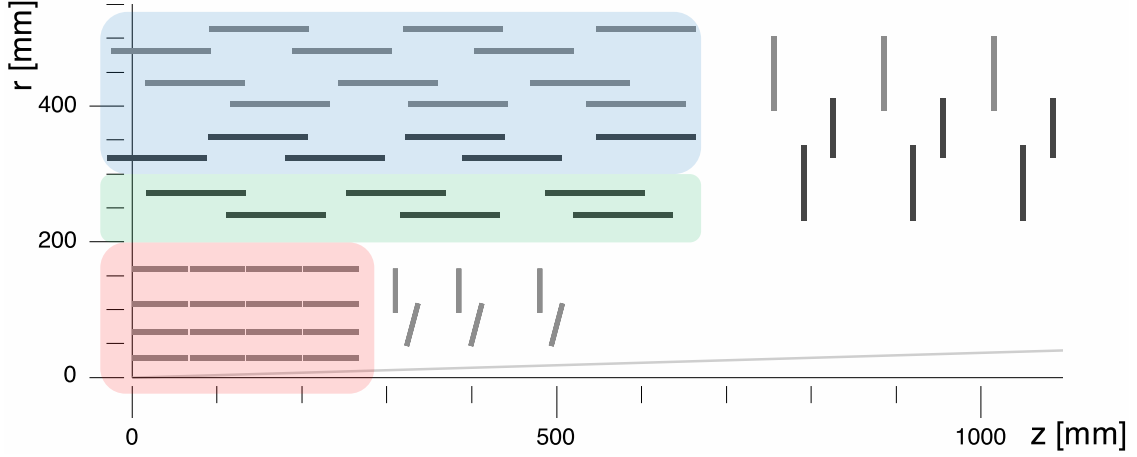
quantity	object	selection
event passes <i>candidate track selection</i>		
≥ 1	tracks	$E_{\text{calo}} < 10 \text{ GeV}$
≥ 1	tracks	missing outer hits ≥ 3

In order to maximize sensitivity to a wide range of lifetimes, we categorize events passing the disappearing track selection by the number of tracker layers with measurement associated to the selected track (n_{layers}). We define three categories: $n_{\text{layers}} = 4$, $n_{\text{layers}} = 5$, and $n_{\text{layers}} \geq 6$, corresponding roughly to decay lengths of $\sim 20 \text{ cm}$, $\sim 30 \text{ cm}$, and $> 35 \text{ cm}$ respectively, as illustrated in Figure 14 for the barrel. As we will see in Section 5, $n_{\text{layers}} = 6$ is approximately where the background from spurious combinations of hits vanishes, leaving only the background from charged leptons.

The distributions of E_{calo} and number of missing outer hits are shown in Figure 15 and 16, respectively, for tracks in simulation that pass the full selection for each of the three signal regions defined above, except for the requirement on the variable being plotted. Very good separation between signal and background is seen for both of these variables, and requiring $E_{\text{calo}} < 10 \text{ GeV}$ and at least three missing outer hits is sufficient to remove almost all background within the albeit limited statistical precision of the simulation. Tuning the requirement on E_{calo} below 10 GeV is not beneficial to the overall sensitivity of the search, since there remains still a narrow distribution of pileup deposits surviving the ρ -based pileup correction.

Finally, the cumulative selection efficiency of each requirement is shown in Figure 17 for sim-

Figure 14: Illustration of the three signal regions defined in terms of n_{layers} , for the barrel region. The $n_{\text{layers}} = 4$ region would correspond to the pixel detector layers (in red), the $n_{\text{layers}} = 5$ region would contain the first TIB layer (in green), and the $n_{\text{layers}} \geq 6$ region would contain the rest of the TIB layers and the TOB sector (in blue).



ulated signal events with various lifetime hypotheses. Several features of this plot are worth noting. For the shortest lifetimes, large drops in efficiency are seen for the requirements on p_T , number of pixel hits, number of valid hits, and track isolation. This is because when the chargino passes through few layers of the inner tracker, the resulting track is unlikely to be reconstructed. Thus, the tracks we select are mostly either fake tracks, and mostly rejected by the requirements on p_T , number of pixel hits, and number of valid hits, or are constituents of the ISR jet, and mostly rejected by the isolation requirement. Event displays demonstrating the latter case are seen in Figures 18 and 19.

For the longest lifetimes, large drops in efficiency are seen for the requirement of $\min \Delta R_{\text{track, muon}} > 0.15$, which is designed to reject reconstructed muons. This is because if the chargino is very long-lived, it may pass through a few muon chambers before decaying, and be reconstructed as a muon. A large drop for the efficiency of the final requirement on number of missing outer hits is also seen for a similar reason: longer lived charginos are more likely to pass through all, or nearly all, layers of the inner tracker before decaying, and so will be more likely to have fewer than three missing outer hits.

Figure 15: Distributions of E_{calo} for tracks in simulation that pass the full *disappearing track selection* with $n_{\text{layers}} = 4$ (top left), 5 (top right), ≥ 6 (bottom) except for the requirement on the variable being plotted. The 2022 samples are shown here.

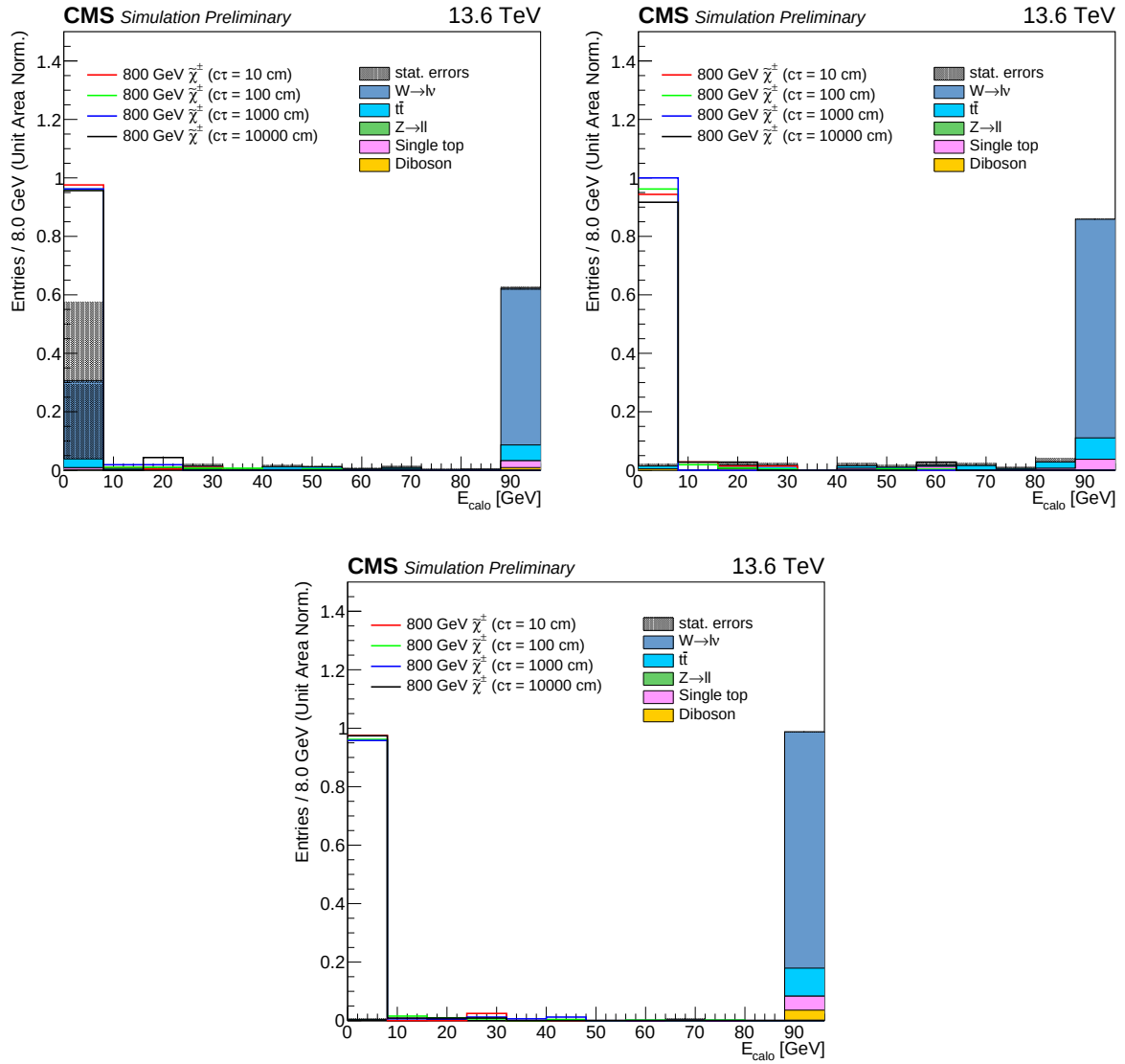


Figure 16: Distributions of number of missing outer hits for tracks in simulation that pass the full *disappearing track selection* with $n_{\text{layers}} = 4$ (top left), 5 (top right), ≥ 6 (bottom) except for the requirement on the variable being plotted. The 2022 samples are shown here.

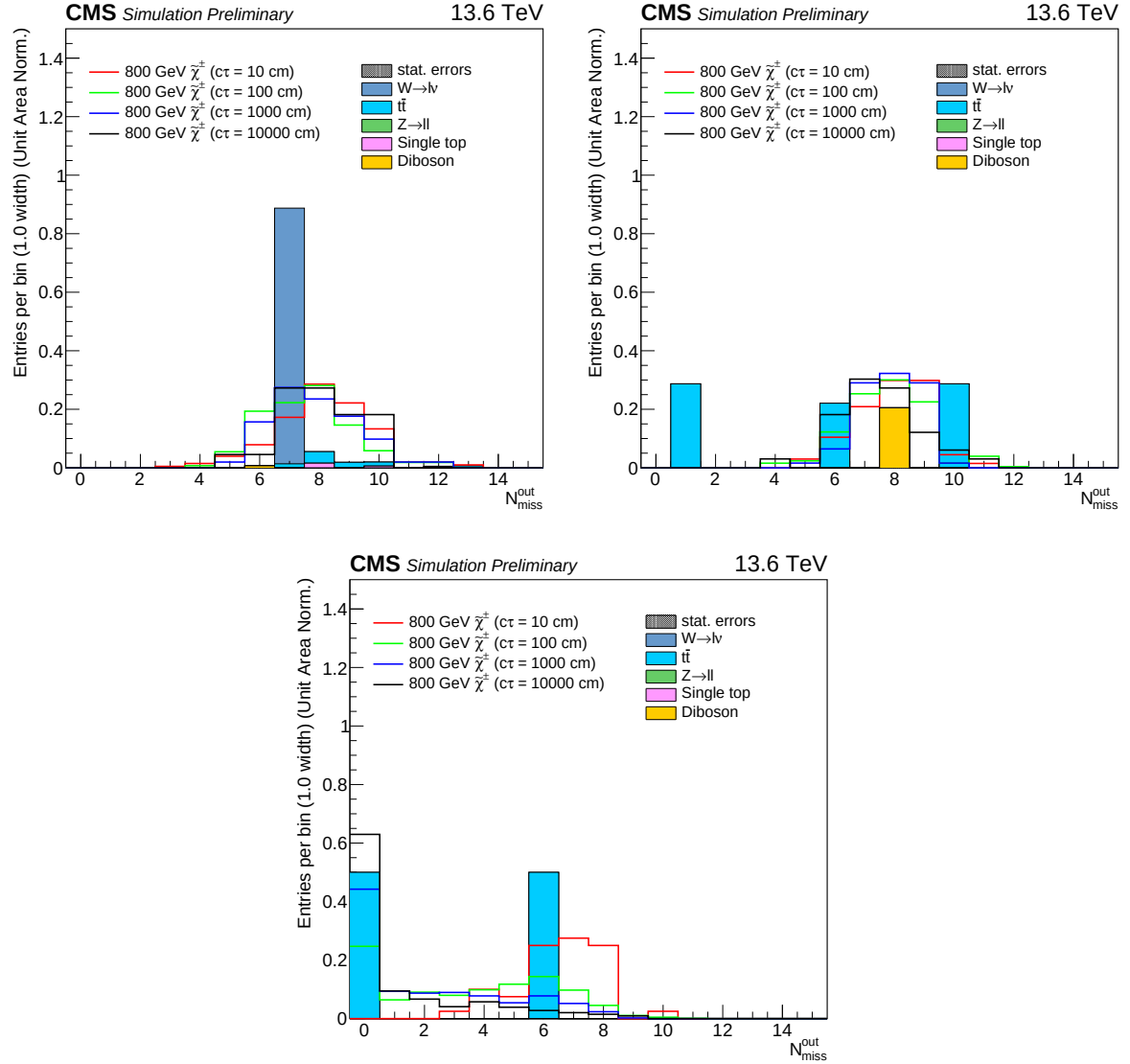


Figure 17: Cumulative selection efficiencies for each selection of the disappearing track selection in $n_{\text{layers}} \geq 6$ category for simulated signal events. The chargino mass in the simulated events is 700 GeV and three lifetime hypotheses are shown: 10 cm, 100 cm, and 1000 cm. The 2022 samples are shown here.

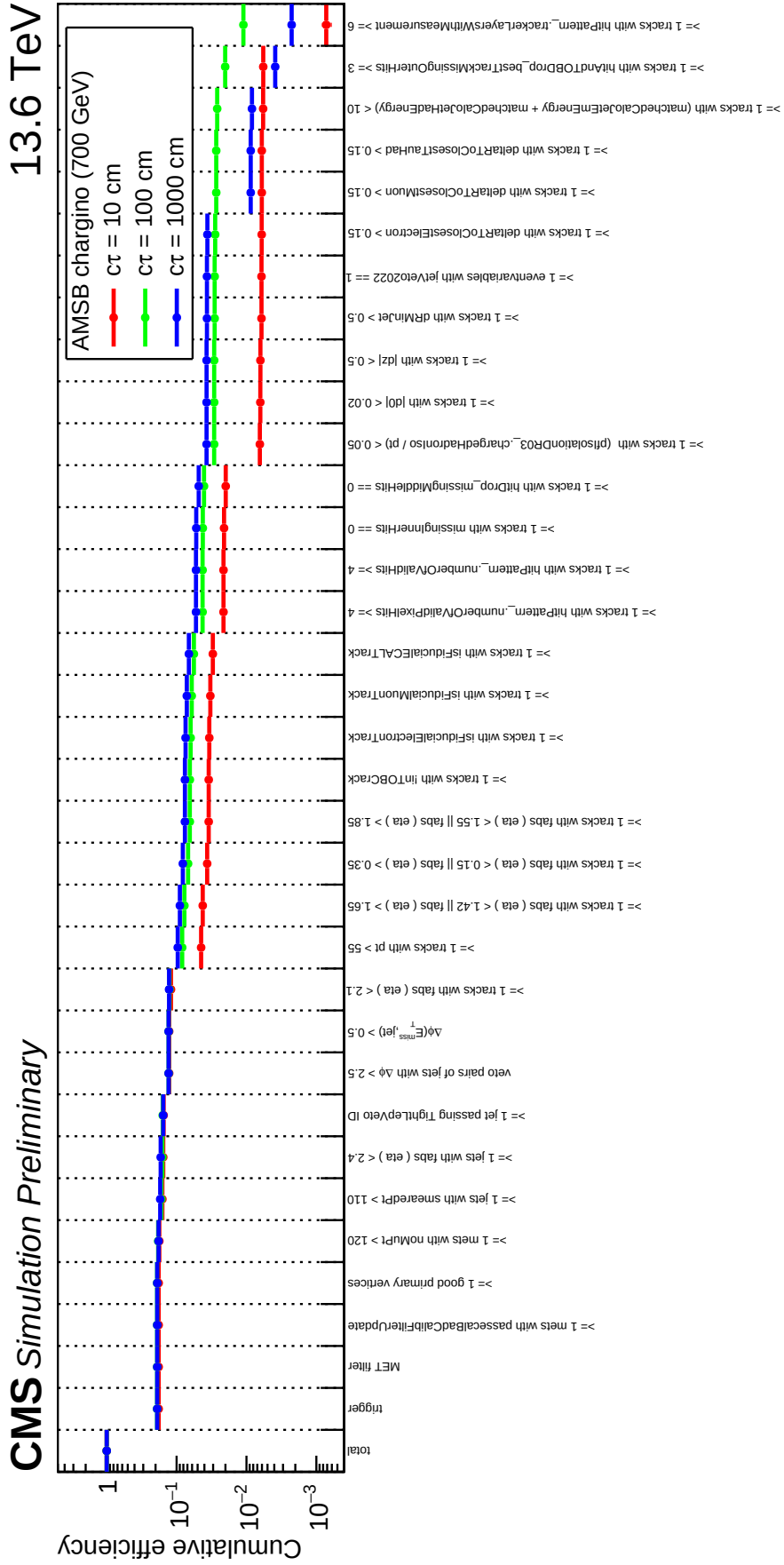


Figure 18: Event display of simulated signal event with a 700 GeV chargino with a lifetime of 10 cm/ c . The highlighted track is rejected by the track isolation requirement.

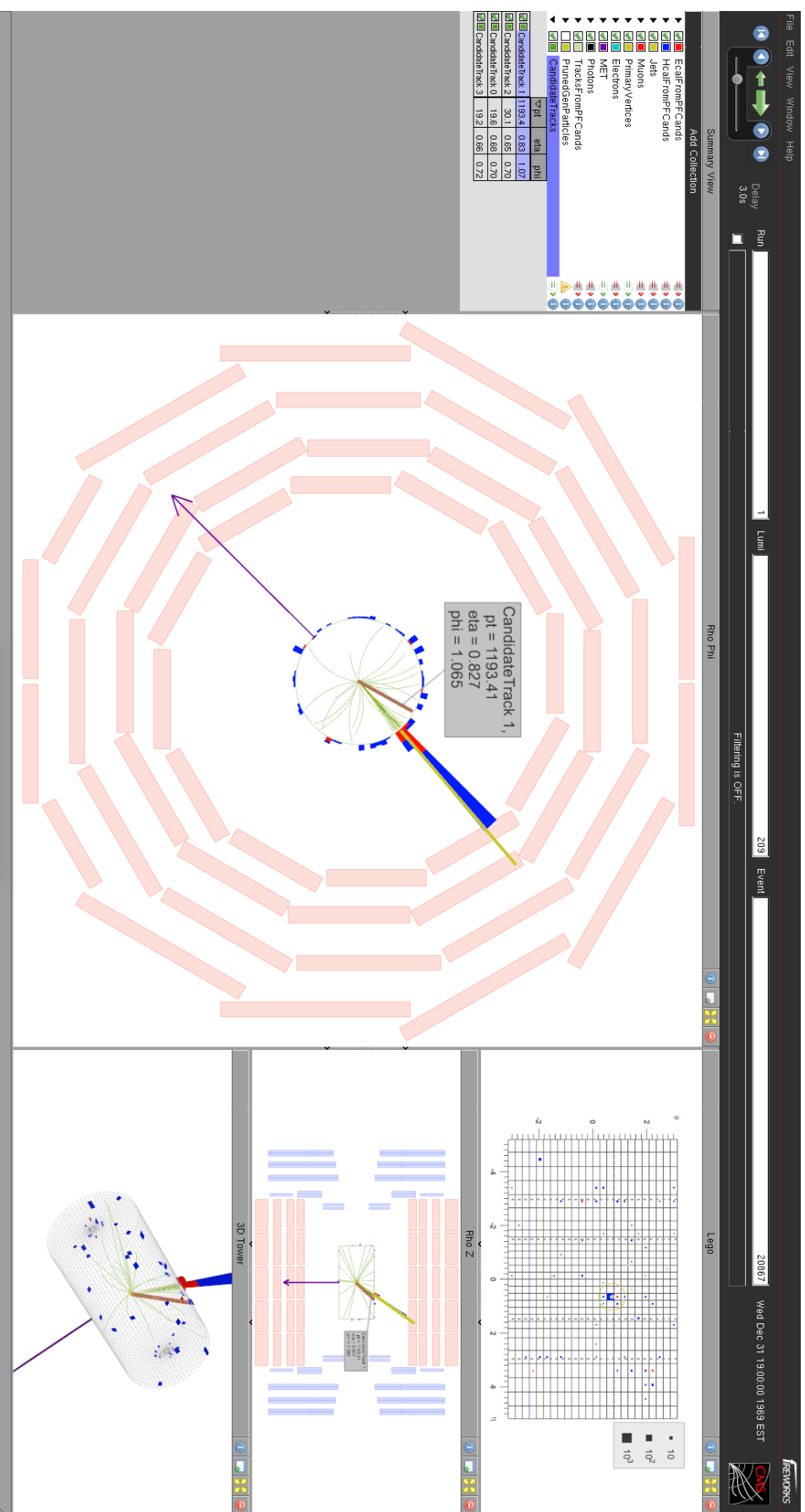
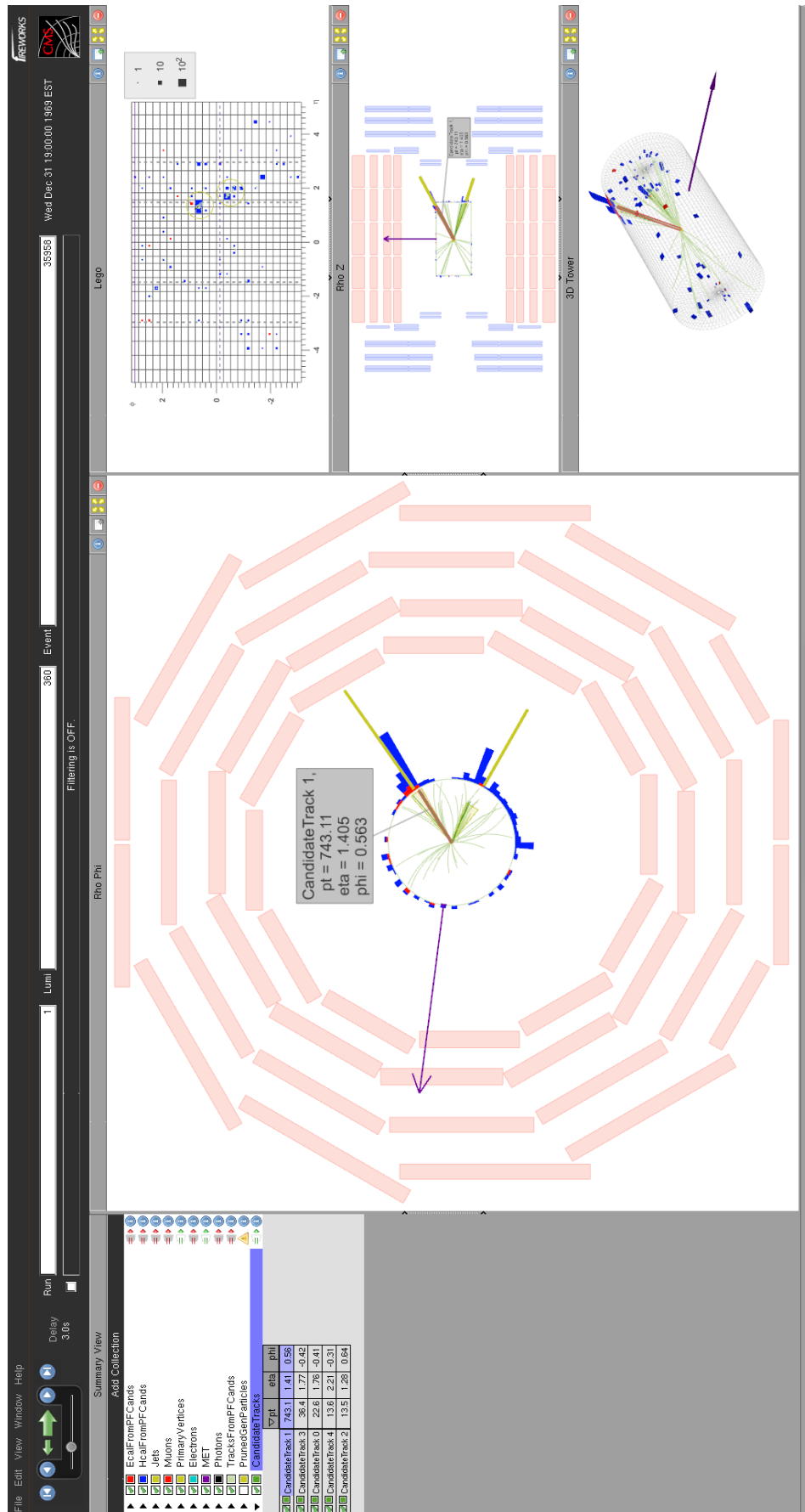


Figure 19: Event display of simulated signal event with a 700 GeV chargino with a lifetime of 10 cm/ c . The highlighted track is rejected by the track isolation requirement.



5 Background estimation

5.1 Charged leptons

The dominant source of high- p_T , isolated tracks from standard model processes is charged leptons from the decay of gauge bosons. In order for events with such tracks to enter our search region, several things must happen:

- the lepton must fail to be reconstructed as such, while still leaving a track in the silicon tracker
- the resulting $p_T^{\text{miss, no } \mu}$ must be large enough for the event to pass our offline $p_T^{\text{miss, no } \mu}$ requirement
- the resulting $p_T^{\text{miss, no } \mu}$ must be large enough for the event to pass our triggers

The key point here is that the $p_T^{\text{miss, no } \mu}$ is correlated with whether the lepton is reconstructed as a lepton or not. A successfully reconstructed lepton will be present in the list of candidates produced by particle flow, and thus, in the case of an electron or a tau, its p_T will count as visible momentum when $p_T^{\text{miss, no } \mu}$ is calculated. Conversely, a lepton that is not successfully reconstructed as such, but which still leaves a track that passes the selections of our search region, will not be present in the list of candidates produced by particle flow, and its p_T will count as invisible momentum when $p_T^{\text{miss, no } \mu}$ is calculated.¹¹ The method used to estimate the background from charged leptons is based on calculating the probability of the events listed above, with each lepton flavor treated independently.

5.1.1 Probability to pass the lepton veto

The first probability we consider is P_{veto} , the probability that the lepton in a single lepton event is not reconstructed as such:

To estimate P_{veto} , we use tag-and-probe studies. We defined electron and muon tag-and-probe selections in Section 4.2 in Tables 60 and 16, respectively, when discussing the method for identifying residual areas of electron and muon reconstruction inefficiency. The tag-and-probe samples we use here are identical except for the additional rejection of tracks near the identified areas of reconstruction inefficiency from that study. We also do not choose only one tag-probe pair per event, but rather use all unique pairs passing the selections. In addition, there are two *tau tag-and-probe samples*: one that attempts to tag the electron from a $\tau \rightarrow e\nu\nu$ decay, and one that attempts to tag the muon from a $\tau \rightarrow \mu\nu\nu$ decay. The selections of these two samples are listed in Tables 22 and 23, respectively.

$$P_{\text{veto}} = P(\text{pass lepton veto}). \quad (4)$$

The tag-and-probe selection criteria defined above are the signal selection criteria described in Section 4.3 with the exception of the lepton ID criteria for the veto under study, which are inverted as defined in Table 21. There is a separate veto for each lepton flavor, and each veto starts with the $\min \Delta R_{\text{track, lepton}} > 0.15$ criterion used in the candidate track selection. Recall from the definition of the candidate track selection in Section 4.3, that the definitions of $\min \Delta R_{\text{track, lepton}}$ include all reconstructed electrons and muons, while hadronic taus passing

¹¹In principle, the lepton could be misreconstructed as a charged hadron and contribute visible momentum in that way. However, this is highly unlikely given the requirement that $E_{\text{calo}} < 10 \text{ GeV}$: particle flow would have to match a track with $p_T > 55 \text{ GeV}$ to an energy deposit of less than 10 GeV.

the selections in Table 14 with the exception of the isolation criterion are used. Depending on the lepton flavor, the $\min \Delta R_{\text{track,lepton}}$ requirement is highly correlated with certain other criteria from the search region; e.g., the electron and tau vetoes are highly correlated with the selection on E_{calo} . For this reason, other selections are included in the lepton vetoes, including that of missing outer hits ≥ 3 , to account for any possible correlations between lepton reconstruction and our track selections. Thus, P_{veto} implicitly includes the probability for a track to have missing outer hits.

Table 21: Definitions of the inverted criteria applied that define the lepton vetoes that are studied with the *tag-and-probe samples* whose selection criteria is described in Section 5.1.1.

flavor	quantity	object	selection
electron	≥ 1	tracks	$\min \Delta R_{\text{track,electron}} > 0.15$
	≥ 1	tracks	$E_{\text{calo}} < 10 \text{ GeV}$
	≥ 1	tracks	missing outer hits ≥ 3
muon	≥ 1	tracks	$\min \Delta R_{\text{track,muon}} > 0.15$
	≥ 1	tracks	missing outer hits ≥ 3
tau	≥ 1	tracks	$\min \Delta R_{\text{track, had. tau}} > 0.15$
	≥ 1	track-jet pairs	$\Delta R_{\text{track,jet}} > 0.5$
	≥ 1	tracks	$E_{\text{calo}} < 10 \text{ GeV}$
	≥ 1	tracks	missing outer hits ≥ 3

Table 22: The criteria defining the *tau tag-and-probe sample* with an electron tag. The requirements are cumulative, with only the objects passing a given criterion being considered in subsequent criteria.

quantity	object	selection
	event passes SingleElectron/EGamma triggers	
≥ 1	electrons	$p_T > 32 \text{ GeV}$
≥ 1	electrons	$ \eta < 2.1$
≥ 1	electrons	passing tight electron ID
≥ 1	electrons	$M_T(p_T^{\text{miss}}, \text{electron}) < 40 \text{ GeV}$
≥ 1	tracks	$p_T > 30 \text{ GeV}$
≥ 1	tracks	passing fiducial selections (see Section 4.2)
≥ 1	tracks	number of pixel hits ≥ 4
≥ 1	tracks	missing inner hits = 0
≥ 1	tracks	missing middle hits = 0
≥ 1	tracks	rel. PF-based iso. < 0.05
≥ 1	tracks	$ d_{xy} < 0.02 \text{ cm}$
≥ 1	tracks	$ d_z < 0.5 \text{ cm}$
≥ 1	tracks	$\min \Delta R_{\text{track,electron}} > 0.15$
≥ 1	tracks	$\min \Delta R_{\text{track,muon}} > 0.15$
= 1	track-electron pairs	$15 \text{ GeV} < M_Z - M_{\text{track,electron}} < 50 \text{ GeV}$
= 1	track-electron pairs	$q_{\text{track}} \cdot q_{\text{electron}} < 0$
≥ 1	track	$n_{\text{layers}} \geq 4$ (three signal region bins)

We require $M_T(p_T^{\text{miss}}, \text{lepton}) < 40 \text{ GeV}$ to reduce the contamination from W +jets where the W decays to an electron or muon and a parton fragments into a single high- p_T hadron. This is the same strategy used to reduce this background in the measurement of the $Z \rightarrow \tau\tau$ cross section at 7 TeV [28]. In these samples there is significant contribution from hadrons not from tau

Table 23: The criteria defining the *tau tag-and-probe sample* with a muon tag. The requirements are cumulative, with only the objects passing a given criterion being considered in subsequent criteria.

quantity	object	selection
event passes SingleMuon triggers		
≥ 1	muons	$p_T > 26 \text{ GeV}$
≥ 1	muons	$ \eta < 2.1$
≥ 1	muons	passing tight muon ID
≥ 1	muons	$M_T(p_T^{\text{miss}}, \text{muon}) < 40 \text{ GeV}$
≥ 1	tracks	$p_T > 30 \text{ GeV}$
≥ 1	tracks	passing fiducial selections (see Section 4.2)
≥ 1	tracks	number of pixel hits ≥ 4
≥ 1	tracks	missing inner hits = 0
≥ 1	tracks	missing middle hits = 0
≥ 1	tracks	rel. PF-based iso. < 0.05
≥ 1	tracks	$ d_{xy} < 0.02 \text{ cm}$
≥ 1	tracks	$ d_z < 0.5 \text{ cm}$
≥ 1	tracks	$\min \Delta R_{\text{track,electron}} > 0.15$
≥ 1	tracks	$\min \Delta R_{\text{track,muon}} > 0.15$
$= 1$	track-muon pairs	$15 \text{ GeV} < M_Z - M_{\text{track,muon}} < 50 \text{ GeV}$
$= 1$	track-muon pairs	$q_{\text{track}} \cdot q_{\text{muon}} < 0$
≥ 1	track	$n_{\text{layers}} \geq 4$ (three signal region bins)

decays, however this is acceptable since isolated hadrons are indistinguishable from hadrons from $\tau \rightarrow \pi^\pm \nu$ decays. We additionally remove the requirement of $\Delta R_{\text{track,jet}} > 0.5$ from these selections to include this source of background¹².

While the tag-and-probe selections are dominated by $Z \rightarrow \ell\ell$ events desired for this measurement, some non- Z events are selected due to continuum Drell-Yan events, and the background of non-Drell-Yan events and events with fake tracks. To subtract the latter two backgrounds which decrease the purity of the lepton flavor under study, we invert the requirement in the tag-and-probe selections that the tag and probe should be oppositely charged. With these *same-sign tag-and-probe selections*, one for each of the four nominal tag-and-probe selections, we remove the $Z \rightarrow \ell\ell$ and Drell-Yan events and are left with an estimate of the charge-symmetric component of the background contamination in the nominal tag-and-probe selections. This applies both before and after the lepton veto is applied to the probe track, so that P_{veto} is defined as

$$P_{\text{veto}} = \frac{N_{\text{T\&P}}^{\text{veto}} - N_{\text{SST\&P}}^{\text{veto}}}{N_{\text{T\&P}} - N_{\text{SST\&P}}}, \quad (5)$$

where $N_{\text{T\&P}}$ ($N_{\text{T\&P}}^{\text{veto}}$) and $N_{\text{SST\&P}}$ ($N_{\text{SST\&P}}^{\text{veto}}$) are the numbers of passing tag-probe pairs from the events in the nominal (opposite-sign) and same-sign tag-and-probe selections, respectively, before (after) the final vetoes are applied to the probe track. The values of each component of P_{veto} in experimental data are shown in Table 24, and the resulting P_{veto} shown in Table 28.

¹²This more accurately makes the “tau estimate” an estimate of both hadronic taus and single hadronic tracks, but the language is kept simpler for readability.

Table 24: The components used in the calculation of P_{veto} , and its value. $N_{\text{T\&P}}$ ($N_{\text{T\&P}}^{\text{veto}}$) and $N_{\text{SST\&P}}$ ($N_{\text{SST\&P}}^{\text{veto}}$) are the numbers of passing tag-probe pairs from the events in the nominal and same-sign tag-and-probe selections, respectively, before (after) the final vetoes are applied to the probe track. Where the numerator for P_{veto} is negative, we set the central value to 0 and the upper uncertainty to 1σ of a normal distribution.

run period	flavor	n_{layers}	$N_{\text{T\&P}}$	$N_{\text{T\&P}}^{\text{veto}}$	$N_{\text{SST\&P}}$	$N_{\text{SST\&P}}^{\text{veto}}$	P_{veto}
2022 CD	electron	4	33198	16	2193	16	$0.0^{+1.82}_{-0.0} \times 10^{-4}$
		5	83475	2	1832	2	$0.0^{+2.45}_{-0.0} \times 10^{-5}$
		≥ 6	1135112	18	3817	0	$(1.59 \pm 0.38) \times 10^{-5}$
2022 CD	muon	4	58	28	20	16	0.32 ± 0.19
		5	27	3	8	6	$(0.0^{+6.00}_{-0.0}) \times 10^{-2}$
		≥ 6	2118297	1	74	1	$(0.0^{+6.68}_{-0.0}) \times 10^{-7}$
2022 CD	tau	4	48	11	24	15	$(0.0^{+4.75}_{-0.0}) \times 10^{-2}$
		5	51	1	11	3	$(0.0^{+2.85}_{-0.0}) \times 10^{-2}$
		≥ 6	3133	0	491	0	$(0.0^{+6.10}_{-0.0}) \times 10^{-4}$
2022 EFG	electron	4	66548	47	4615	42	$8.07^{+15.23}_{-8.07} \times 10^{-5}$
		5	172270	12	4318	8	$2.38^{+2.66}_{-2.38} \times 10^{-5}$
		≥ 6	2425839	32	8064	0	$(1.32 \pm 0.23) \times 10^{-5}$
2022 EFG	muon	4	413	38	149	34	$(1.5^{+3.2}_{-1.5}) \times 10^{-2}$
		5	249	5	52	8	$(0.0^{+5.78}_{-0.0}) \times 10^{-3}$
		≥ 6	11365245	2	418	0	$(1.76 \pm 1.24) \times 10^{-7}$
2022 EFG	tau	4	95	21	36	20	$(1.69^{+10.86}_{-1.69}) \times 10^{-2}$
		5	102	5	9	2	$(3.23^{+2.87}_{-2.43}) \times 10^{-2}$
		≥ 6	4396	2	700	0	$(5.41^{+4.91}_{-3.83}) \times 10^{-4}$
2023 C	electron	4	58118	32	4222	35	$(0.0^{+2.11}_{-0.0}) \times 10^{-5}$
		5	166567	19	4453	10	$(5.55 \pm 3.32) \times 10^{-5}$
		≥ 6	2380183	34	8473	2	$(1.35 \pm 0.25) \times 10^{-5}$
2023 C	muon	4	179	92	86	81	$0.12^{+0.14}_{-0.12}$
		5	110	18	18	15	$(3.26^{+6.26}_{-3.26}) \times 10^{-2}$
		≥ 6	4047502	3	149	1	$(4.94 \pm 4.94) \times 10^{-7}$
2023 C	tau	4	92	25	37	55	$(0.0^{+3.08}_{-0.0}) \times 10^{-2}$
		5	126	10	32	15	$(0.0^{+1.21}_{-0.0}) \times 10^{-2}$
		≥ 6	6822	1	1139	0	$((1.76^{+2.67}_{-1.76}) \times 10^{-4}$
2023 D	electron	4	28627	16	2134	24	$(0.0^{+4.30}_{-0.0}) \times 10^{-5}$
		5	82860	9	2275	9	$(0.0^{+5.26}_{-0.0}) \times 10^{-5}$
		≥ 6	1233525	4	4315	1	$(2.44 \pm 1.82) \times 10^{-6}$
2023 D	muon	4	108	55	62	60	$(0.0^{+2.48}_{-0.0}) \times 10^{-2}$
		5	71	17	18	16	$(1.89^{+10.84}_{-1.89}) \times 10^{-2}$
		≥ 6	2195110	2	111	0	$(9.11 \pm 6.44) \times 10^{-7}$
2023 D	tau	4	70	28	35	25	$(8.57^{+20.95}_{-6.86}) \times 10^{-2}$
		5	53	5	19	8	$(0.0^{+3.35}_{-0.0}) \times 10^{-2}$
		≥ 6	3296	0	599	2	$(0.0^{+4.22}_{-0.0}) \times 10^{-4}$

5.1.2 Probability to pass the offline p_T^{miss} requirements

We define P_{offline} as the probability of a single lepton event to pass our offline requirements of $p_T^{\text{miss, no } \mu} > 120 \text{ GeV}$ and $|\Delta\Phi(\text{leading jet}, p_T^{\text{miss, no } \mu})| > 0.5$ given that the lepton is not reconstructed as such. That is,

$$P_{\text{offline}} = P(\text{pass offline } p_T^{\text{miss}} \text{ cuts} | \text{pass lepton veto}). \quad (6)$$

To estimate this probability, we use *single-lepton control regions* that are designed to count the leptons that could enter the search region, were they fail to be reconstructed. Each involves selecting a reconstructed lepton matched to a track passing the selections of the *isolated track selection* in an event passing the jet criteria of the *basic selection*. The selections of the *single-electron control sample*, *single-muon control sample*, and *single-tau control sample* are listed in Tables 25, 26, and 27, respectively.¹³¹⁴

Table 25: The criteria defining the *single-electron control sample*. The requirements are cumulative, with only the objects passing a given criterion being considered in subsequent criteria.

quantity	object	selection
		event passes SingleElectron/EGamma triggers
≥ 1	electrons	$p_T > 32 \text{ GeV}$
≥ 1	electrons	$ \eta < 2.1$
≥ 1	electrons	passes tight electron ID
		exactly one passing electron chosen randomly
		event passes jet criteria of <i>basic selection</i>
		event passes track criteria of <i>isolated track selection</i>
		track selected in <i>isolated track selection</i> matched to selected electron within $\Delta R < 0.1$

Table 26: The criteria defining the *single-muon control sample*. The requirements are cumulative, with only the objects passing a given criterion being considered in subsequent criteria.

quantity	object	selection
		event passes SingleMuon triggers
≥ 1	muons	$p_T > 26 \text{ GeV}$
≥ 1	muons	$ \eta < 2.1$
≥ 1	muons	passes tight muon ID
		exactly one passing muon chosen randomly
		event passes jet criteria of <i>basic selection</i>
		event passes track criteria of <i>isolated track selection</i>
		track selected in <i>isolated track selection</i> matched to selected muon within $\Delta R < 0.1$

Now to estimate P_{offline} , the main assumption we make is that if a lepton is not reconstructed, then it contributes no visible momentum to the event.¹⁵ Thus, for electrons and taus, we intro-

¹³The *isolated track selection* includes the requirement $p_T > 55 \text{ GeV}$. So although we include a selection on the lepton p_T in these control regions, we later impose a tighter requirement on the track p_T .

¹⁴Although the *single-tau control sample* is defined using the hadronic tau ID, this sample also includes isolated hadrons from other sources that are mistaken for single-prong hadronic tau decays. This is because the pion from a $\tau \rightarrow \pi^\pm \nu$ decay is indistinguishable from an isolated hadron from any other source.

¹⁵In practice, an unreconstructed lepton cannot contribute more than 10 GeV in the search region because we explicitly require the associated calorimeter energy of selected tracks to be under this threshold. The calculation of a systematic uncertainty associated with this assumption is in progress

Table 27: The criteria defining the *single-tau control sample*. The requirements are cumulative, with only the objects passing a given criterion being considered in subsequent criteria.

quantity	object	selection
		event passes Tau triggers
≥ 1	taus	$p_T > 50 \text{ GeV}$
≥ 1	taus	$ \eta < 2.1$
≥ 1	taus	passes hadronic tau ID
		exactly one passing tau chosen randomly
		event passes jet criteria of <i>basic selection</i>
		event passes track criteria of <i>isolated track selection</i>
		track selected in <i>isolated track selection</i> matched to selected tau within $\Delta R < 0.1$

duce modified p_T^{miss} distributions that represent what $p_T^{\text{miss, no } \mu}$ would look like if the lepton in these events were not reconstructed:

$$p_T^{\text{miss, no } \mu} \longrightarrow |\vec{p}_T^{\text{miss, no } \mu} + \vec{p}_T^{\text{lepton}}| \quad \text{for electrons and taus.} \quad (7)$$

No such modification is necessary in the case of muons since we always count the p_T of muons as invisible. Then, using the events in the single-lepton control regions defined above, we create two-dimensional distributions of $p_T^{\text{miss, no } \mu}$ vs. $|\Delta\Phi(\text{leading jet}, p_T^{\text{miss, no } \mu})|$ with these modifications to $p_T^{\text{miss, no } \mu}$. These distributions, before and after the modification to $p_T^{\text{miss, no } \mu}$, are shown in Figure 20 for electrons. The distributions for muons, which receive no such modification to $p_T^{\text{miss, no } \mu}$, and taus are shown in Figure 21.

Finally, we estimate P_{offline} by counting the fraction of events with $p_T^{\text{miss, no } \mu} > 120 \text{ GeV}$ and $|\Delta\Phi(\text{leading jet}, p_T^{\text{miss, no } \mu})| > 0.5$ in these modified $p_T^{\text{miss, no } \mu}$ distributions. This is illustrated in Figure 20 where P_{offline} is the fraction of events in the plot that fall to the upper right of the blue lines. The values we calculate for P_{offline} in experimental data are shown below in Table 28.

Figure 20: Distributions of $p_T^{\text{miss, no } \mu}$ vs. $|\Delta\Phi(\text{leading jet}, p_T^{\text{miss, no } \mu})|$ in single electron events with $n_{\text{layers}} = 4$ (top), $n_{\text{layers}} = 5$ (middle), and $n_{\text{layers}} \geq 6$ (bottom) for the 2022 postEE data set. The left (right) column shows the $p_T^{\text{miss, no } \mu}$ distribution with the momentum of the electron counted as visible (invisible). The red lines denote the requirements on $p_T^{\text{miss, no } \mu}$ and $|\Delta\Phi(\text{leading jet}, p_T^{\text{miss, no } \mu})|$ in the signal selection, and P_{offline} is calculated as the fraction of events falling in the upper right region.

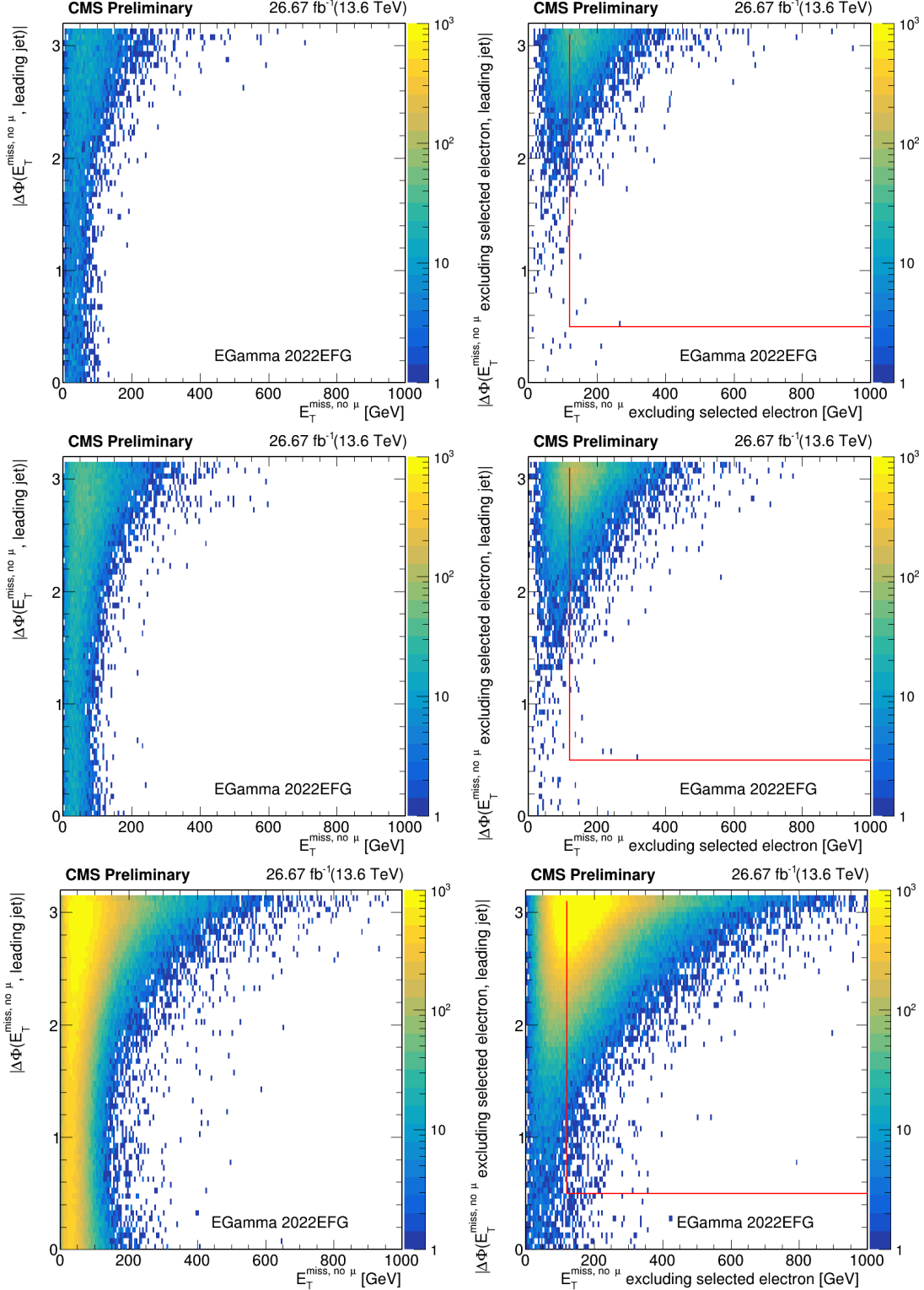
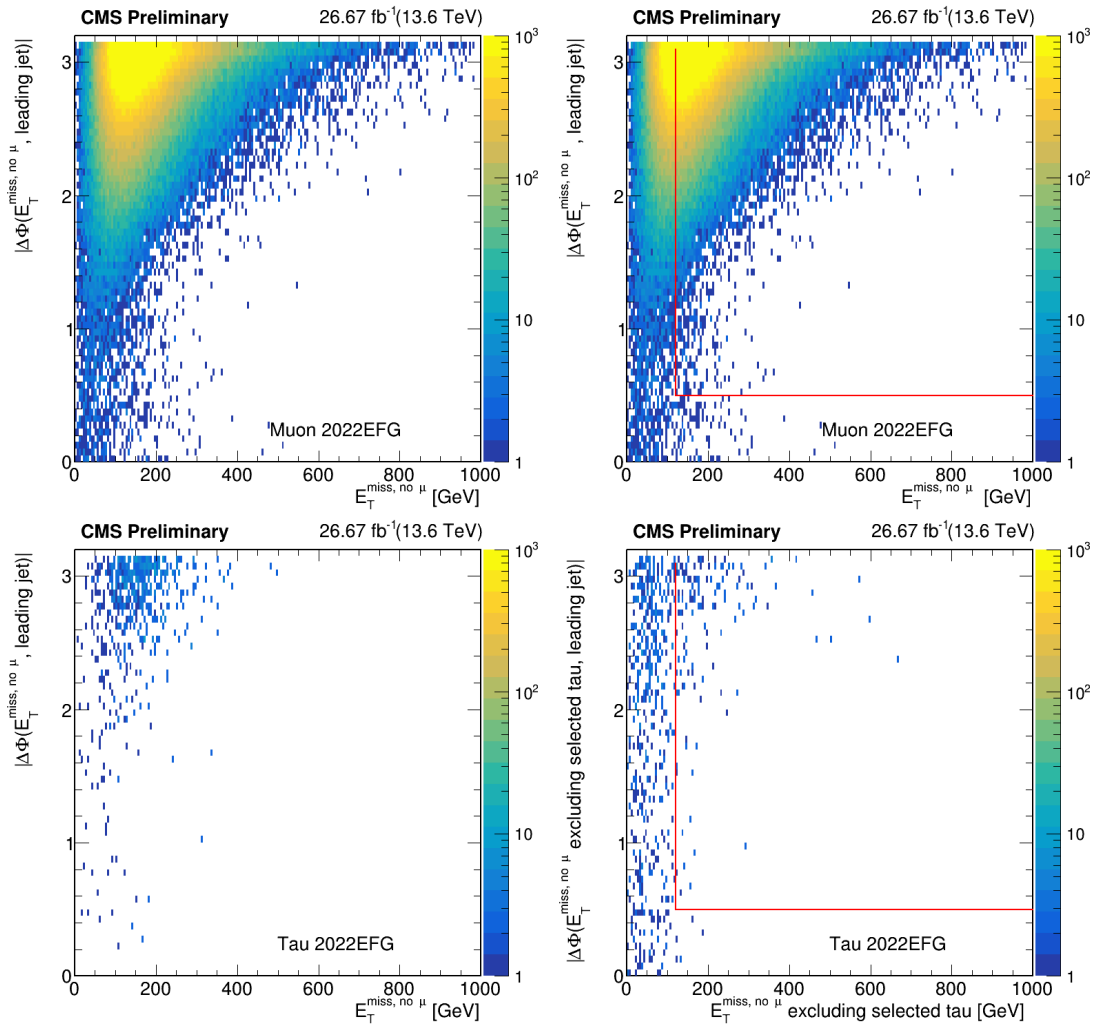


Figure 21: Distributions of $p_T^{\text{miss, no } \mu}$ vs. $|\Delta\Phi(\text{leading jet}, p_T^{\text{miss, no } \mu})|$ in single muon (top) and tau (bottom) events with $n_{\text{layers}} \geq 6$ for the 2022 postEE data set. The left (right) column shows the $p_T^{\text{miss, no } \mu}$ distribution with the momentum of the electron counted as visible (invisible). The red lines denote the requirements on $p_T^{\text{miss, no } \mu}$ and $|\Delta\Phi(\text{leading jet}, p_T^{\text{miss, no } \mu})|$ in the signal selection, and P_{offline} is calculated as the fraction of events falling in the upper right region.



5.1.3 Probability to pass the p_T^{miss} triggers

We define P_{trigger} as the probability of a single lepton event to pass the p_T^{miss} triggers used in the search given that the lepton is not reconstructed as such and the event passes our offline requirements of $p_T^{\text{miss, no } \mu} > 120 \text{ GeV}$ and $|\Delta\Phi(\text{leading jet}, p_T^{\text{miss, no } \mu})| > 0.5$; i.e.,

$$P_{\text{trigger}} = P(\text{pass } p_T^{\text{miss}} \text{ triggers} | \text{pass lepton veto} \cap \text{pass } p_T^{\text{miss}} \text{ cuts}). \quad (8)$$

To estimate P_{trigger} , we first calculate the efficiency $\epsilon(x)$ of the p_T^{miss} triggers used in the search as a function of $p_T^{\text{miss, no } \mu}$ using the events in the single-lepton control regions defined above in Section 5.1.1. This efficiency is calculated simply as the probability to pass the signal trigger requirement, as described below in Section 6.3. This efficiency is shown in Figures 22–24 for the single-lepton control regions. Because of the very low number of events in the single-tau control region, we use the trigger efficiency as measured in the single-electron control region for the estimate of P_{trigger} for taus.

We then project the events passing $p_T^{\text{miss, no } \mu} > 120 \text{ GeV}$ and $|\Delta\Phi(\text{leading jet}, p_T^{\text{miss, no } \mu})| > 0.5$ shown in Figure 20 above from the two-dimensional modified $p_T^{\text{miss, no } \mu}$ distribution onto the horizontal axis ($p_T^{\text{miss, no } \mu}$). This projection is illustrated in Figure 25. Calling this projection $n(x)$, P_{trigger} is estimated as

$$P_{\text{trigger}} = \frac{\int_{120 \text{ GeV}}^{\infty} n(x) \epsilon(x) dx}{\int_{120 \text{ GeV}}^{\infty} n(x) dx}, \quad (9)$$

where $\epsilon(x)$ is the trigger efficiency calculated previously. The values calculated for P_{trigger} , P_{veto} , and P_{offline} in experimental data are given below in Tables 28, 29, and 30.

5.1.4 Background estimates

With the probabilities defined above, we can then define the probability of an event with a charged lepton to enter our search region:

$$\begin{aligned} &P(\text{enter search region}) \\ &= P(\text{pass lepton veto} \cap \text{pass offline } p_T^{\text{miss}} \text{ cut} \cap \text{pass } p_T^{\text{miss}} \text{ triggers}) \\ &= P_{\text{veto}} P_{\text{offline}} P_{\text{trigger}}, \end{aligned}$$

where we use the chain rule of probability to factor the probabilities in the last line.

The probability of an event with a charged lepton to enter our search region must be normalized to the number of events with charged leptons of each flavor. These are counted as the number of events passing the single lepton control regions defined in Section 5.1.2 (Tables 25–27), and are referred to as N_{ctrl}^l . Recall that the disappearing track selection requires tracks to have $p_T > 55 \text{ GeV}$, and the single lepton control regions make this requirement as well.

A final consideration in the normalization of the lepton background estimates is that the trigger requirements in the single lepton control regions may not be completely efficient. This leads to the underestimation of the number of events with charged leptons, and N_{ctrl}^l is corrected by dividing by the measured single lepton trigger efficiency for each flavor. The trigger efficiencies are measured using the tag-and-probe samples defined in Section 5.1.1 (Tables 60–16 and 22–23), with the additional requirement of the probe track having $p_T > 55 \text{ GeV}$. In these samples

Figure 22: Efficiency of the p_T^{miss} triggers used in the search as a function of $p_T^{\text{miss, no } \mu}$ as measured in the single-electron control region with $n_{\text{layers}} = 4$ (top), $n_{\text{layers}} = 5$ (middle), and $n_{\text{layers}} \geq 6$ (bottom). The 2022 preEE data set is shown on the left, and the 2022 postEE data set is shown on the right.

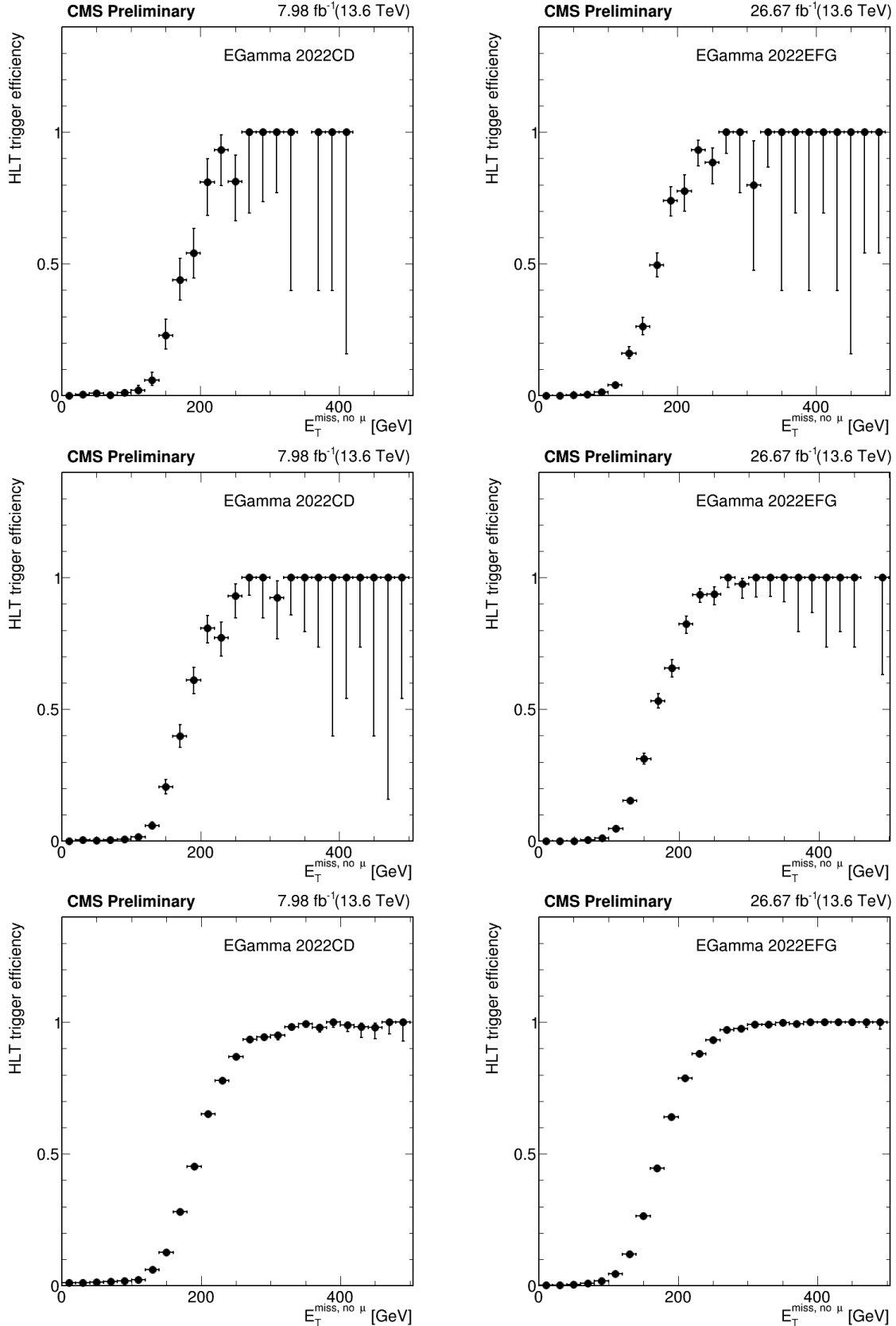


Figure 23: Efficiency of the p_T^{miss} triggers used in the search as a function of $p_T^{\text{miss, no } \mu}$ as measured in the single-electron control region with $n_{\text{layers}} = 4$ (top), $n_{\text{layers}} = 5$ (middle), and $n_{\text{layers}} \geq 6$ (bottom). The 2023 preBpix data set is shown on the left, and the 2023 postBpix data set is shown on the right.

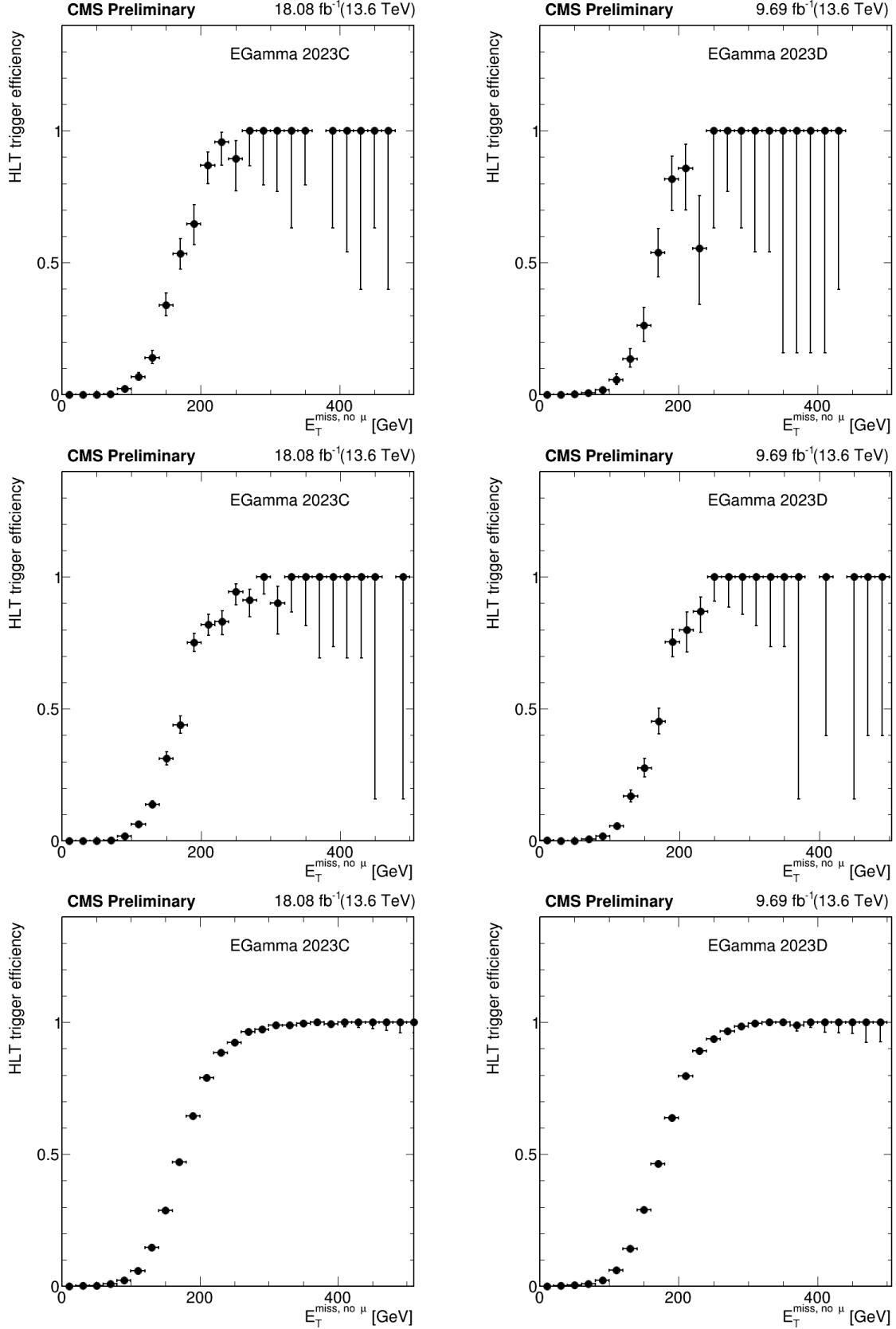


Figure 24: Efficiency of the p_T^{miss} triggers used in the search as a function of $p_T^{\text{miss, no } \mu}$ as measured in the single-muon control region for $n_{\text{layers}} \geq 6$. The 2022 preEE data set (top left), the 2022 postEE data set (top right), the 2023 preBpix (bottom left), and the 2023 postBpix (bottom right).

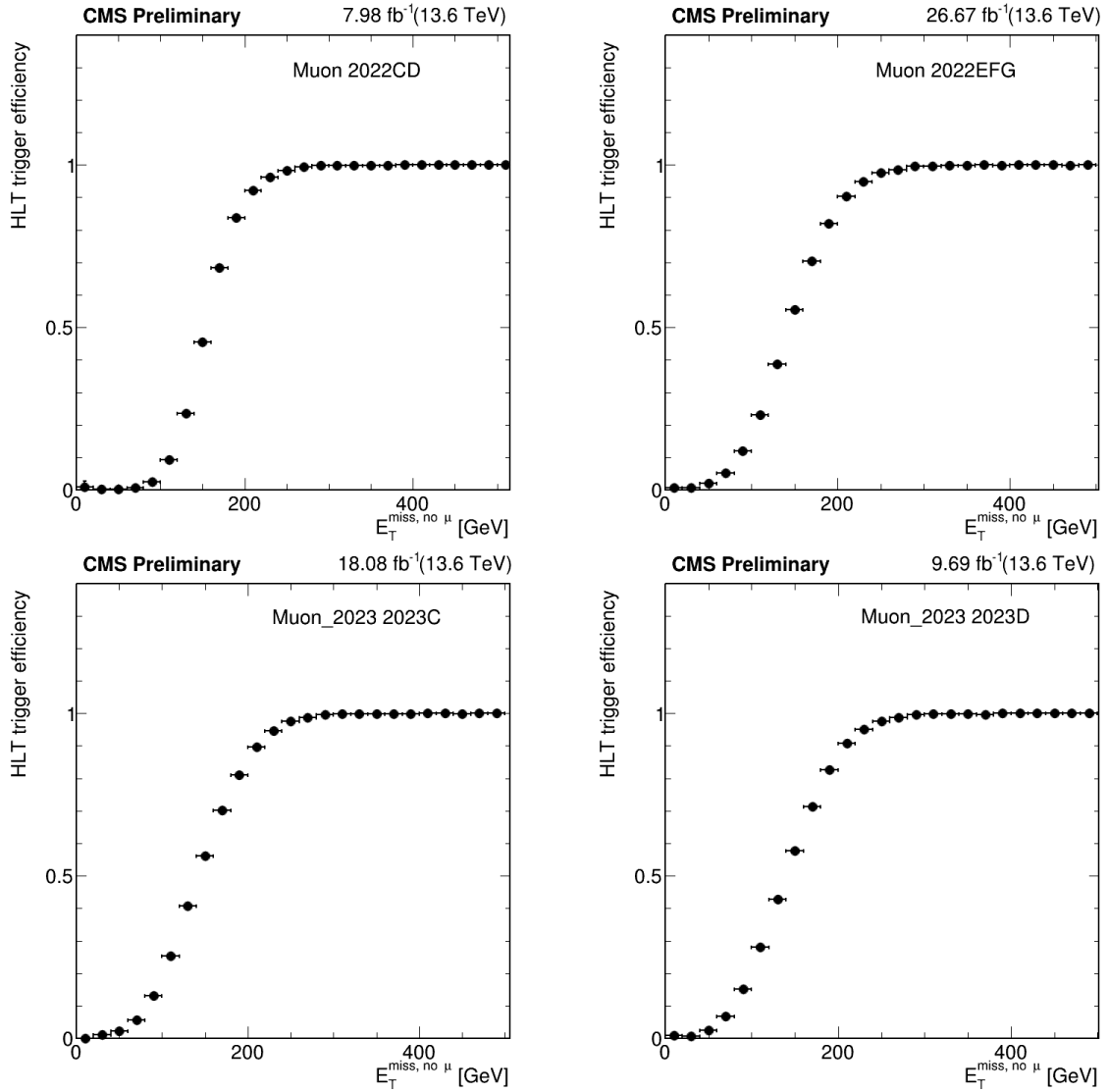
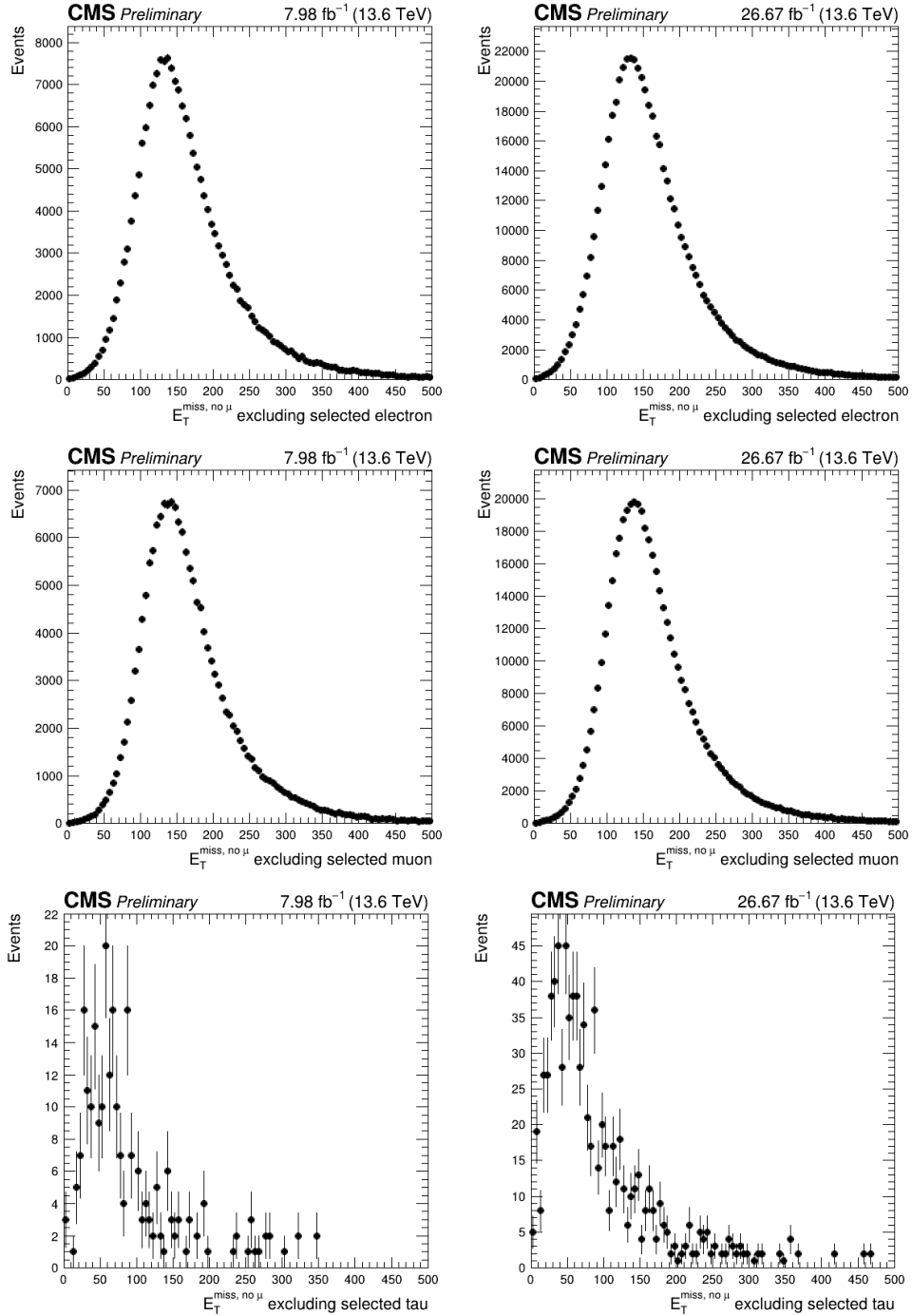


Figure 25: The projections used in the calculation of P_{trigger} using the two-dimensional $p_T^{\text{miss, no } \mu}$ distributions from Section 5.1.1, modified in the electron and tau cases by excluding the selected electron or tau. The events passing $|\Delta\Phi(\text{leading jet}, p_T^{\text{miss, no } \mu})| > 0.5$ are projected onto the horizontal axis, which is then multiplied by the corresponding trigger efficiency. The relative area of these projections after this multiplication passing $p_T^{\text{miss, no } \mu} > 120 \text{ GeV}$ then forms P_{trigger} . Shown are the single electron selections (top), single muon selections (middle) and tau selections (bottom) with $n_{\text{layers}} \geq 6$ with the 2022 preEE data set (top left), the 2022 postEE dataset (top right).



the trigger efficiency, $\epsilon_{\text{trigger}}^l$, is defined as the probability for a probe track to be matched within $\Delta R < 0.3$ of an HLT primitive of the appropriate path for the flavor under study (see Table 9). Applying the same-sign subtraction technique as used for P_{veto} :

$$\epsilon_{\text{trigger}}^l = \frac{N_{\text{T\&P } p_T > 55}^{\text{matched}} - N_{\text{SS T\&P } p_T > 55}^{\text{matched}}}{N_{\text{T\&P } p_T > 55} - N_{\text{SS T\&P } p_T > 55}} \quad (10)$$

where $N_{\text{T\&P } p_T > 55}$ ($N_{\text{T\&P } p_T > 55}^{\text{matched}}$) and $N_{\text{SS T\&P } p_T > 55}$ ($N_{\text{SS T\&P } p_T > 55}^{\text{matched}}$) are the numbers of passing passing tag-probe pairs from the events in the opposite- and same-sign tag-and-probe selections with the probe track having $p_T > 55$ GeV, respectively, before (after) requiring that the probe track is matched to the appropriate HLT primitive. In run 2 this efficiency was found to be 0.84. Currently this is the value we are using as a flat efficiency across all eras and signal bins. Studies are ongoing to set values for $\epsilon_{\text{trigger}}^l$ in each of our datasets in run 3.

For the tau lepton background there is an additional scaling to estimate the number of single tau's that would pass a single tau HLT trigger. We estimate this quantity using two available triggers `HLT_IsoMu20_v*` (defined as the muon trigger) and our nominal tau trigger `HLT_IsoMu20_eta2p1_LooseDeepTauPFTauHPS27_eta2p1_CrossL1_v*` (defined as the muon+tau trigger). We estimate the probability to pass tau only HLT as the ratio of events passing the tau + muon HLT to the muon only HLT as in equation 11. This ratio estimating the probability to pass the single tau HLT ($P(\text{tau})$) is then used to scale the number of events in our lepton control region N_{ctrl}^l . Because the probability to pass the tau+muon HLT is already contained in N_{ctrl}^l we multiply by $1/P(\text{muon}) = N_{\text{total}}/N_{\text{pass, muon}}$ to obtain the number of events passing the tau only HLT.

$$\frac{P(\text{muon+tau})}{P(\text{muon})} = P(\text{tau}) \quad (11)$$

With this final piece, all that remains is to multiply the number of events with charged leptons by the probability for those events to pass into the search region:

$$N_{\text{est}}^l = \frac{N_{\text{ctrl}}^l}{\epsilon_{\text{trigger}}^l} P_{\text{veto}} P_{\text{offline}} P_{\text{trigger}}. \quad (12)$$

The resulting electron, muon, and tau background estimates are ongoing but a preliminary result for the electrons is summarized in table 28, along with each of the quantities used to calculate the estimates.

Table 28: Electron background estimates for each n_{layers} category in the search, along with the quantities used to calculate each estimate. The rightmost column is simply the product of the middle five columns. The combined-category values of P_{offline} and P_{trigger} are included to provide a systematic uncertainty for muon and tau estimates in categories without sufficient statistics to estimate these values. The uncertainties listed are statistical only.

run period	n_{layers}	electron background					estimate
		$\epsilon_{\text{trigger}}^1$	N_{ctrl}^1	P_{veto}	P_{offline}	P_{trigger}	
2022 CD	4	0.728 ± 0.026	4239 ± 65	$(8.83^{+14.98}_{-8.83}) \times 10^{-5}$	0.642 ± 0.016	0.373 ± 0.041	$0.12^{+0.21}_{-0.12}$
	5	0.846 ± 0.015	11926 ± 109	$(0.0^{+9.48}_{-0.0}) \times 10^{-6}$	0.667 ± 0.010	0.406 ± 0.025	$0.0^{+0.04}_{-0.0}$
	≥ 6	0.905 ± 0.004	295183 ± 543	$(1.62 \pm 0.31) \times 10^{-5}$	0.738 ± 0.002	0.402 ± 0.005	1.57 ± 0.30
	combined	0.899 ± 0.004	311348 ± 558	$(1.64 \pm 0.49) \times 10^{-5}$	0.734 ± 0.002	0.403 ± 0.005	1.68 ± 0.50
2022 EFG	4	0.671 ± 0.017	8664 ± 93	$(4.76^{+11.53}_{-4.76}) \times 10^{-5}$	0.658 ± 0.011	0.474 ± 0.032	$0.19^{+0.46}_{-0.19}$
	5	0.824 ± 0.011	24859 ± 158	$(2.60 \pm 2.12) \times 10^{-5}$	0.667 ± 0.007	0.500 ± 0.019	0.26 ± 0.21
	≥ 6	0.882 ± 0.003	628397 ± 793	$(1.21 \pm 0.19) \times 10^{-5}$	0.741 ± 0.001	0.500 ± 0.004	3.18 ± 0.51
	combined	0.876 ± 0.003	661920 ± 814	$(1.38 \pm 0.35) \times 10^{-5}$	0.737 ± 0.001	0.500 ± 0.004	3.85 ± 0.98
2023 C	4	0.683 ± 0.021	5303 ± 73	$(0.0^{+1.38}_{-0.00}) \times 10^{-4}$	0.648 ± 0.014	0.504 ± 0.041	$0.0^{+0.35}_{-0.0}$
	5	0.802 ± 0.012	15652 ± 125	$(6.81 \pm 3.01) \times 10^{-5}$	0.655 ± 0.008	0.539 ± 0.025	0.47 ± 0.21
	≥ 6	0.876 ± 0.003	422301 ± 650	$(1.95 \pm 0.28) \times 10^{-5}$	0.725 ± 0.002	0.553 ± 0.005	3.77 ± 0.55
	combined	0.867 ± 0.003	448559 ± 670	$(2.17 \pm 0.52) \times 10^{-5}$	0.721 ± 0.002	0.553 ± 0.005	4.48 ± 1.07
2023 D	4	0.671 ± 0.029	2484 ± 50	$(0.0^{+3.67}_{-0.0}) \times 10^{-5}$	0.653 ± 0.021	0.462 ± 0.069	$0.00^{+0.04}_{-0.00}$
	5	0.801 ± 0.017	7372 ± 86	$(2.13^{+5.00}_{-2.13}) \times 10^{-5}$	0.655 ± 0.012	0.554 ± 0.039	$0.07^{+0.17}_{-0.07}$
	≥ 6	0.878 ± 0.004	212453 ± 461	$(4.29 \pm 2.02) \times 10^{-6}$	0.718 ± 0.002	0.562 ± 0.008	0.42 ± 0.20
	combined	0.866 ± 0.004	224793 ± 474	$(0.0^{+7.33}_{-0.0}) \times 10^{-7}$	0.715 ± 0.002	0.562 ± 0.007	$0.00^{+0.07}_{-0.00}$
2024 CEFghi	4	0.708 ± 0.016	10109.0 ± 100.5	$(0.00^{+1.10}_{-0.10}) \times 10^{-5}$	0.571 ± 0.009	0.410 ± 0.027	$0.00^{+0.037}_{-0.00}$
	5	0.822 ± 0.010	25894.0 ± 160.9	$(0.00^{+4.30}_{-0.00}) \times 10^{-6}$	0.583 ± 0.006	0.402 ± 0.016	$0.00^{+0.00}_{-0.00}$
	≥ 6	0.892 ± 0.002	843214.0 ± 918.3	$(3.87 \pm 1.02) \times 10^{-6}$	0.679 ± 0.001	0.422 ± 0.003	$1.046 \pm .278$
	combined	0.886 ± 0.002	879216.0 ± 937.7	$(2.00^{+3.57}_{-2.00}) \times 10^{-6}$	0.675 ± 0.001	0.422 ± 0.003	$0.564^{+1.009}_{-0.564}$

Table 29: Muon background estimates for each n_{layers} category in the search, along with the quantities used to calculate each estimate. In the $n_{\text{layers}} = 4$ and 5 categories, sufficient statistics were not available to estimate P_{offline} and P_{trigger} , so the values measured in the combined-category sample are used instead. The rightmost column is simply the product of the middle four columns. The uncertainties listed are statistical only.

run period	n_{layers}	muon background					estimate
		$\epsilon_{\text{trigger}}^1$	N_{ctrl}^1	P_{veto}	P_{offline}	P_{trigger}	
2022 CD	4	0.94 ± 0.004	$0.0^{+1.1}_{-0.0}$	0.32 ± 0.19	-	-	$0.0^{+0.20}_{-0.0}$
	5	0.94 ± 0.004	1.0 ± 1.0	$(0.0^{+6.00}_{-0.0}) \times 10^{-2}$	-	-	$(0.0^{+3.11}_{-0.0}) \times 10^{-2}$
	≥ 6	0.94 ± 0.004	205108 ± 453	$(0.0^{+6.88}_{-0.0}) \times 10^{-7}$	0.778 ± 0.004	0.628 ± 0.004	$(0.0^{+7.11}_{-0.0}) \times 10^{-2}$
	combined	0.94 ± 0.004	205109 ± 453	$(4.25 \pm 3.50) \times 10^{-6}$	0.777 ± 0.003	0.628 ± 0.004	0.45 ± 0.37
2022 EFG	4	0.94 ± 0.004	5.0 ± 2.2	$(1.6^{+3.2}_{-1.6}) \times 10^{-2}$	-	-	$(4.2^{+9.1}_{-4.2}) \times 10^{-2}$
	5	0.94 ± 0.004	3.0 ± 1.7	$(0.0^{+5.78}_{-0.0}) \times 10^{-2}$	-	-	$(0.0^{+9.56}_{-0.0}) \times 10^{-3}$
	≥ 6	0.94 ± 0.004	640761 ± 800	$(1.76 \pm 1.24) \times 10^{-7}$	0.771 ± 0.001	0.672 ± 0.002	$(6.2 \pm 4.4) \times 10^{-2}$
	combined	0.94 ± 0.004	640769 ± 800	$(2.64^{+8.21}_{-2.64}) \times 10^{-7}$	0.771 ± 0.001	0.672 ± 0.002	$0.093^{+0.290}_{-0.093}$
2023 C	4	0.94 ± 0.002	1.0 ± 1.0	$0.12^{+0.14}_{-0.12}$	-	-	$(6.66^{+10.44}_{-6.66}) \times 10^{-2}$
	5	0.94 ± 0.002	1.0 ± 1.0	$(3.26^{+6.26}_{-3.26}) \times 10^{-2}$	-	-	$(1.84^{+3.97}_{-1.84}) \times 10^{-2}$
	≥ 6	0.94 ± 0.003	307476 ± 555	$(4.94 \pm 4.94) \times 10^{-7}$	0.747 ± 0.002	0.708 ± 0.003	$(8.55 \pm 8.55) \times 10^{-2}$
	combined	0.94 ± 0.002	307478 ± 555	$(3.95 \pm 3.58) \times 10^{-6}$	0.747 ± 0.002	0.708 ± 0.003	0.68 ± 0.62
2023 D	4	0.945 ± 0.003	$0.0^{+1.1}_{-0.0}$	$(0.0^{+1.81}_{-0.0}) \times 10^{-2}$	-	-	$(0.0^{+8.98}_{-0.0}) \times 10^{-7}$
	5	0.945 ± 0.003	$0.0^{+1.1}_{-0.0}$	$(0.0^{+1.90}_{-0.0}) \times 10^{-2}$	-	-	$(0.0^{+1.22}_{-0.0}) \times 10^{-6}$
	≥ 6	0.945 ± 0.003	209274 ± 457	$(7.47 \pm 5.28) \times 10^{-7}$	0.760 ± 0.003	0.703 ± 0.004	0.088 ± 0.062
	combined	0.945 ± 0.003	209274 ± 457	$(0.0^{+4.26}_{-0.0}) \times 10^{-7}$	0.760 ± 0.003	0.703 ± 0.004	$0.0^{+0.05}_{-0.0}$

Table 30: Tau background estimates for each n_{layers} category in the search, along with the quantities used to calculate each estimate. In the $n_{\text{layers}} = 4$ and 5 categories, sufficient statistics were not available to estimate P_{offline} and P_{trigger} , so the values measured in the combined-category sample are used instead. The rightmost column is simply the product of the middle four columns. The data in the single-tau control sample were collected with a prescaled trigger, and the statistical uncertainties in N_{ctrl}^1 include the effective prescale for each run period. The HLT trigger used for tau selection requires the presence of both a tau and a muon. The uncertainties listed are statistical only.

run period	n_{layers}	$\epsilon_{\text{trigger}}^1$	$P(\text{tau})$	tau background		P_{veto}	P_{offline}	P_{trigger}	estimate
				N_{ctrl}					
2022 CD	4	$0.90 \pm 6.00 \times 10^{-3}$	$1.46 \pm 2.39 \times 10^{-3}$	8.00 ± 2.83		$0.0^{+0.04}_{-0.0}$	0.50 ± 0.31	0.69 ± 0.72	$0.0^{+0.06}_{-0.0}$
	5	$0.90 \pm 6.00 \times 10^{-3}$	$1.46 \pm 2.39 \times 10^{-3}$	4.00 ± 2.00		$0.0^{+0.02}_{-0.0}$	0.50 ± 0.43	0.76 ± 0.77	$0.0^{+0.02}_{-0.0}$
	6	$0.90 \pm 6.00 \times 10^{-3}$	$1.46 \pm 2.39 \times 10^{-3}$	385.00 ± 19.62		$0.0^{+5.31 \times 10^{-4}}_{-0.0}$	0.32 ± 0.03	0.57 ± 0.08	$0.0^{+12.75}_{-0.0}$
	combined	$0.90 \pm 5.00 \times 10^{-3}$	$1.46 \pm 2.39 \times 10^{-3}$	397.00 ± 19.92		$0.0^{+3.66 \times 10^{-4}}_{-0.0}$	0.33 ± 0.03	0.57 ± 0.08	$0.0^{+0.04}_{-0.0}$
2022 EFG	4	$0.90 \pm 6.00 \times 10^{-3}$	$1.53 \pm 1.21 \times 10^{-3}$	15.00 ± 3.87		$0.15^{+0.10}_{-0.09}$	0.47 ± 0.21	0.69 ± 0.45	$0.61^{+0.43}_{-0.41}$
	5	$0.90 \pm 6.00 \times 10^{-3}$	$1.53 \pm 1.21 \times 10^{-3}$	20.00 ± 4.47		$0.02^{+0.03}_{-0.01}$	0.45 ± 0.18	0.67 ± 0.34	$0.09^{+0.15}_{-0.06}$
	≥ 6	$0.90 \pm 6.00 \times 10^{-3}$	$1.53 \pm 1.21 \times 10^{-3}$	822.00 ± 28.67		$0.0^{+7.97 \times 10^{-5}}_{-0.0}$	0.35 ± 0.02	0.68 ± 0.06	$0.0^{+5.60}_{-0.0}$
	combined	$0.90 \pm 5.00 \times 10^{-3}$	$1.53 \pm 1.21 \times 10^{-3}$	856.99 ± 29.27	$1.51 \times 10^{-3} \pm 9.39 \times 10^{-4}$		0.36 ± 0.02	0.68 ± 0.06	$0.53^{+0.02}_{-0.34}$

5.1.5 Closure tests in simulated events

We perform a closure test of the lepton background estimation method in samples of simulated events in the (2022) conditions, detailed in Table 7. Because of the limited event counts afforded by the full search selection, the test is performed with slightly relaxed selection criteria:

- all jet-related selections are removed from the basic selection (see Table 17)
- the track p_T requirement is lowered from 55 GeV to 35 GeV (see Table 18) for the electron and muon flavors
- the final selection on missing outer hits for tracks is removed (see Table 20)

The first two modifications are made to the single-lepton control regions used to determine N_{ctrl}^l , and the missing outer hits requirement is removed from the vetoes of the lepton tag-and-probe samples (see Table 21). The P_{veto} for each lepton flavor is then measured in the $Z \rightarrow l\bar{l}$ samples. The full lepton background estimation procedure is remade from scratch, including the calculation of $\epsilon(x)$, since the different track p_T requirement may have an effect on it. The final modification is to utilize the truth information available in the selected (observed) events to assign a flavor to each event, allowing for a comparison of each flavor separately.

With the exception of measuring P_{veto} in the $Z \rightarrow l\bar{l}$ samples and the above modifications, the closure test is performed in the simulated samples of $t\bar{t}$ in the same manner as in data. The $t\bar{t}$ samples are chosen for this purpose as they provide the largest raw number of events passing the modified search selection, where other samples do not contribute significantly. The results of this closure test are shown in Table 31.

Table 31: Results of lepton background closure test in simulated $t\bar{t}$ events (2022) with relaxed selection criteria. The numbers of events are normalized to 34.7 fb^{-1} , the luminosity of the 2022 data set.

n_{layers}	flavor	N_{est}^l	N_{obs}^l	$ N_{\text{est}}^l - N_{\text{obs}}^l $
4	electron	0.05 ± 0.03	$0.0^{+0.07}_{-0.0}$	0.66σ
	muon	0.01 ± 0.01	0.08 ± 0.07	1.01σ
	tau	$0.06^{+0.04}_{-0.03}$	0.12 ± 0.08	0.65σ
5	electron	$0.0^{+0.03}_{-0.0}$	$0.04^{+0.06}_{-0.04}$	0.59σ
	muon	$0.02^{+0.03}_{-0.02}$	$0.04^{+0.06}_{-0.04}$	0.24σ
	tau	insufficient statistics		
≥ 6	electron	$0.0^{+0.06}_{-0.0}$	0.12 ± 0.08	1.19σ
	muon	$0.0^{+0.06}_{-0.0}$	0.12 ± 0.08	1.19σ
	tau	$0.0^{+0.02}_{-0.0}$	$0.04^{+0.06}_{-0.04}$	0.63σ

5.2 Fake tracks

Fake tracks, i.e., tracks that do not correspond to actual charged particle trajectories but rather to errors in the pattern recognition of the track reconstruction algorithm, are an obvious source of background. Because they do not correspond to actual particles, they are generally missing associated calorimeter energy and hits in the muon chambers. They would also be expected to have missing hits in the inner tracker in a random pattern. Although this contribution to the background should be largely suppressed by our requirements on the impact parameters of the tracks and that the tracks be missing no inner or middle hits in the inner tracker, especially in the $n_{\text{layers}} \geq 6$ search category, we estimate the surviving fake tracks as a background to the search.

We use two different control regions as representative samples of standard model events: the $Z \rightarrow \mu\mu$ control region defined in Table 32, and the $Z \rightarrow ee$ control region defined in Table 33. Since the $Z \rightarrow \mu\mu$ and $Z \rightarrow ee$ control regions are completely free of signal, we can be confident that any additional tracks in these samples satisfying our disappearing track requirements are fake tracks.

Table 32: The criteria defining the $Z \rightarrow \mu\mu$ control sample. The requirements are cumulative, with only the objects passing a given criterion being considered in subsequent criteria.

quantity	object	selection
	event passes SingleMuon triggers	
= 2	muons	$p_T > 26 \text{ GeV}$
= 2	muons	$ \eta < 2.1$
= 2	muons	passes tight muon ID
= 1	muon pairs	$q_{\text{muon}} \cdot q_{\text{muon}} < 0$
= 1	muon pairs	$ M_{\text{muon,muon}} - M_Z < 10 \text{ GeV}$

Table 33: The criteria defining the $Z \rightarrow ee$ control sample. The requirements are cumulative, with only the objects passing a given criterion being considered in subsequent criteria.

quantity	object	selection
	event passes SingleElectron triggers	
≥ 1	electrons	$p_T > 32 \text{ GeV}$
= 2	electrons	$p_T > 25 \text{ GeV}$
= 2	electrons	$ \eta < 2.1$
= 2	electrons	passes tight electron ID
= 1	electron pairs	$q_{\text{electron}} \cdot q_{\text{electron}} < 0$
= 1	electron pairs	$ M_{\text{electron,electron}} - M_Z < 10 \text{ GeV}$

The number of fake tracks in the search regions is estimated by calculating the raw probability of an event to contain a disappearing track candidate with a given n_{layers} , and applying this probability to the number of events in the basic selection, $N_{\text{ctrl}}^{\text{basic}}$. To enhance the purity of fake tracks, a sideband selection is employed replacing the nominal $|d_{xy}|$ requirement:

$$|d_{xy}| < 0.02 \text{ cm} \quad \longrightarrow \quad 0.05 \text{ cm} < |d_{xy}| < 0.50 \text{ cm} \quad (13)$$

which selects tracks not associated with the primary vertex. Assuming this sideband is composed entirely of fake tracks¹⁶, the raw fake track event probability is defined as:

¹⁶The validity of this assumption was explored in chapter 6.2.1 the Run 2 analysis note AN-18-311.

$$P_{\text{fake}}^{\text{raw}} \equiv \frac{N(\geq 1 \text{ disappearing track in sideband} \mid \text{control region})}{N(\text{control region})}. \quad (14)$$

This is evaluated for both $Z \rightarrow \mu\mu$ and $Z \rightarrow ee$ control regions. Since the $n_{\text{layers}} = 5, \geq 6$ categories are statistically limited, a transfer factor (labeled ζ) is calculated with the $n_{\text{layers}} = 4$ category events to normalize the sideband d_{xy} region to the signal region. This normalization is necessary both as the sideband is wider in d_{xy} than the signal region and as the distribution is not uniform, as shown in Figure 26.

The transfer factor ζ is extracted from a fit to the $|d_{xy}|$ distribution in the $n_{\text{layers}} = 4$ control regions, restricting the range of the fit to be within the sideband, $0.1 \text{ cm} \leq |d_{xy}| < 0.5 \text{ cm}$. The fit function used is a gaussian with mean fixed to zero, plus a constant. The transfer factor is then defined as:

$$\zeta \equiv \frac{\int_0^{0.02 \text{ cm}} \text{fit } d(|d_{xy}|)}{\int_{0.05 \text{ cm}}^{0.50 \text{ cm}} \text{fit } d(|d_{xy}|)} \quad (15)$$

which gives a value of 0.10 ± 0.06 ($0.11^{+0.16}_{-0.11}$)(fit) in the 2022 preEE dataset and 0.09 ± 0.04 (0.10 ± 0.04)(fit) in the postEE dataset using the $Z \rightarrow \mu\mu$ ($Z \rightarrow ee$) control region. In 2023 for era C the transfer factor is calculated to be 0.08 ± 0.04 (0.07 ± 0.07) and in era D it is 0.09 ± 0.05 (0.09 ± 0.04) using the $Z \rightarrow \mu\mu$ ($Z \rightarrow ee$) control region. The results of these fits are shown in Figure 26.

Combining the transfer factor with the measured values of $P_{\text{fake}}^{\text{raw}}$, the fake track event probability and background estimate are then:

$$P_{\text{fake}} = \zeta \cdot P_{\text{fake}}^{\text{raw}},$$

$$N_{\text{est}}^{\text{fake}} = P_{\text{fake}} N_{\text{ctrl}}^{\text{basic}},$$

where $N_{\text{ctrl}}^{\text{basic}}$ is the number of events in the basic selection.

The estimate is calculated in both the $Z \rightarrow \mu\mu$ and $Z \rightarrow ee$ control regions, using the $Z \rightarrow \mu\mu$ estimate as the central value and the difference between the two as a systematic uncertainty. The fake track event probabilities for each signal category and control region are listed in Table 34.

In the development of the analysis, the fake track estimate was taken as the average of multiple, smaller sideband definitions. The results of this method are described in Appendix G of the Run 2 analysis note AN-18-311.

5.2.1 Validation of fake track background estimate

The fake track background estimate rests on several assumptions. One is that the fake track event probability can be taken from any control region, which is tested by comparing the fake track background estimates from the $Z \rightarrow \mu\mu$ and $Z \rightarrow ee$ control regions in Table 34. The differences between these two estimates are employed as systematic uncertainties. The calculation of these uncertainties is in progress.

A second assumption is that the process of fitting the sideband to a gaussian distribution accurately describes the signal region of the d_{xy} distribution. Given that the number of tracks with

Figure 26: Results of the fit used to extract the transfer factor for the $Z \rightarrow \mu\mu$ (left) and $Z \rightarrow ee$ (right) control regions, for tracks with $n_{\text{layers}} = 4$ in the 2022 (top) and 2023 (bottom) datasets. The resulting gaussian function fits to each year's data points are shown as smooth curves, having restricted the range of the fit to $0.1 \text{ cm} < |d_{xy}| < 0.5 \text{ cm}$ and fixing the mean to be zero. The transfer factor ζ is calculated from the ratio of integrals of these functions in the signal and sideband regions.

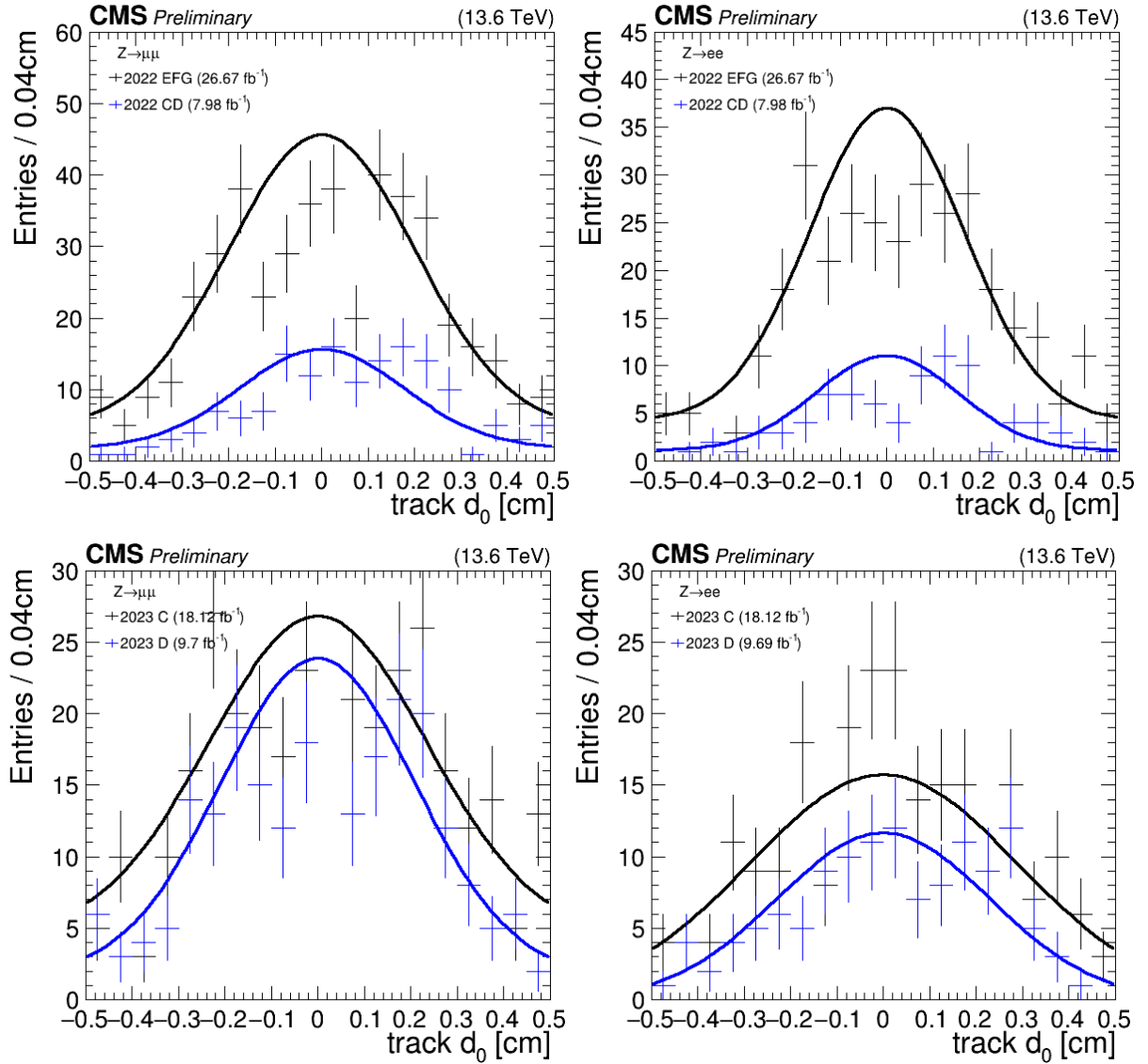


Table 34: Comparison of the fake track event probabilities P_{fake} , and fake track background estimates obtained from the $Z \rightarrow ee$ and $Z \rightarrow \mu\mu$ control region. The listed uncertainties represent the statistical uncertainties only.

run period	n_{layers}	P_{fake}		$N_{\text{est}}^{\text{fake}}$	
		$Z \rightarrow \mu\mu$	$Z \rightarrow ee$	$Z \rightarrow \mu\mu$	$Z \rightarrow ee$
2022 CD	4	$(3.93 \pm 2.37) \times 10^{-6}$	$(4.19 \pm 0.49) \times 10^{-6}$	5.28 ± 3.18	5.63 ± 0.66
	5	$(4.40 \pm 2.88) \times 10^{-7}$	$(2.87 \pm 1.28) \times 10^{-7}$	0.59 ± 0.39	0.39 ± 0.17
	≥ 6	$(3.14^{+3.66}_{-3.14}) \times 10^{-8}$	$(5.74 \pm 5.74) \times 10^{-8}$	$0.04^{+0.05}_{-0.04}$	0.08 ± 0.08
2022 EFG	4	$(3.15 \pm 1.25) \times 10^{-6}$	$(5.17 \pm 0.31) \times 10^{-6}$	15.31 ± 6.09	25.08 ± 1.52
	5	$(4.40 \pm 1.84) \times 10^{-7}$	$(6.67 \pm 1.13) \times 10^{-7}$	2.13 ± 0.89	3.24 ± 0.55
	≥ 6	$(1.69 \pm 1.37) \times 10^{-8}$	$(1.91 \pm 1.91) \times 10^{-8}$	0.08 ± 0.07	0.09 ± 0.09
2023 C	4	$(2.79 \pm 1.64) \times 10^{-6}$	$(2.84 \pm 2.69) \times 10^{-6}$	8.21 ± 4.81	8.34 ± 7.90
	5	$(6.88 \pm 4.09) \times 10^{-7}$	$(7.72 \pm 7.37) \times 10^{-7}$	2.02 ± 1.20	2.27 ± 2.17
	≥ 6	$0^{+1.15}_{-0.0} \times 10^{-8}$	$(1.58^{+2.17}_{-1.58}) \times 10^{-8}$	$0^{+0.03}_{-0}$	$0.05^{+0.06}_{-0.05}$
2023 D	4	$(4.24 \pm 2.52) \times 10^{-6}$	$(4.53 \pm 2.32) \times 10^{-6}$	6.62 ± 3.93	7.06 ± 3.62
	5	$(1.09 \pm 0.66) \times 10^{-6}$	$(7.10 \pm 3.98) \times 10^{-7}$	1.70 ± 1.03	1.11 ± 0.62
	≥ 6	$(4.35 \pm 4.01) \times 10^{-8}$	$(1.33 \pm 1.02) \times 10^{-7}$	0.07 ± 0.06	0.21 ± 0.16

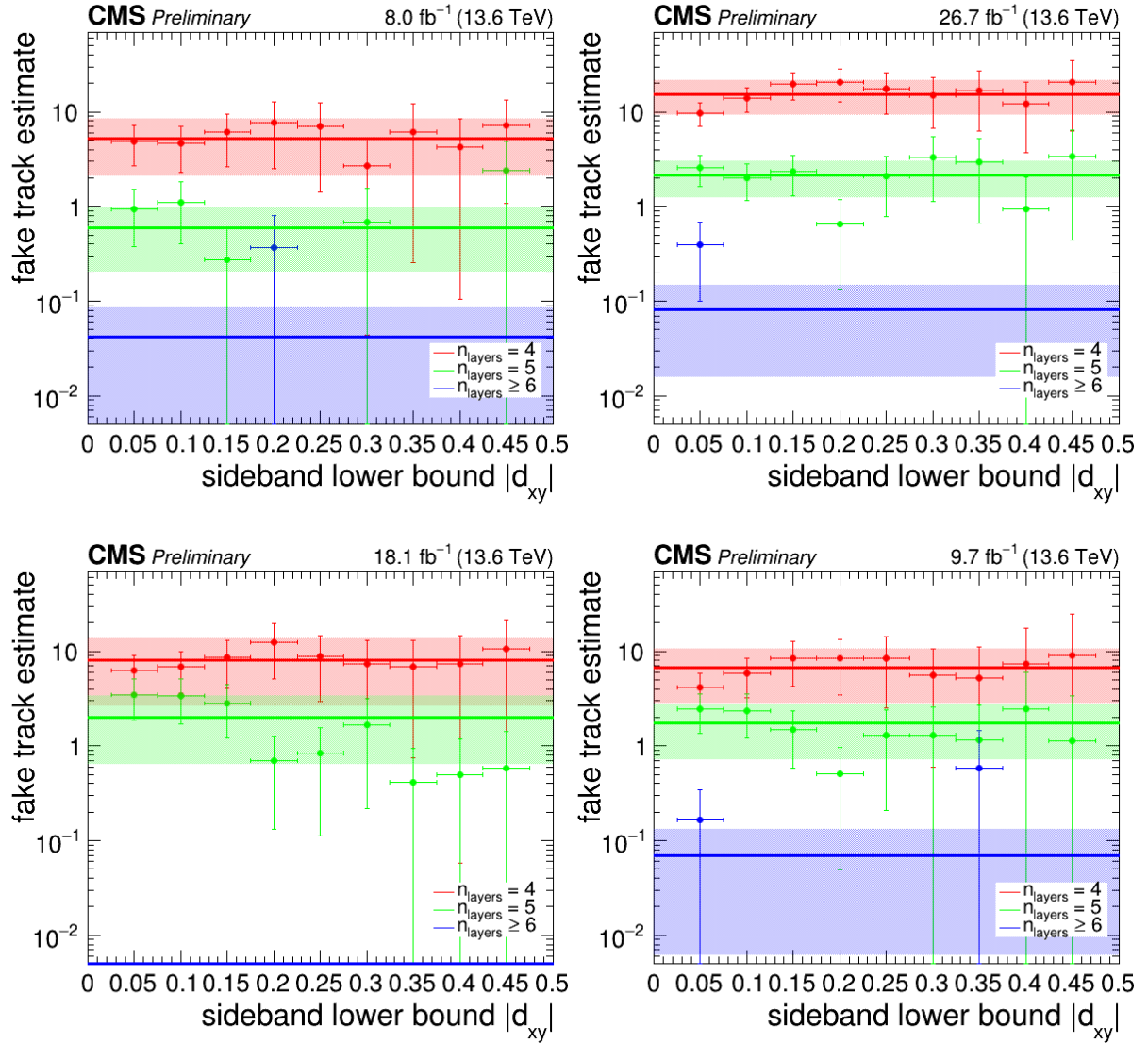
$|d_{xy}| < 0.02$ cm in the $n_{\text{layers}} = 4$ category can be directly counted in Figure 26, we compare the observed number of signal-like tracks to the projected number using the sideband. Good agreement is found in all cases, and no systematic uncertainty is applied for this assumption. The results of this comparison are described in Appendix D.

A third assumption is that the fake track estimate is not dependent on the definition of the d_{xy} sideband, outside of statistical fluctuations of the experimental data. As can be seen from Figure 26 the fit's description of the d_{xy} distribution is overall good, but if only a single bin were chosen as the sideband definition the estimate will vary. To examine this assumption, we ask not how the estimate changes with sideband definition, but how significant the variations are with respect to purely statistical fluctuations. To answer this, nine independent sub-regions of the $|d_{xy}|$ sideband are defined: (0.05, 0.10), (0.10, 0.15), (0.15, 0.20), (0.20, 0.25), (0.25, 0.30), (0.30, 0.35), (0.35, 0.40), (0.40, 0.45), and (0.45, 0.50). For each sub-region the fake track estimate is calculated as described above, but with the d_{xy} range altered. Full details of each estimate are shown in Appendix G of the Run 2 analysis note AN-18-311. Figure 27 compares the nominal estimate with the estimate from each of these sub-regions. The maximal difference of the nominal estimate from each sub-region is within 1.1, 1.7, and 0.5 standard deviations (statistical only) respectively for $n_{\text{layers}} = 4, 5, \geq 6$ in the $Z \rightarrow \mu\mu$ control region. These values are well within the expected range of statistical fluctuations, given the look-elsewhere effect among 27 independent measurements. As such no systematic uncertainty is applied for these differences.

5.2.2 Closure tests in simulated events

To further test that the method for estimating the background from fake tracks is sound, we perform a closure test in simulated background events. The method is performed exactly as it is in experimental data with the $Z \rightarrow \mu\mu$ control region, using all the samples listed in Section 2.3, and the result is compared to the actual number of events in these samples that contain a track that pass the disappearing track selections and is separated by $\Delta R > 0.1$ from all generator particles with $p_T > 10$ GeV. This is done separately for each of the three signal regions, except for the $n_{\text{layers}} \geq 6$ region, where there are insufficient statistics to perform a meaningful comparison. The results are listed in Table 35. In each region, closure is achieved to within

Figure 27: Fake track background estimates for different definitions of the $|d_{xy}|$ sideband in the 2022 CD (top left), EFG (top right) periods and the 2023 data for era C (bottom left) and era D (bottom right). The data points span the defined range of the 9 smaller sub-ranges as described in the text. The solid line is the estimate from the nominal sideband definition, and the shaded region shows its statistical uncertainty.

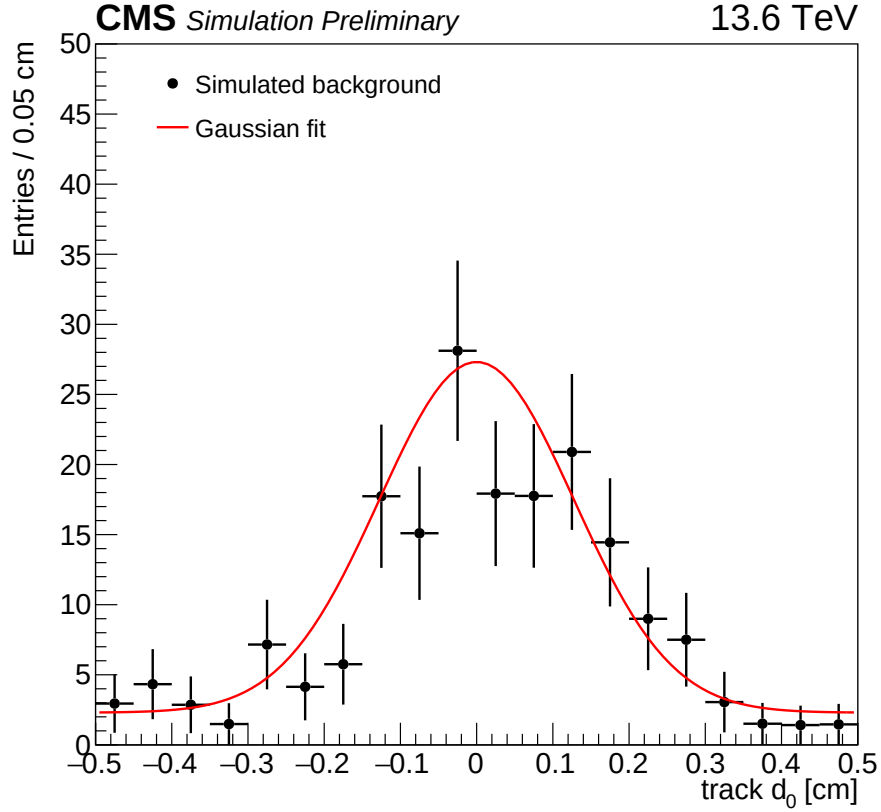


766 2.2σ , not including the 40% uncertainty from the fit used to extract the transfer factor. The d_{xy}
 767 distribution for tracks in the $n_{\text{layers}} = 4$ control region is shown in Figure 28, along with the
 768 result of the Gaussian fit.

Table 35: Results of the closure test for the fake track background estimate in 2022 simulated background events. The uncertainties are statistical and do not include the 40% uncertainty from the fit used to extract the transfer factor.

n_{layers}	$N_{\text{est}}^{\text{fake}}$	$N_{\text{obs}}^{\text{fake}}$	$ N_{\text{est}}^{\text{fake}} - N_{\text{obs}}^{\text{fake}} $
4	5.3 ± 2.0	2.7 ± 0.4	1.3σ
5	1.8 ± 0.9	0.3 ± 0.1	1.7σ
≥ 6	insufficient statistics		

Figure 28: Distribution of d_{xy} for tracks in the $n_{\text{layers}} = 4$ control region in 2022 simulated background events. The red line shows the result of the Gaussian fit to this distribution.

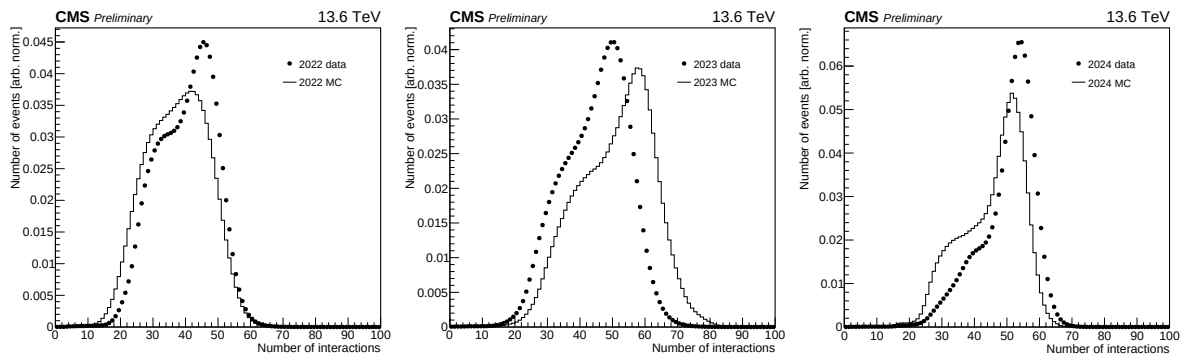


6 Corrections to the simulated signal samples

6.1 Pileup scale factors

Pileup scale factors are applied to simulated signal events to correct for differences in the number of pileup interactions between the simulated and experimental data. These scale factors are calculated by dividing the PDF for the actual number of pileup interactions by that used during the production of the simulated events. These PDFs are shown in Figure 29 for experimental data and the corresponding simulated data.

Figure 29: Distributions of the numbers of interactions for the 2022 (left), 2023 (middle), and 2024 (right) data sets, both experimental (points) and simulated (lines).



6.2 Missing hits corrections

As was discussed in Section 4, the selections used in the search utilize the number of missing middle and outer hits in candidate tracks to discriminate against tracks from background sources. However, as can be seen in Figure 30, taken from the *muon control region*, defined as the candidate track selection without the muon veto, these are not well-modeled in the simulation, especially at and above three missing outer hits which defines the search region. The overall probability for tracks to be missing middle and outer hits is slightly higher in the experimental data than in simulation, with the majority of the disagreement being near six missing outer hits. This is known to be caused by a mis-modeling of the extrapolation of tracks from the TIB to the TOB, the failure of which can create several missing outer hits as the TOB has six layers.

This mis-modeling does not affect the background estimates since they are taken exclusively from experimental data, but it does affect the estimated signal selection efficiencies, which are taken directly from simulation. We attempt to correct the missing outer hits distribution in the simulation in order to get more realistic estimates of the signal acceptances. This is done by randomly removing hits from the nominal hit pattern at a rate that is determined by fitting the resulting missing outer hits distribution to those from experimental data.

Because nothing as sophisticated as refitting tracks is done, there are four hit removal probabilities that must be determined:

- $P_0 \equiv P(\text{missing extra middle hit})$
- $P_{\text{pre-TOB}} \equiv P(\text{missing extra outer hits})$
- $P_{\text{TOB}} \equiv P(\text{missing all TOB hits})$
- $P_{\text{post-TOB}} \equiv P(\text{missing extra outer hits} | \text{missing all TOB hits})$

Figure 30: Uncorrected distributions of missing middle (left) and outer (right) hits in 2022 for tracks selected in the muon control region from in experimental data and simulated background processes. In the case of the missing middle hits distribution, the usual requirement of zero missing middle hits is not applied.

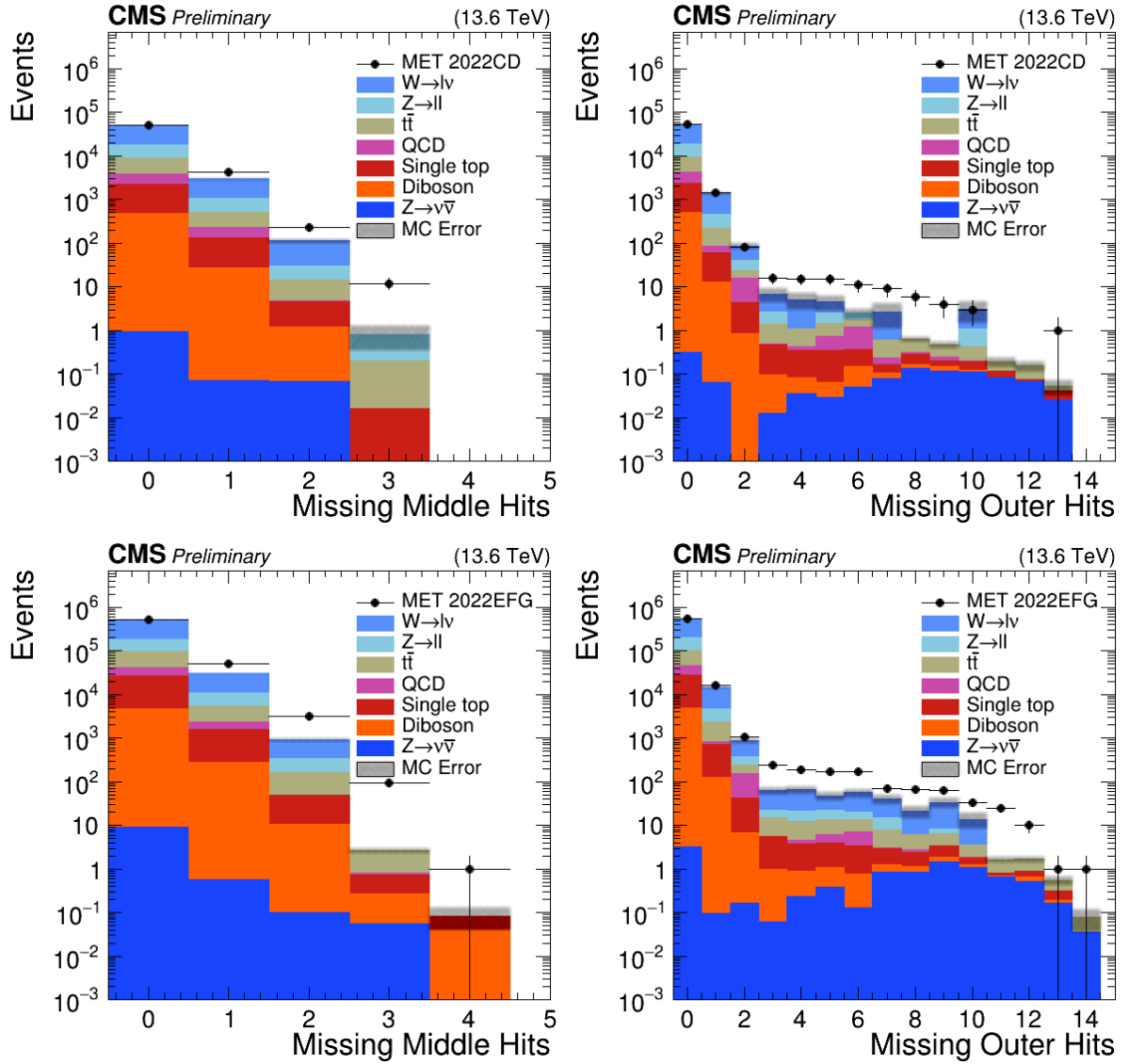
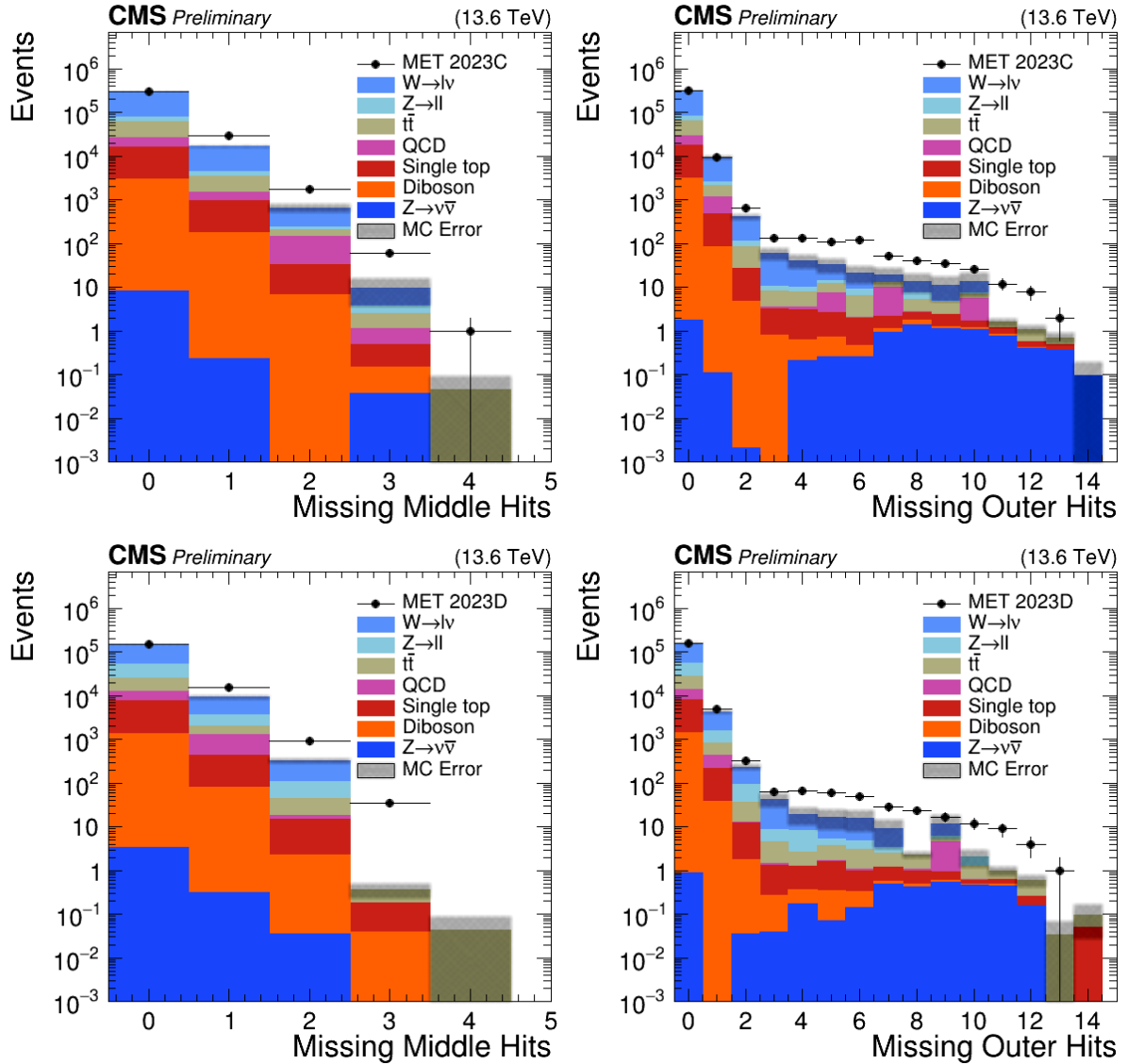


Figure 31: Uncorrected distributions of missing middle (left) and outer (right) hits in 2023 for tracks selected in the muon control region from in experimental data and simulated background processes. In the case of the missing middle hits distribution, the usual requirement of zero missing middle hits is not applied.



The first parameter is determined by fitting the missing middle hits distribution in simulated background events to that in experimental data, selecting events in the muon control region with the requirement on missing middle hits omitted. The fit is performed by creating a grid of potential values for P_0 , and applying each value to the distribution in simulated events. For each value the χ^2 difference between the corrected distribution and the distribution in experimental data is calculated, and the value of P_0 which minimizes this test statistic is chosen¹⁷.

The other three parameters are determined simultaneously by fitting the missing outer hits distribution in the muon control region where the criterion on missing middle hits has been applied. The fit is performed by correcting the missing outer hits distribution in simulated events for each point in a grid of values for $(P_{\text{pre-TOB}}, P_{\text{TOB}}, P_{\text{post-TOB}})$ and determining the values which minimize the χ^2 difference with the distribution in experimental data¹⁸.

The values for these four probabilities are listed in Table 36 for the run periods considered in the search.

Table 36: Values for the hit removal probabilities obtained by fitting the missing middle and outer hit distributions from simulated background events to those distributions from 2022-2023 experimental data. These values are used to randomly remove hits from the nominal hit patterns of tracks in simulated signal events.

Probability	2022 CD	2022 EFG	2023 C	2023 D	2024
P_0	3.203×10^{-3}	5.060×10^{-3}	5.095×10^{-3}	5.651×10^{-3}	7.619×10^{-3}
P_{TOB}	4.241×10^{-4}	6.651×10^{-4}	9.330×10^{-4}	7.755×10^{-4}	1.105×10^{-3}
$P_{\text{pre-TOB}}$	3.231×10^{-10}	3.368×10^{-3}	2.641×10^{-3}	4.298×10^{-3}	5.393×10^{-3}
$P_{\text{post-TOB}}$	0.786	0.779	0.593	0.706	0.674

After correcting the missing middle and outer hits distributions in this way, good agreement between the simulated and experimental data is achieved, as can be seen in Figures 32- 33. A systematic uncertainty on the signal selection efficiencies is calculated to account for any residual disagreement, as discussed in Section 7.2.

6.3 Trigger efficiency scale factors

Trigger efficiency scale factors are applied to simulated signal events to correct for differences in the trigger efficiency between simulated and experimental data.

In principle, since the triggers used in the search select events based on p_T^{miss} and isolated track p_T , it would be logical to bin the trigger efficiencies in p_T^{miss} and track p_T . But, as can be seen in Figure 34, signal and standard model events populate very different regions of this plane, and it is not possible to get trigger efficiencies in the region relevant to the signal.

However, we are really only interested in the trigger efficiency on the plateau of the track leg of `HLT_MET105_IsoTrk50_v*` since in the search region, we require track $p_T > 55 \text{ GeV}$, which is just on the plateau. And because we see that the difference between experimental and simulated data does not change dramatically as a function of track p_T on this plateau, we can marginalize out the track p_T from the trigger efficiency:

$$\varepsilon(p_T^{\text{miss, no } \mu}) = \int_{55 \text{ GeV}}^{\infty} \varepsilon(p_T^{\text{miss, no } \mu}, p_T) dp_T. \quad (16)$$

¹⁷See Appendix B.1 for details on how the corrected distribution is calculated.

¹⁸See Appendix B.2 for details on how the corrected missing outer hits distribution is calculated.

Figure 32: Corrected distributions of missing middle (left) and outer (right) hits in 2022 for tracks selected in the muon control region from in experimental data and simulated background processes. In the case of the missing middle hits distribution, the usual requirement of zero missing middle hits is not applied.

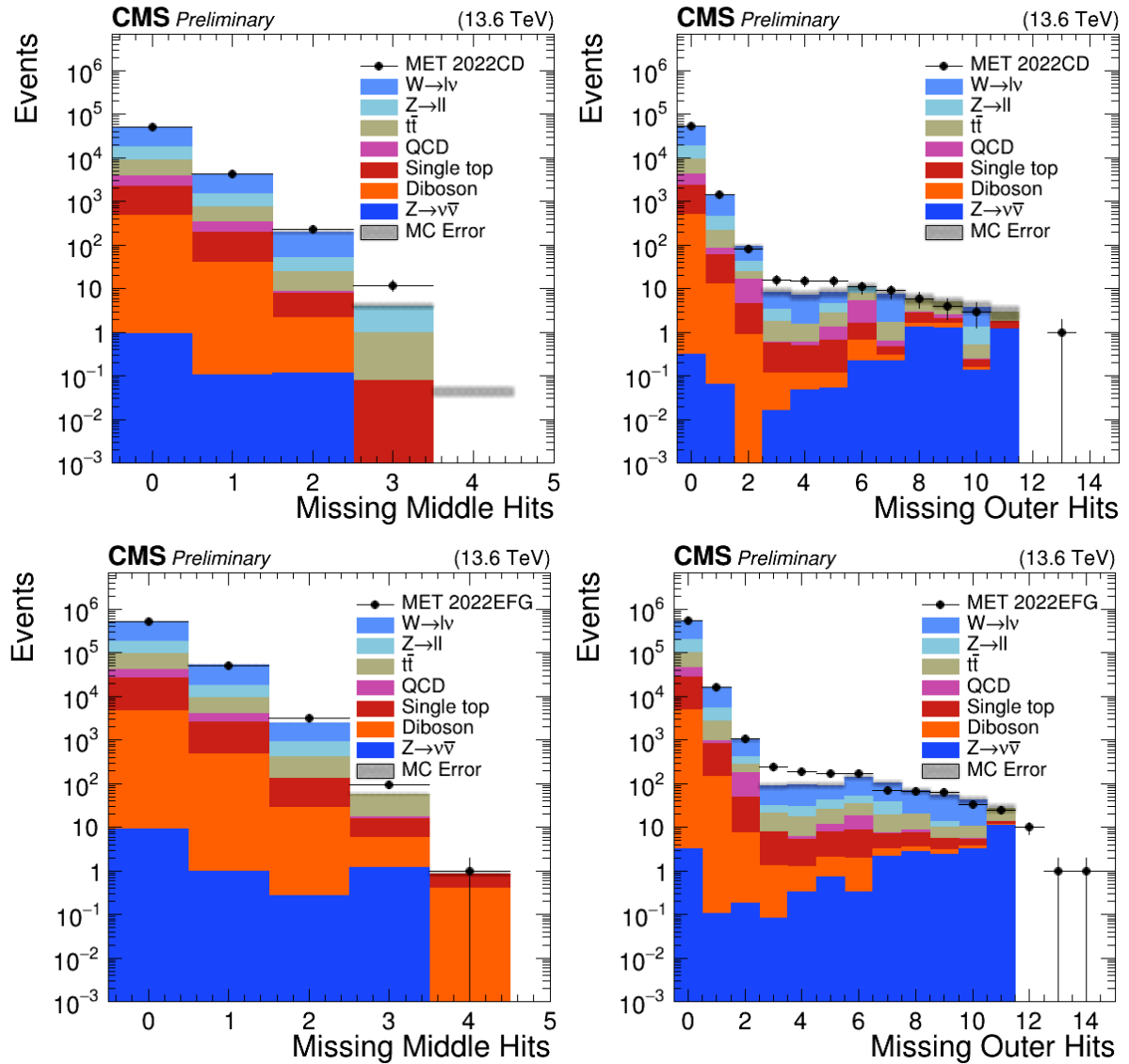


Figure 33: Corrected distributions of missing middle (left) and outer (right) hits in 2023 for tracks selected in the muon control region from in experimental data and simulated background processes. In the case of the missing middle hits distribution, the usual requirement of zero missing middle hits is not applied.

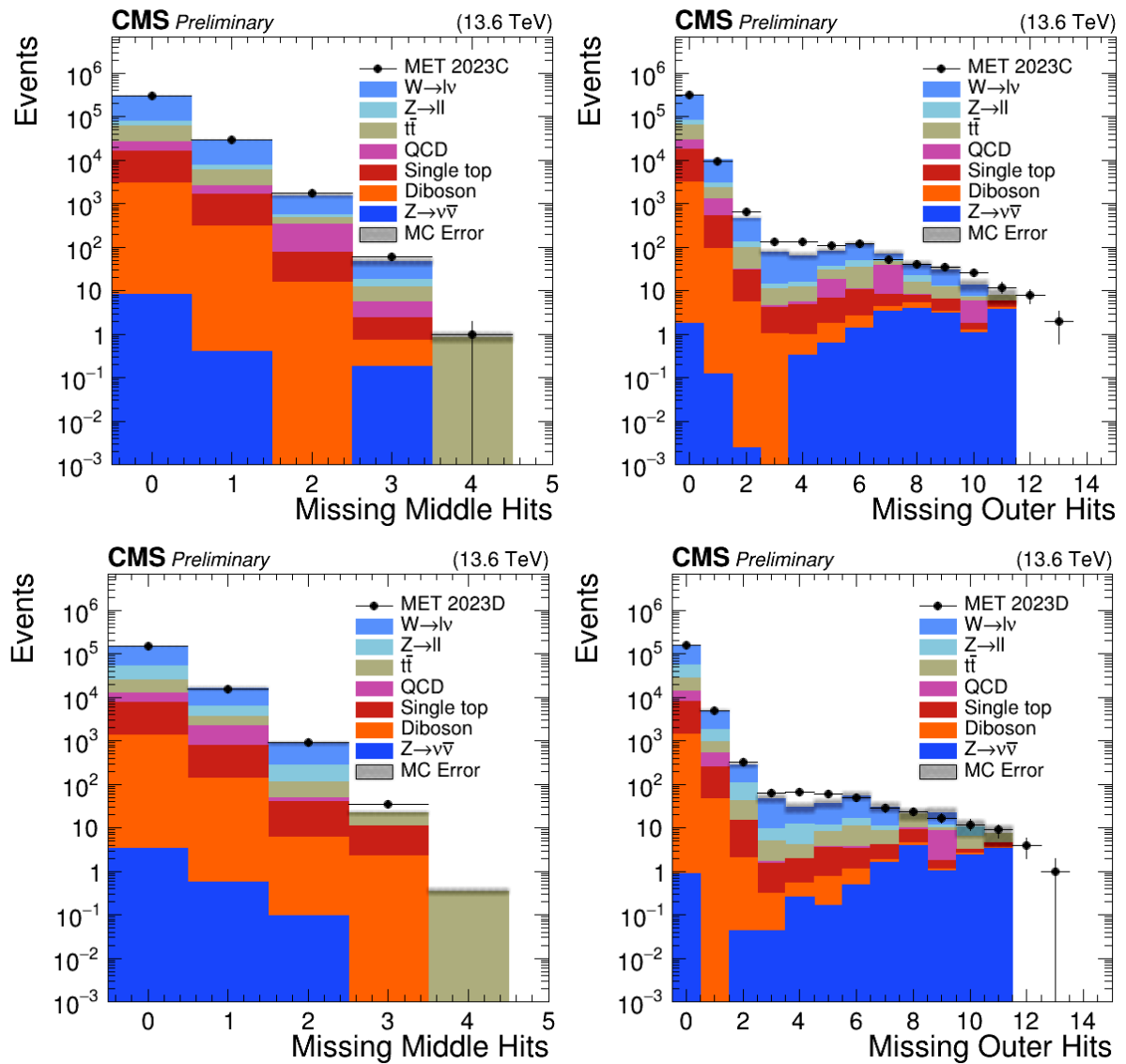
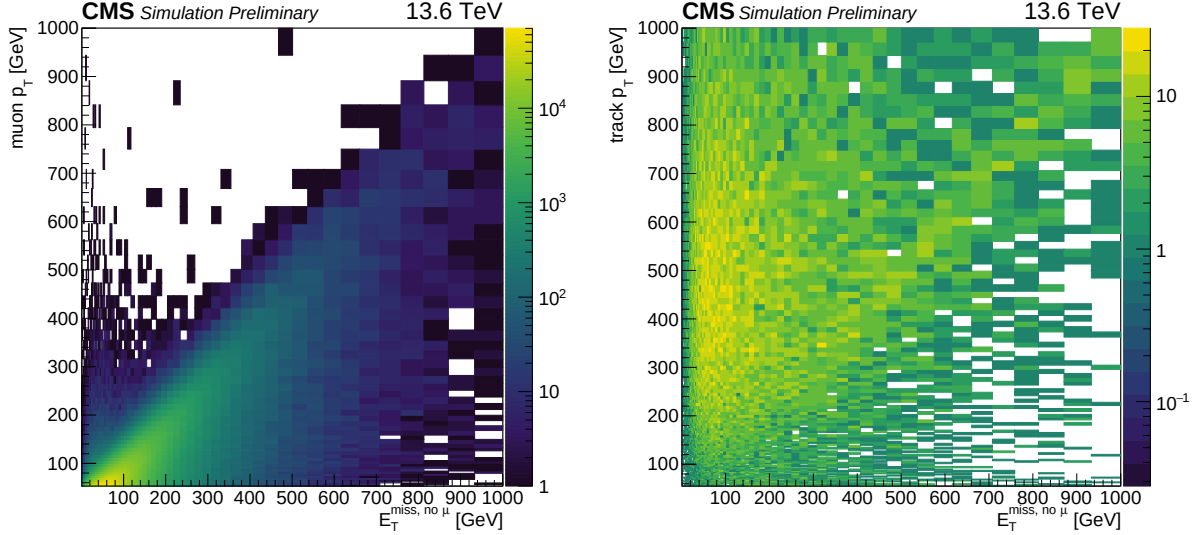


Figure 34: Distribution of events in the plane of muon (left) or track (right) p_T vs. $p_T^{\text{miss, no } \mu}$ for simulated $W \rightarrow l\nu$ (left) or simulated signal (right) events. For the signal events, the chargino has a mass of 700 GeV and a lifetime of 100 cm.



Now, the largest source of events from standard model processes with significant p_T^{miss} and a high- p_T , isolated track is $W \rightarrow l\nu$. For this reason, when evaluating the trigger efficiency, we target $W \rightarrow \mu\nu$ and use events collected with the `SingleMuon` triggers listed in Table 9. Offline, we require the events to have a well-reconstructed, isolated muon with no missing inner or middle hits in the silicon tracker. The marginalization of the track p_T discussed above is achieved by requiring this muon to have $p_T > 55$ GeV. This *base trigger selection* is detailed in Table 37.

Table 37: *Base trigger selection* used to select events used to measure the efficiency of the `HLT MET105_IsoTrk50_v*` trigger.

quantity	object	selection
	event passes <code>SingleMuon</code> triggers	
≥ 1	muons	$p_T > 55$ GeV
≥ 1	muons	$ \eta < 2.1$
≥ 1	muons	passing tight muon ID (see Table 13)
≥ 1	muons	missing inner hits = 0
≥ 1	muons	missing middle hits = 0

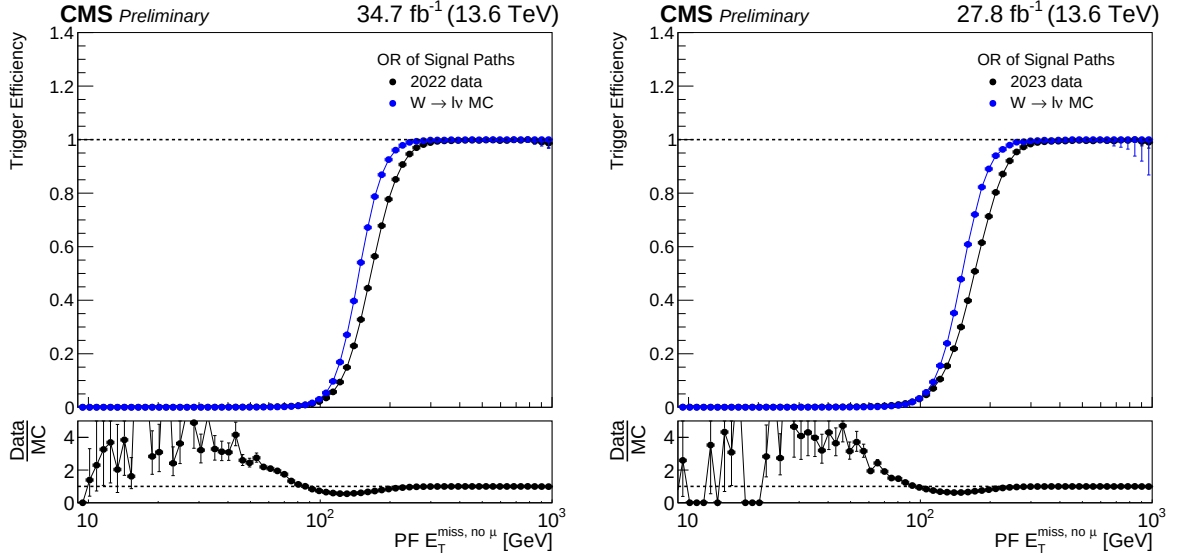
The trigger efficiency is then defined as

$$\varepsilon(p_T^{\text{miss, no } \mu}) = \frac{\text{events from denom. passing combination of } p_T^{\text{miss}} \text{ triggers}}{\text{events passing base trigger selection}}, \quad (17)$$

where the combination of p_T^{miss} triggers is that listed in Table 9. The resulting efficiency for experimental and simulated data is shown in Figure 35. Finally, the ratio of the trigger efficiency for experimental and simulated data defines the trigger efficiency scale factors that are used to correct the simulated signal events.

The effect of the trigger efficiency scale factors on the simulated signal is shown in Figure 36.

Figure 35: Efficiency of the combination of p_T^{miss} triggers used in the search for simulated $W \rightarrow l\nu$ and experimental data from 2022 (left) and 2023 (right). The efficiency is plotted only with respect to $p_T^{\text{miss, no } \mu}$, with the dependence on the track p_T integrated out, as described in the text. The ratio between data and MC is shown in the subplot.



Only the trigger decision is applied to the data in the plots, and the $p_T^{\text{miss, no } \mu}$ distributions are compared with and without the scale factors applied. The signal yield passing the triggers is reduced by $\sim 25\%$ in the case of the 100 GeV chargino and by $\sim 15\%$ for the 700 GeV chargino.

6.4 ISR scale factors

Because the search depends on jets from ISR for triggering and the initial basic selection, we correct for any mismodeling of this feature in the simulation. Because the production mechanism for AMSB charginos/neutralinos is not truly known, we make the assumption that the hadronic recoil of the electroweak-ino pair system is similar to that of a Z boson.

The correction is made by comparing the p_T distribution of $Z \rightarrow \mu\mu$ candidates in the $Z \rightarrow \mu\mu$ control region in simulated data to those in experimental data. The ratio of these distributions provides scale factors that are used to correct the simulated signal events. Since the p_T of $Z \rightarrow \mu\mu$ candidates is usually generated by the recoil of the Z off an ISR jet, this is a way of correcting the modeling of ISR jets that is independent of the uncertainties on jet energy scale and resolution. The p_T and $p_T^{\text{miss, no } \mu}$ distributions of $Z \rightarrow \mu\mu$ candidates is shown in Figure 37, with the resulting ISR scale factors taken as the ratio of the p_T distribution in data to simulation.

Figure 36: Distributions of $p_T^{\text{miss, no } \mu}$ with and without the trigger efficiency scale factors for the 2022 period applied with no cuts required. Simulated signal events passing the triggers are plotted, with a 100 GeV chargino in the left plot and a 700 GeV chargino in the right plot.

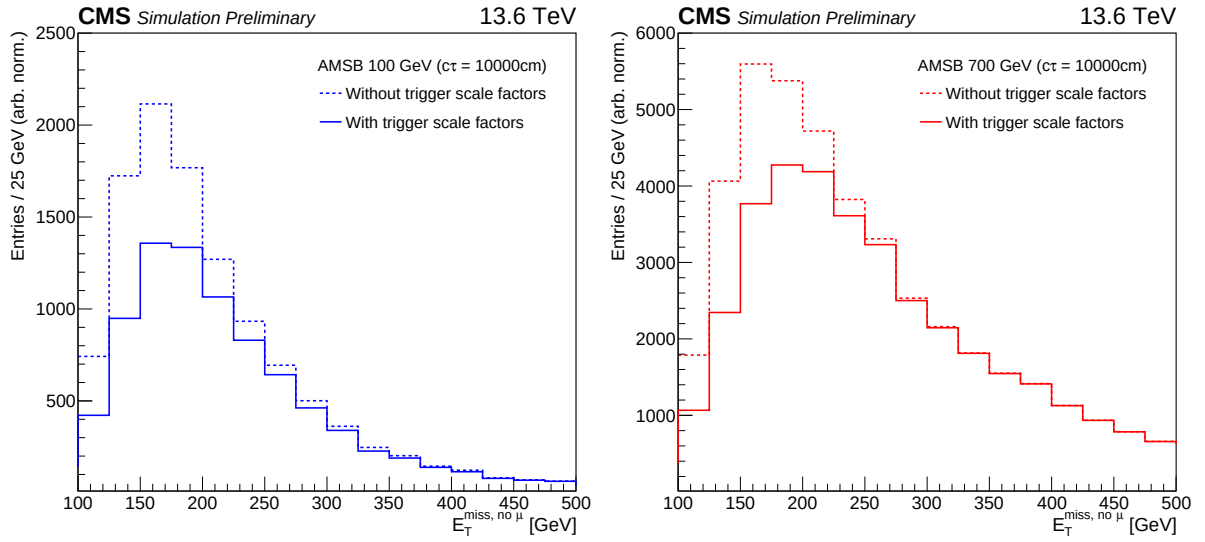
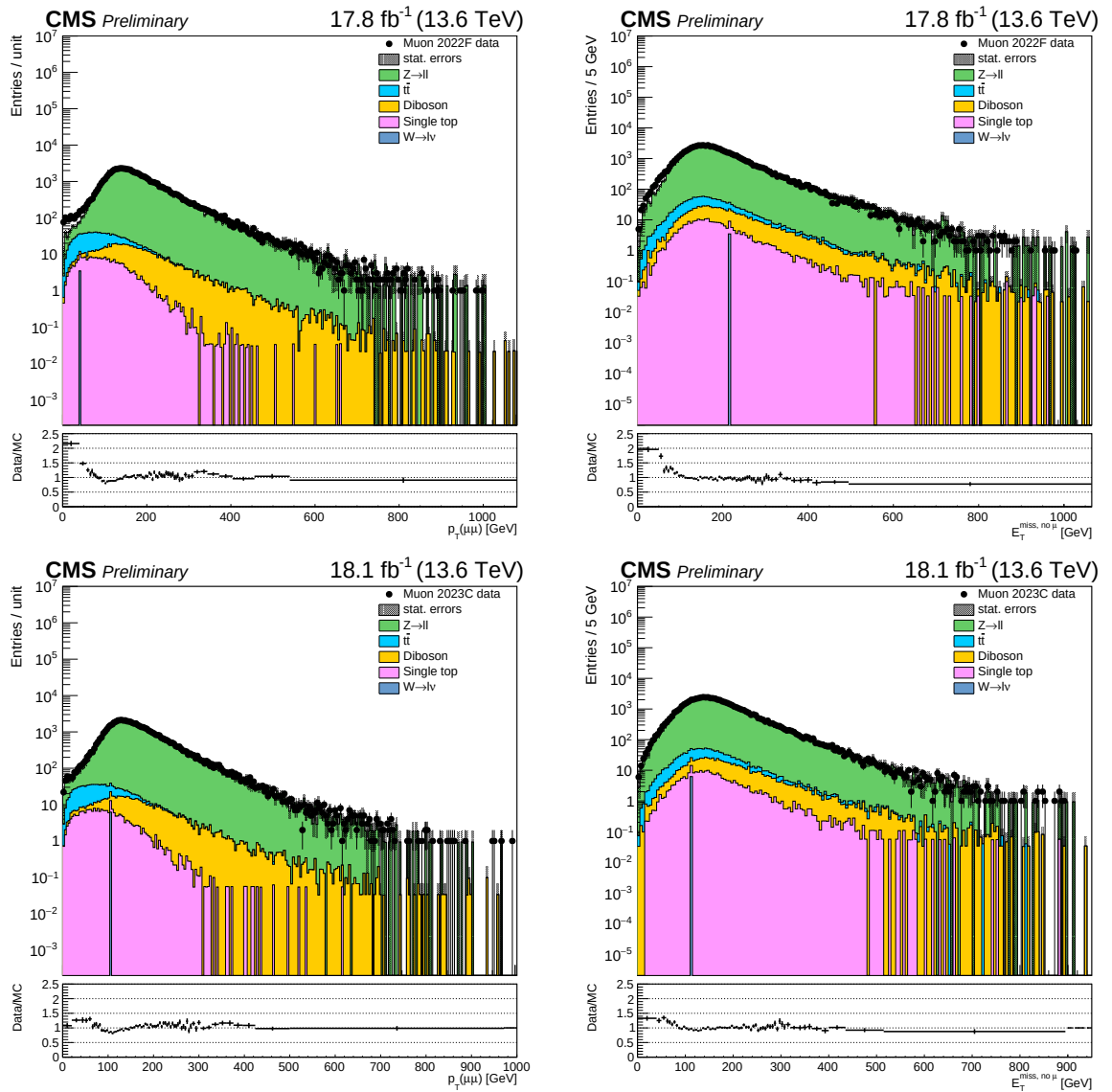


Figure 37: Distributions of the p_T of $Z \rightarrow \mu\mu$ candidates (left) and $p_T^{\text{miss, no } \mu}$ (right) in the $Z \rightarrow \mu\mu$ control region for experimental data and simulated data before the ISR weighting has been applied. The ratio plot shown below the left plots are taken as the ISR scale factors to be applied to simulated signal events. The 2022 data is shown on the top and the 2023 data is shown on the bottom.



7 Systematic uncertainties

The results of this analysis are subject to systematic uncertainties in the integrated luminosity, the estimated background contributions, and in the efficiency of the signal. The uncertainty on the integrated luminosity is 1.4% for the 2022 data set [29] and 1.3% for the 2023 data set [30]. The uncertainties in the background estimates and in the signal selection efficiency are described in the sections below.

7.1 Uncertainties on background estimates

7.1.1 Systematic uncertainty in fake track background estimate

We assign a systematic uncertainty corresponding to the first assumption of the fake track background estimation method discussed in Section 5.2.1. We assign the difference between the estimates obtained from the $Z \rightarrow \mu\mu$ and $Z \rightarrow ee$ control regions (see Table 34), as an uncertainty resulting from potential dependence of P_{fake} on the control region in which it is calculated. While in some cases this appears to be a large systematic, in the same cases the statistical uncertainties are even larger, and the fake track estimate is still dominated by statistical uncertainties.

The uncertainty in the transfer factor ζ is evaluated by fluctuating the resulting parameters of the gaussian function fit up and down by their estimated uncertainties, and ζ is re-estimated.

We do not assign systematic uncertainties for the second or third assumptions, respectively that the transfer factor fit procedure correctly extrapolates from the d_{xy} sidebands to the signal region and that our choice of sideband definition is statistically consistent with other choices. In both cases we find any deviations to be well within purely statistical fluctuations of the data, and any additional systematic uncertainty would be negligible and redundant with the statistical uncertainties.

The systematic uncertainties on the fake track background estimate are summarized in Table 38.

Table 38: Summary of systematic uncertainties on the fake track background estimate.

run period	n_{layers}	uncertainty source		total
		control region	transfer factor	
2022 CD	4	6.6%	76.0%	76.3%
	5	33.9%	76.0%	83.2%
	≥ 6	100%	76.0%	125.6%
2022 EFG	4	63.8%	46.0%	78.7%
	5	52.1%	46.0%	69.5%
	≥ 6	12.5%	46.0%	47.7%
2023 C	4	1.6%	182.2%	182.2%
	5	12.4%	182.2%	182.6%
	≥ 6	166.7%	182.2%	247.0%
2023 D	4	6.6%	113.8%	114.0%
	5	34.7%	113.8%	119.0%
	≥ 6	200.0%	113.8%	230.1%

7.1.2 Systematic uncertainty in lepton background estimates

The lepton background estimates rely on the assumption that when a lepton is not reconstructed as such, it contributes exactly zero visible momentum to the event. We test the impact

of this assumption by replacing the nominal $|\vec{p}_T^{\text{miss, no } \mu} + \vec{p}_T^{\text{lepton}}|$ distribution used to calculate P_{offline} and P_{trigger} with a “scaled down” version,

$$\left| \vec{p}_T^{\text{miss, no } \mu} + \frac{p_T^{\text{lepton}} - 10 \text{ GeV}}{p_T^{\text{lepton}}} \vec{p}_T^{\text{lepton}} \right|, \quad (18)$$

and recalculating P_{offline} and P_{trigger} . In other words, we assume that unreconstructed leptons contribute 10 GeV of visible momentum to the event. The value of 10 GeV is chosen because candidate tracks are required to have $E_{\text{calo}} < 10 \text{ GeV}$ in the disappearing tracks search region. A comparison of the nominal and scaled down distributions is shown in Figures 38 and Figures 39.¹⁹

Finally, the difference from unity of the ratio

$$\frac{(P_{\text{offline}} P_{\text{trigger}})_{\text{scaled down}}}{(P_{\text{offline}} P_{\text{trigger}})_{\text{nominal}}} \quad (19)$$

is taken as the systematic uncertainty. In the case of taus for $n_{\text{layers}} = 4$ and 5, sufficient statistics are not available to calculate P_{offline} or P_{trigger} . As described above in Section 5.1.4, the values derived from the combined category ($n_{\text{layers}} \geq 4$) are used instead.

The nominal and scaled down values of the constituent probabilities, as well as the resulting systematic uncertainties, are listed in Table 39 for electrons and in Table 40 for taus.

Table 39: P_{offline} and P_{trigger} calculated with the nominal and scaled down versions of the $|\vec{p}_T^{\text{miss, no } \mu} + \vec{p}_T^{\text{lepton}}|$ distribution described in the text for events in the single electron control sample, as well as the resulting systematic uncertainty.

run period	n_{layers}	nominal		scaled down		systematic uncertainty
		P_{offline}	P_{trigger}	P_{offline}	P_{trigger}	
2022 CD	4	0.644 ± 0.019	0.370 ± 0.048	0.566 ± 0.018	0.371 ± 0.048	11.8%
	5	0.655 ± 0.012	0.425 ± 0.033	0.583 ± 0.011	0.409 ± 0.031	14.5%
	≥ 6	0.724 ± 0.003	0.402 ± 0.006	0.654 ± 0.002	0.388 ± 0.006	12.8%
2022 EFG	4	0.635 ± 0.012	0.476 ± 0.034	0.563 ± 0.011	0.465 ± 0.034	13.5%
	5	0.645 ± 0.007	0.500 ± 0.021	0.574 ± 0.006	0.485 ± 0.020	13.8%
	≥ 6	0.712 ± 0.001	0.509 ± 0.004	0.642 ± 0.001	0.493 ± 0.004	12.6%
2023 C	4	0.634 ± 0.015	0.501 ± 0.045	0.572 ± 0.014	0.482 ± 0.043	13.2%
	5	0.657 ± 0.009	0.491 ± 0.024	0.589 ± 0.008	0.474 ± 0.024	13.6%
	≥ 6	0.710 ± 0.002	0.531 ± 0.005	0.645 ± 0.002	0.514 ± 0.005	12.1%
2023 D	4	0.637 ± 0.022	0.428 ± 0.062	0.580 ± 0.021	0.412 ± 0.059	12.3%
	5	0.651 ± 0.013	0.504 ± 0.041	0.587 ± 0.012	0.488 ± 0.039	12.8%
	≥ 6	0.698 ± 0.003	0.532 ± 0.008	0.635 ± 0.002	0.515 ± 0.008	12.1%

This uncertainty is not evaluated for muons, since even successfully reconstructed muons are not expected to contribute substantial visible calorimeter energy to an event.

As mentioned above and in Section 5.1.4, the lack of statistics in the single-muon and single-tau control regions with $n_{\text{layers}} = 4$ and 5 make calculating P_{offline} and P_{trigger} difficult (if not impossible in the case of zero selected events). To study the effect of averaging these probabilities over all three categories of n_{layers} , P_{offline} and P_{trigger} are calculated for electrons for the combined category ($n_{\text{layers}} \geq 4$). Plentiful statistics are available in the single-electron control region in all n_{layers} categories, so the $n_{\text{layers}} = 4$ and 5 values for P_{offline} and P_{trigger} are compared to the combined values. The opposite of the differences between these values are taken as one-sided systematic uncertainties in the muon and tau background estimates in the respective n_{layers} categories, and are listed in Table 41.

¹⁹The data in the single-tau control sample were collected with a prescaled trigger as mentioned in Section 2.1, and the effective integrated luminosities reported on the plots include this prescale.

Figure 38: Comparison of the nominal and scaled down versions of the $|\vec{p}_T^{\text{miss, no } \mu} + \vec{p}_T^{\text{lepton}}|$ distribution described in the text for events in the single electron control sample (left) and tau control sample (right) with $n_{\text{layers}} = 6$ for the 2022 data-taking periods. The mean (μ) of each distribution is given in the legends.

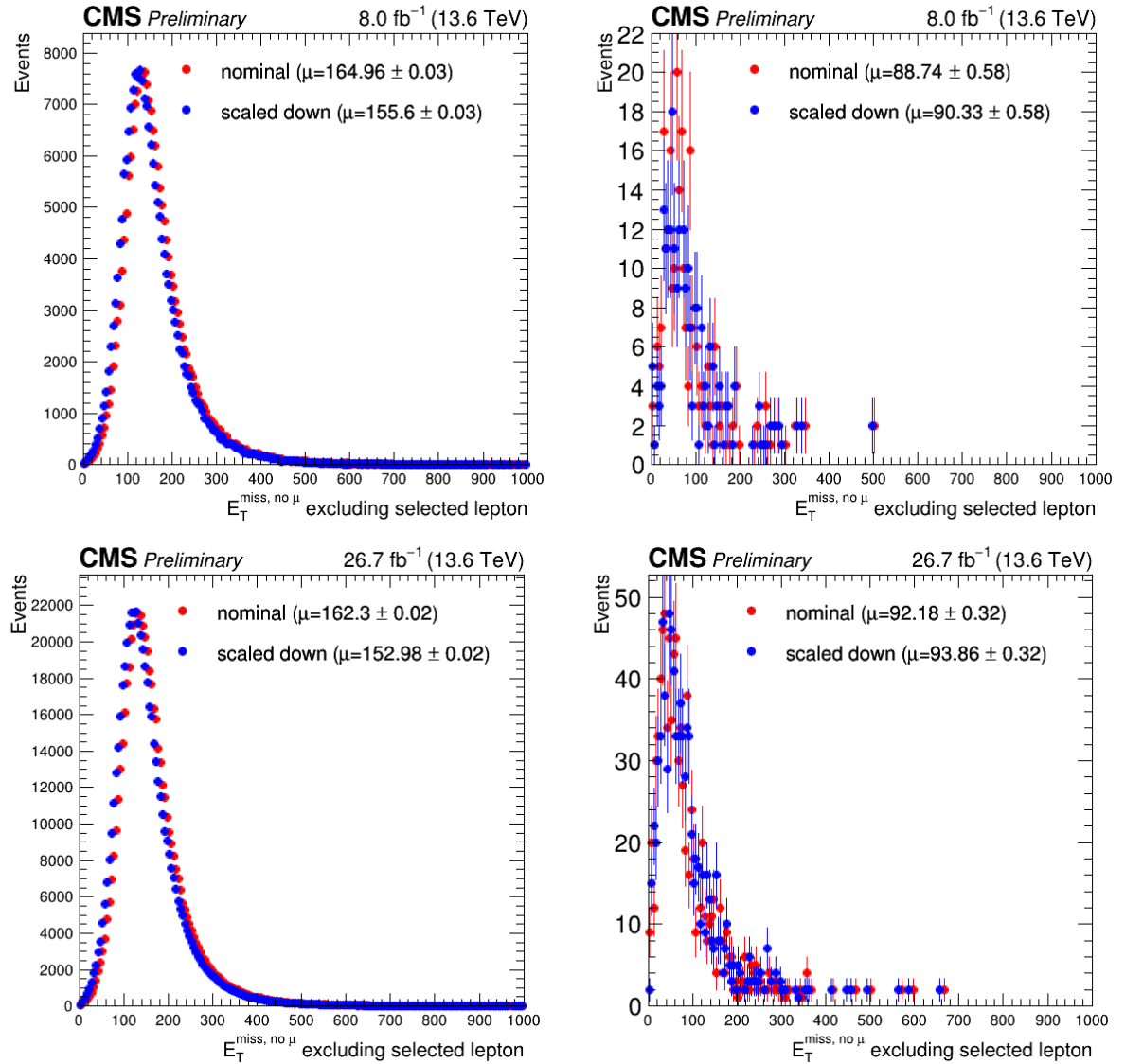


Figure 39: Comparison of the nominal and scaled down versions of the $|\vec{p}_T^{\text{miss, no } \mu} + \vec{p}_T^{\text{lepton}}|$ distribution described in the text for events in the single electron control sample (left) and tau control sample (right) with $n_{\text{layers}} = 6$ for the 2023 data-taking periods. The mean (μ) of each distribution is given in the legends.

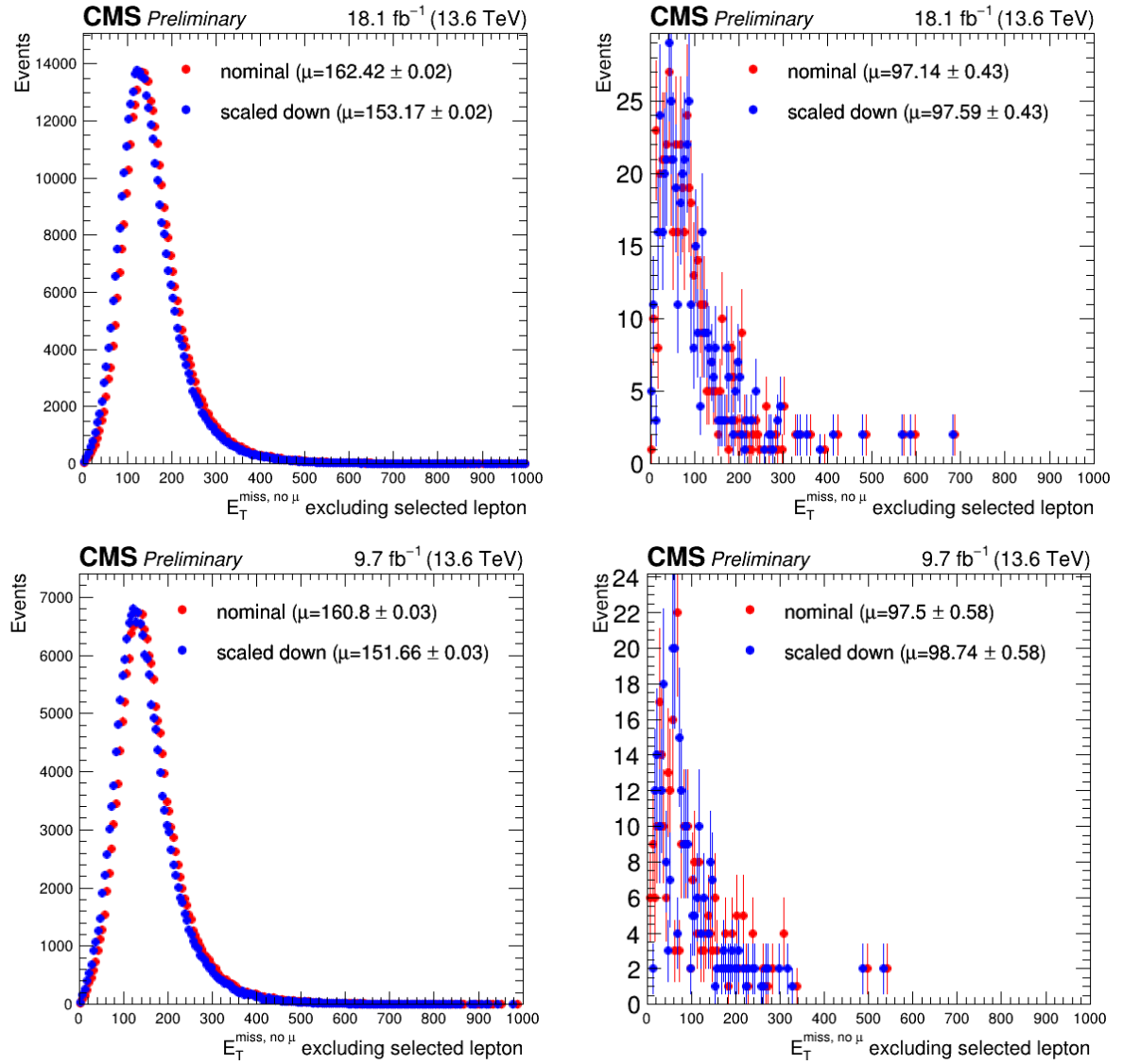


Table 40: P_{offline} and P_{trigger} calculated with the nominal and scaled down versions of the $|\vec{p}_T^{\text{miss, no } \mu} + \vec{p}_T^{\text{lepton}}|$ distribution described in the text for events in the single tau control sample, as well as the resulting systematic uncertainty. In the $n_{\text{layers}} = 4$ and 5 categories, sufficient statistics were not available to estimate P_{offline} and P_{trigger} , so the values measured in the combined-category sample are used instead.

run period	n_{layers}	nominal		scaled down		systematic uncertainty
		P_{offline}	P_{trigger}	P_{offline}	P_{trigger}	
2022 CD	4	-	-	-	-	3.1%
	5	-	-	-	-	3.1%
	≥ 6	0.208 ± 0.031	0.449 ± 0.101	0.208 ± 0.031	0.435 ± 0.099	3.1%
2022 EFG	4	-	-	-	-	0.3%
	5	-	-	-	-	0.3%
	≥ 6	0.235 ± 0.018	0.543 ± 0.059	0.238 ± 0.018	0.534 ± 0.058	0.3%
2023 C	4	-	-	-	-	2.1%
	5	-	-	-	-	2.1%
	≥ 6	0.257 ± 0.025	0.580 ± 0.080	0.262 ± 0.025	0.556 ± 0.076	2.1%
2023 D	4	-	-	-	-	3.1%
	5	-	-	-	-	3.1%
	≥ 6	0.262 ± 0.034	0.580 ± 0.106	0.266 ± 0.034	0.555 ± 0.101	3.1%

Table 41: Comparison of P_{offline} and P_{trigger} in the single-electron control region between $n_{\text{layers}} = 4$, $n_{\text{layers}} = 5$, and a combined sample of $n_{\text{layers}} \geq 4$. The opposite of the differences between each category and the combined result, shown in parentheses, are applied as one-sided systematic uncertainties in the muon and tau background estimates, where insufficient statistics exist to properly measure these values.

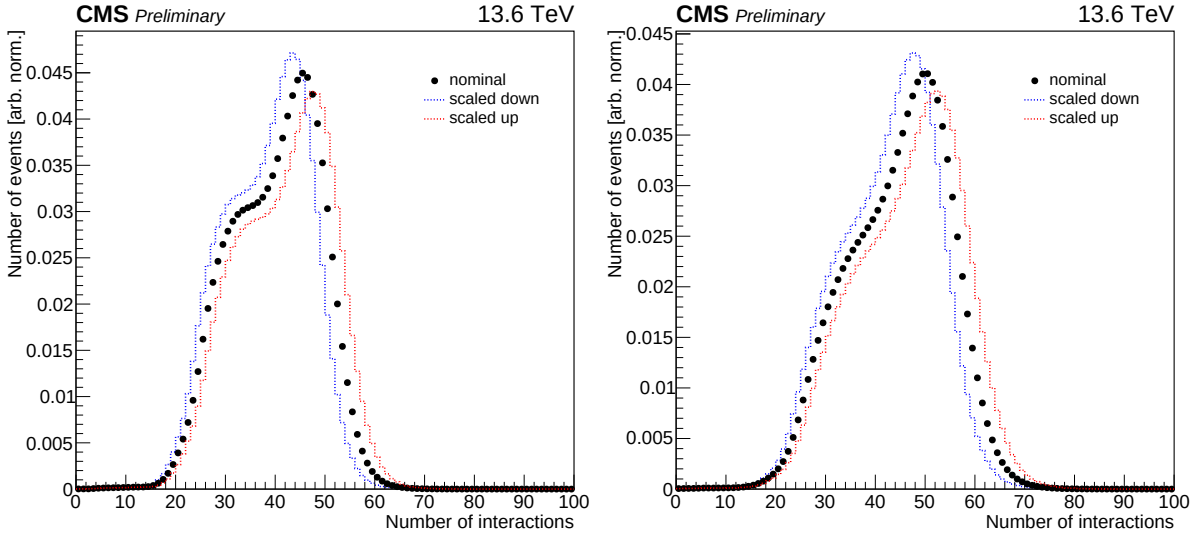
run period	n_{layers}	electron background	
		P_{offline}	P_{trigger}
2022 CD	4	0.644 ± 0.019 (−10.68%)	0.370 ± 0.048 (−8.64%)
	5	0.655 ± 0.012 (−9.15%)	0.425 ± 0.033 (+4.94%)
	combined	0.721 ± 0.002	0.405 ± 0.006
2022 EFG	4	0.635 ± 0.012 (−10.45%)	0.476 ± 0.034 (−6.48%)
	5	0.645 ± 0.007 (−8.90%)	0.500 ± 0.021 (−1.77%)
	combined	0.708 ± 0.001	0.509 ± 0.004
2023 C	4	0.634 ± 0.015 (−10.45%)	0.501 ± 0.045 (−5.47%)
	5	0.651 ± 0.013 (−8.05%)	0.491 ± 0.024 (−7.36%)
	combined	0.708 ± 0.002	0.530 ± 0.005
2023 D	4	0.637 ± 0.022 (−8.48%)	0.428 ± 0.062 (−19.55%)
	5	0.651 ± 0.013 (−6.47%)	0.504 ± 0.041 (−5.26%)
	combined	0.696 ± 0.002	0.532 ± 0.008

7.2 Uncertainties on signal selection efficiencies

7.2.1 Pileup

There is an uncertainty associated with the modeling of pileup interactions in the samples of simulated signal events, the distribution of which may differ from that present in the experimental data. We evaluate this uncertainty by using different target distributions for pileup reweighting, produced using different assumptions for the minimum bias cross section. The best-fit value for this cross section was determined to be 69 200 mb with an uncertainty of 4.6% in 2022. The difference in signal efficiency when using the distributions produced with the best-fit value plus or minus its uncertainty is taken as the systematic uncertainty from the modeling of pileup. These distributions are compared to the nominal target distributions in Figure 40 for the 2022 and 2023 data, and the resulting uncertainty is 2.9 – 6.4% on average.

Figure 40: Target pileup distributions used for evaluating the systematic uncertainty from the modeling of pileup in the simulated signal samples compared to the nominal distributions for the 2022 (left) and 2023 (right) data.



7.2.2 Missing inner and middle hits

The uncertainty associated with the modeling of missing inner and middle hits in simulation is evaluated using three control regions. The first is the muon control region, defined in Section 6.2 as the candidate track selection with the muon veto (defined in Table 21) removed. The second and third are the $Z \rightarrow \mu\mu + \tau_h$ and $Z \rightarrow ee + \tau_h$ control regions, combining both into a single sample of events. These are respectively defined as the $Z \rightarrow \mu\mu$ and $Z \rightarrow ee$ selections, detailed in Tables 32 and 33, with the additional requirement of a track passing the candidate track selection with the tau veto (defined in Table 21) removed.

The $Z + \tau_h$ control regions are introduced because when a reconstructed muon is present near to a track, additional out-in iterations of the track fitting algorithm are performed. These iterations widen the hit-finding parameters, and can associate additional hits to the track that would not have been without them. This extra hit-finding efficiency may perform differently between data and simulation, given that candidate disappearing tracks in data are expressly required to be well-separated from reconstructed muons. Therefore the $Z + \tau_h$ control regions

are introduced to examine this difference in hit efficiencies, using samples of events dominated by hadronically decaying tau leptons and fake tracks, yet still similar to the search region.

These selections are run on samples of data and simulated standard model processes without including the track criteria requiring zero missing inner and middle hits. The efficiencies for these criteria are evaluated, and a systematic uncertainty is calculated as the relative difference between the efficiency in experimental data and the efficiency in simulated data.

Combining the $Z \rightarrow \mu\mu + \tau_h$ and $Z \rightarrow ee + \tau_h$ control regions into one control region, the larger of the two systematics between that and the muon control region is applied as the signal systematic. In the muon control region, the $n_{\text{layers}} \geq 6$ systematic is used for all categories, where the $Z + \tau_h$ control region evaluates these separately for each category.

Table 42 below lists the systematic uncertainties obtained in this way, and Figure 41 compares data to $Z \rightarrow ll$ simulation for the $Z + \tau_h$ control regions.

Table 42: Summary of signal systematic uncertainties on the missing inner and middle hits requirements. For each the larger of the muon (μ) and $Z + \tau_h$ control regions are applied. In the muon control region, the $n_{\text{layers}} \geq 6$ category is used for all three signal categories.

run period	n_{layers}	systematic uncertainty			
		inner		middle	
		μ	$Z + \tau_h$	μ	$Z + \tau_h$
2022	4	0.28%	$\sim 0.00\%$	2.81%	2.86%
	5	0.28%	$\sim 0.00\%$	2.81%	1.13%
	≥ 6	0.28%	0.89%	2.81%	2.05%
2022EE	4	$\sim 0.00\%$	0.03%	3.83%	1.59%
	5	$\sim 0.00\%$	1.48%	3.83%	1.17%
	≥ 6	$\sim 0.00\%$	0.65%	3.83%	3.35%
2023	4	0.15%	1.78%	4.20%	5.88%
	5	0.15%	$\sim 0.00\%$	4.20%	2.08%
	≥ 6	0.15%	0.57%	4.20%	0.12%
2023BPix	4	0.30%	2.02%	4.21%	$\sim 0.00\%$
	5	0.30%	4.17%	4.21%	4.17%
	≥ 6	0.30%	0.5%	4.21%	0.34%

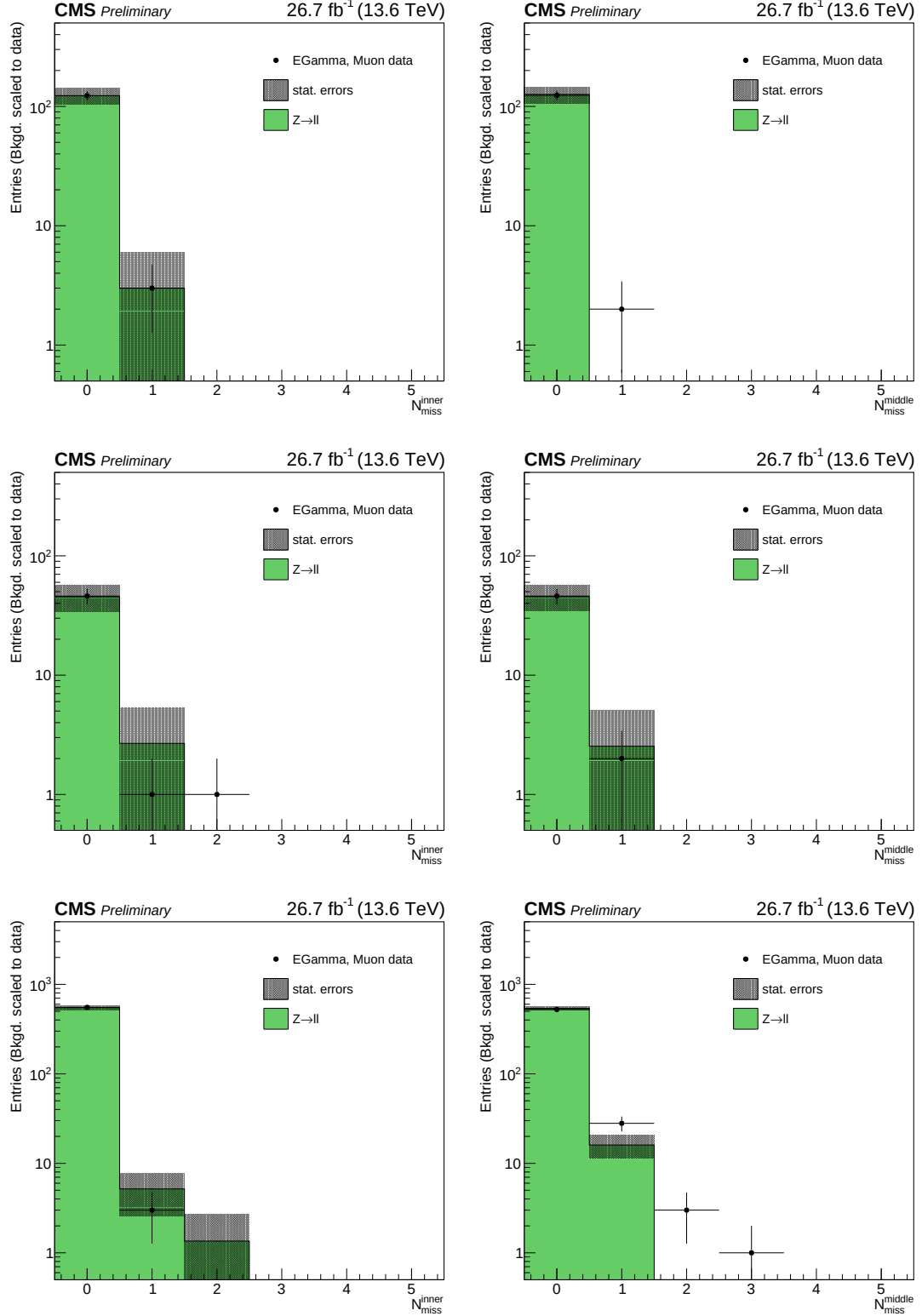
7.2.3 Missing outer hits

The uncertainty associated with the modeling of missing outer hits in simulation is also evaluated using the muon control region. First, we must distinguish between *true* and *accidental* missing outer hits:

- *True* missing outer hits result from a charged particle not traversing the outermost tracker layers. This is the type of missing outer hit expected to be present on tracks in signal events.
- *Accidental* missing outer hits result from detector inefficiencies or mistakes of the tracking software. For muons especially, this type is expected to account for most missing outer hits.

The uncertainties due to the modeling of true missing outer hits is negligible, so only a systematic uncertainty on the modeling of accidental missing outer hits is assigned. Since the vast majority of the tracks in the muon control region come from muons, we assume none of these

Figure 41: Number of missing inner (left) and middle (right) hits for tracks selected in the $Z + \tau_h$ control regions, but without the usual criteria requiring zero missing inner and middle hits. Shown are the $n_{\text{layers}} = 4$ (top), $n_{\text{layers}} = 5$ (middle), and $n_{\text{layers}} \geq 6$ (bottom) categories for the 2022 data set.



tracks have any true missing outer hits in the inner tracker, and indeed this is true for over 97% of these tracks, as can be seen in Figures 30 and 31. Thus, these distributions represent the number of accidental missing outer hits we can expect for tracks in experimental and simulated data.

Now, the signal yield is proportional to the efficiency for a track to have at least three missing outer hits, which can be written as

$$\varepsilon^{\geq 3} = \varepsilon_{\text{true}}^{\geq 3} + \varepsilon_{\text{true}}^{\geq 2} \cdot \varepsilon_{\text{acc.}}^{\geq 1} + \varepsilon_{\text{true}}^{\geq 1} \cdot \varepsilon_{\text{acc.}}^{\geq 2} + \varepsilon_{\text{true}}^{\geq 0} \cdot \varepsilon_{\text{acc.}}^{\geq 3}, \quad (20)$$

where $\varepsilon_{\text{true}}^p$ ($\varepsilon_{\text{acc.}}^p$) are the efficiencies for selected tracks to have p true (accidental) missing outer hits. The efficiencies $\varepsilon_{\text{true}}^p$ depend on the signal point in question and can be estimated directly from the missing outer hits distribution for selected tracks in simulated signal events passing the disappearing track selection except the criterion on number of missing outer hits.²⁰ Per the discussion above, the efficiencies $\varepsilon_{\text{acc.}}^p$ can be estimated from the missing outer hits distributions from the muon control region, and can apply either to the simulation or to the experimental data, depending on the source of the distribution.

Thus, we assign the relative difference between $\varepsilon^{\geq 3}$ for experimental and simulated data as a systematic on the signal acceptance. The values of this uncertainty are very small, being zero for many chargino masses and lifetimes in the $n_{\text{layers}} = 4$ and 5 categories, and at a maximum of 0.2% for $n_{\text{layers}} \geq 6$.

7.2.4 Trigger efficiency

There is an uncertainty associated with the trigger efficiency scale factors due to the finite sample sizes used to calculate them, discussed in Section 6.3. We estimate this by shifting the measured trigger efficiencies in data and simulation up and down by their statistical uncertainties, treating the data and simulation fluctuations as independent uncertainties. The scale factors are recalculated from these shifted efficiencies and applied to the signal events, and the resulting changes to the signal yields relative to the nominal values contribute to the systematic uncertainty.

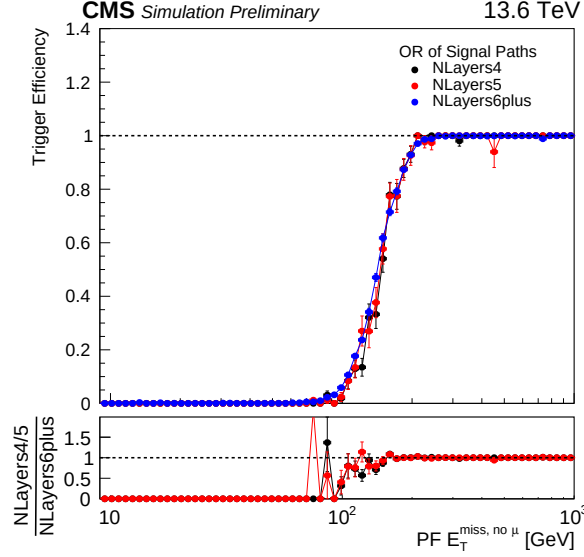
Additionally there is an uncertainty in the trigger efficiency for the $n_{\text{layers}} = 4, 5$ search categories due to the requirement of the `ISOTrk50` leg of the trigger. The online requirement of this leg is for tracks to have at least five valid hits, which reduces the trigger efficiency at low $p_{\text{T}}^{\text{miss, no } \mu}$ in these categories.

To study the effect of these considerations, the trigger efficiency is studied in signal simulation for all deployed combinations of paths used in data-taking, and for each n_{layers} category separately. The available triggers in each period are summarized in Table 9, and the measured efficiencies for a chargino mass of 700 GeV shown in Figure 42.

To quantify the systematic uncertainty based on these varying efficiencies, the ratios of efficiencies between the $n_{\text{layers}} = 4, 5$ categories to the $n_{\text{layers}} \geq 6$ category are calculated. These ratios are then convoluted with the $p_{\text{T}}^{\text{miss, no } \mu}$ distribution of each signal sample to study the effect of this loss in efficiency on the signal yields. This is done for each chargino mass, averaging over chargino lifetimes as such short tracks will not contribute to the calorimeter energy (and thus $p_{\text{T}}^{\text{miss, no } \mu}$) of the events or the trigger efficiency.

²⁰This distribution includes accidental missing outer hits, but should be dominated by true missing outer hits since the former only occur at a rate of $\sim 3\%$. Thus, this distribution is a good approximation of the distribution of true missing outer hits.

Figure 42: Efficiency of all deployed combinations of p_T^{miss} triggers used in the search for simulated signal events with a chargino mass of 700 GeV, measured in each category of n_{layers} . In the subplot, the ratio between the $n_{\text{layers}} = 4$ and $n_{\text{layers}} = 5$ with $n_{\text{layers}} = 6$ or more is being plotted. All chargino lifetimes are considered, as the chargino decay length does not affect the $p_T^{\text{miss, no } \mu}$ of each event. Period 2022EE is shown.



The combined systematic uncertainty on the trigger efficiency, averaged over all chargino masses and lifetimes, is 0.8% for $n_{\text{layers}} = 4$, 0.4% for $n_{\text{layers}} = 5$, and 0.2% for $n_{\text{layers}} \geq 6$ in 2022 conditions, 0.5% for $n_{\text{layers}} = 4$, 0.5% for $n_{\text{layers}} = 5$, and 0.2% for $n_{\text{layers}} \geq 6$ in 2022EE conditions, 0.9% for $n_{\text{layers}} = 4$, 0.8% for $n_{\text{layers}} = 5$, and 0.5% for $n_{\text{layers}} \geq 6$ in 2023 conditions, 0.9% for $n_{\text{layers}} = 4$, 0.8% for $n_{\text{layers}} = 5$, and 0.5% for $n_{\text{layers}} \geq 6$ in 2023BPix conditions²¹. While the trigger efficiencies themselves may vary considerably between the shorter and longer n_{layers} categories at lower $p_T^{\text{miss, no } \mu}$, the overall uncertainty remains fairly small because only $\sim 10\%$ of the selected signal events fall on the turn-on region where these larger differences are relevant.

7.2.5 ISR

As discussed in Section 6.4, the production mechanism for AMSB charginos/neutralinos is not truly known, and it is assumed that the hadronic recoil of the electroweak-ino pair system is similar to that of a Z boson. Therefore the only systematic uncertainty evaluated for the ISR weights applied to signal simulation is due to the limited statistics in the samples used to derive the weights. This is evaluated by fluctuating the weights up and down 1σ according to their statistical uncertainties, and recalculating the relative changes in signal yield in each case. The average value of this uncertainty is 6.6–7.1%.

7.2.6 Jet energy scale and resolution

The jets used in the search have the standard set of jet energy corrections (JEC) applied. These corrections are designed to remove the energy from pileup events (L1), make the jet response uniform over p_T and η (L2L3), and to correct for any remaining small differences within the jet

²¹The 2023 and 2023BPix signal trigger systematic values are only the same after rounding to one decimal place.

response in data and simulation (L2L3Residuals) [31].²² The JEC payload versions used are:

- Data:

- 2022: Run2_Combined_UL_2017_2018_DATA_AK4PFchs
- 2022EE: Run2_Run3_DATA_AK4PFchs_offline_v2
- 2023: Run2_Run3_DATA_AK4PFchs_offline_v2

- Simulation:

- 2022: Winter22Run3_V1_MC_AK4PFchs
- 2022EE: Summer22EE_AK4PFchs_offline_v2
- 2023: Winter23_AK4PFchs_offline_v1

The effect of the uncertainty on the jet energy scale is evaluated by shifting the p_T of the jets up and down by the 1σ uncertainty on the cumulative jet energy corrections. The resulting relative change in signal yield is taken as a systematic uncertainty, on average 0.7–4.8%.

The effect of the uncertainty on the jet energy resolution is evaluated according to the scaling and smearing methods recommended by the JME group²³:

- **scaling method**: reconstructed jet p_T is scaled with a factor c_{JER} based on the difference in p_T between the reconstructed jet and a well-matched generator jet, i.e.:

$$c_{\text{JER}} = 1 + (s_{\text{JER}} - 1) \frac{p_{T,\text{reco}} - p_{T,\text{gen}}}{p_{T,\text{reco}}}, \quad (21)$$

where s_{JER} is the ratio of jet p_T resolutions measured by the JME group in experimental and simulated data (the resolution scale factor)

- **stochastic smearing**: reconstructed jet p_T is scaled with a randomized factor based on s_{JER} :

$$c_{\text{JER}} = 1 + N(0, \sigma_{\text{JER}}) \sqrt{\max(s_{\text{JER}}^2 - 1, 0)}, \quad (22)$$

where $N(0, \sigma_{\text{JER}})$ is a normal distribution with mean zero and variance equal to the square of the relative p_T resolution in simulated data as measured by the JME group

Reconstructed jets are considered well-matched to generator jets according to the criteria

$$\Delta R < R/2 \quad \text{and} \quad |p_{T,\text{reco}} - p_{T,\text{gen}}| < 3\sigma_{\text{JEC}} p_{T,\text{reco}}, \quad (23)$$

where $R = 0.4$ is the anti- k_T cone parameter of the jets. We apply the scaling method to the p_T of well-matched jets and stochastic smearing to the p_T of all others, as recommended in the “hybrid” method. The resulting relative change in signal yield after applying the hybrid method is taken as a systematic uncertainty. This uncertainty is on average 0.4–4.6%.

7.2.7 p_T^{miss}

The effect of the uncertainty on the p_T^{miss} is determined by varying the p_T of each object that contributes visible momentum to the event according to the uncertainty in the energy scale of the object, as prescribed by the JME group. The resulting relative change in signal yield is taken as a systematic uncertainty. The average values of this uncertainty for each object type whose p_T is varied are listed in Table 43.

²²See also the “Introduction to Jet Energy Corrections at CMS” from the JME POG.

²³See “Smearing Procedures” from the JME POG.

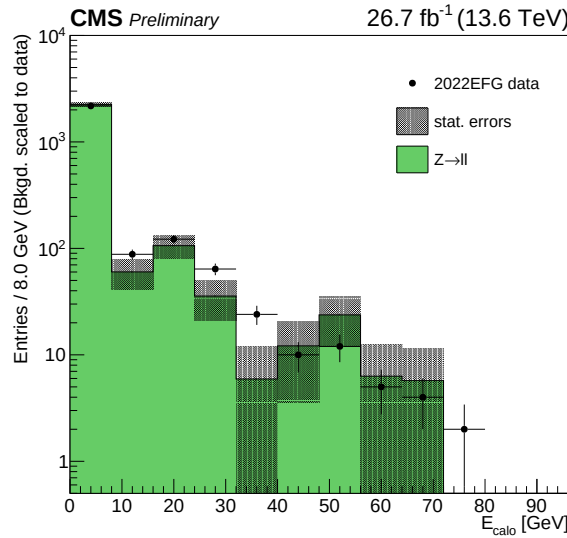
Table 43: Average value of the systematic uncertainty resulting from the variation of p_T^{miss} due to each type of object energy scale or resolution. The electron energy scale is also varied, but results in a negligible uncertainty.

type	average sys. uncertainty			
	2022	2022EE	2023	2023BPix
jet resolution, energy scale	0.3%	0.4%	0.2%	0.3%
tau energy scale	0.2%	0.1%	0.1%	0.2%
photon energy scale	1.5%	1.7%	1.4%	1.7%
unclustered energy scale	0.3%	0.2%	0.5%	0.4%

7.2.8 E_{calo}

The uncertainty associated with the modeling of the E_{calo} distribution in simulation is assessed by comparing the efficiency of the $E_{\text{calo}} < 10 \text{ GeV}$ selection for a sample of fake tracks in experimental data and simulation. For this, disappearing tracks with at most 4 tracker layers with measurement are selected from the $Z \rightarrow \mu\mu$ control region, which provides a sample overrun with fake tracks, the idea being to provide a uniform random sampling of the calorimeter for comparison. The E_{calo} distribution in experimental data and simulation is shown in Figure 43. The efficiency of the $E_{\text{calo}} < 10 \text{ GeV}$ selection is shown in Table 44 along with the resulting systematic uncertainty, which is taken as the relative difference between the efficiencies.

Figure 43: Comparison of E_{calo} in experimental data and simulation for tracks with at most 4 tracker layers with measurement in the $Z \rightarrow \mu\mu$ control region. Shown is 2022EFG data and simulation.



7.2.9 Lepton veto efficiency

As disappearing track candidates are required to be well separated from any electron and muon candidates, there is a systematic uncertainty due to the efficiency of these vetoes, and how they differ between data and simulation. While the POGs measure scale factors to correct simulation to data for several different lepton ID working points, no such scale factors are produced for the stricter veto definition used in this analysis. Consequently, for example, a stray muon detector segment or calorimeter hit could produce a PF muon or electron that we would veto against,

Table 44: Efficiency of the $E_{\text{calo}} < 10 \text{ GeV}$ criterion for tracks with at most 4 tracker layers with measurement in the $Z \rightarrow \mu\mu$ control region in experimental data and simulation, as well as the resulting systematic uncertainty.

run period	eff. of $E_{\text{calo}} < 10 \text{ GeV}$ data	simulation	systematic uncertainty
2022	$(89.9 \pm 1.3)\%$	$(87.8 \pm 2.9)\%$	2.3%
2022EE	$(86.8 \pm 0.7)\%$	$(89.8 \pm 1.5)\%$	3.5%
2023	$(86.7 \pm 0.9)\%$	$(70.5 \pm 1.4)\%$	18.8%
2023BPix	$(86.1 \pm 1.4)\%$	$(68.3 \pm 1.8)\%$	20.7%

yet the quality of such an object would be too poor to pass even the loosest EGamma or Muon POG-approved selection working point.

We account for this by computing a scale factor appropriate for the electron and muon definitions used in our veto, by measuring data and simulation efficiencies of our veto relative to that of the loosest POG-approved working points. The full expression for the scale factor is thus:

$$\begin{aligned}
 \text{SF} &\equiv \frac{\epsilon_{\text{veto}}^{\text{data}}}{\epsilon_{\text{veto}}^{\text{MC}}} \\
 &= \frac{\epsilon_{\text{loose}}^{\text{data}} \cdot (\epsilon_{\text{veto}}^{\text{data}} / \epsilon_{\text{loose}}^{\text{data}})}{\epsilon_{\text{loose}}^{\text{MC}} \cdot (\epsilon_{\text{veto}}^{\text{MC}} / \epsilon_{\text{loose}}^{\text{MC}})} \\
 &= (\text{POG-measured loose ID SF}) \cdot \frac{\epsilon_{\text{veto}}^{\text{data}}}{\epsilon_{\text{loose}}^{\text{data}}} \cdot \frac{\epsilon_{\text{loose}}^{\text{MC}}}{\epsilon_{\text{veto}}^{\text{MC}}}
 \end{aligned} \tag{24}$$

where the expression has been factorized into three terms. Here the subscript “veto” refers to the veto used in this analysis, and “loose” refers to the loosest approved working point criteria produced by the EGamma and Muon POGs. The “POG-measured loose ID SF” are the POG-measured scale factors for these loosest-possible working points. To measure the ratios $(\epsilon_{\text{veto}} / \epsilon_{\text{loose}})$, the typical veto against “any” electron or muon is replaced with a veto against a loose ID electron or muon and the relative selection efficiencies are compared. In signal MC this is done using the search selection for each chargino mass and lifetime. In data, the tag-and-probe samples used in the lepton background estimate (see Tables 60 and 16) are used—the muon veto efficiencies are measured using the electron tag-and-probe sample, and the electron veto efficiencies are measured using the muon tag-and-probe sample. In this way the efficiencies of each veto are measured in a pure sample of tracks well-separated from the object being vetoed against. The ratio $(\epsilon_{\text{veto}}^{\text{data}} / \epsilon_{\text{loose}}^{\text{data}})$ and resulting scale factors are summarized in Table 45.

The systematic uncertainty on the signal acceptance is calculated by applying these scale factors to any veto electron or muon candidates in the signal simulation. As veto leptons are infrequent in the signal hypothesis, the value of the systematic uncertainty listed in Table 46 is smaller than the scale factors themselves.

7.2.10 Track reconstruction efficiency

The uncertainty in the track reconstruction efficiency is assessed by the tracking POG using a tag-and-probe study [32]. The track reconstruction efficiency is measured for probe muons, which are identified using only information from the muon system and matched to tracks in

Table 45: Values of the ratio ($\epsilon_{\text{veto}}^{\text{data}}/\epsilon_{\text{loose}}^{\text{data}}$) for electron and muon vetoes, and the resulting scale factors on the lepton veto efficiencies. The scale factors listed are the average over all chargino masses and lifetimes. A signal systematic is calculated by applying these scale factors to any veto lepton in the simulated signal events. This is still in progress, and these values will be updated.

run period	n_{layers}	$(\epsilon_{\text{veto}}^{\text{data}}/\epsilon_{\text{loose}}^{\text{data}})$		average SF	
		e	μ	e	μ
2022	4	$1.00^{+0}_{-0.05}$	$1.00^{+0}_{-0.08}$	0.99	1.00
	5	$1.00^{+0}_{-0.6}$	$1.00^{+0}_{-0.17}$	1.01	1.00
	≥ 6	$0.88^{+0.10}_{-0.23}$	$1.00^{+0}_{-0.05}$	0.95	1.00
2022EE	4	$0.94^{+0.03}_{-0.04}$	$1.00^{+0}_{-0.04}$	0.96	1.00
	5	$1.00^{+0}_{-0.15}$	$1.00^{+0}_{-0.15}$	1.01	1.00
	≥ 6	$1.00^{+0}_{-0.18}$	$1.00^{+0}_{-0.05}$	1.01	1.00
2023	4	$0.96^{+0.02}_{-0.03}$	$1.00^{+0}_{-0.05}$	0.97	1.00
	5	$0.96^{+0.03}_{-0.08}$	$1.00^{+0}_{-0.06}$	0.98	1.00
	≥ 6	$1.00^{+0}_{-0.07}$	$0.98^{+0.01}_{-0.02}$	1.01	0.98
2023BPix	4	$0.98^{+0.01}_{-0.04}$	$1.00^{+0}_{-0.11}$	1.01	1.00
	5	$0.95^{+0.04}_{-0.11}$	$1.00^{+0}_{-0.15}$	0.96	1.00
	≥ 6	$1.00^{+0}_{-0.46}$	$1.00^{+0}_{-0.23}$	1.01	1.00

the inner tracker that are reconstructed without the benefit of muon seeds. The resulting track reconstruction efficiencies for experimental and simulated data are shown in Figure 44. We assign a systematic uncertainty to cover to the largest difference between the efficiencies in experimental and simulated data. It is around 0.7% for both data taking periods of 2022 and 2023. The systematic uncertainty is assigned as 2.0% in the period after the BPix issue in 2023.

7.2.11 Summary

A summary of the systematic uncertainties described above in the signal efficiency for simulated samples of charginos with masses in the range 100-1200 GeV and lifetimes of 0.2-10 000 cm/c is given in Tables 46 and 47 for the wino-like LSP case and in Tables 48 and 49 for the higgsino-like LSP case.

Figure 44: Track reconstruction efficiency as measured in a muon tag-and-probe study by the Tracking POG as a function of muon $|\eta|$ (left) and p_T (right). The performance in the data from 2022 (top), 2023 (middle) and 2023BPix (bottom) is compared to that in simulation.

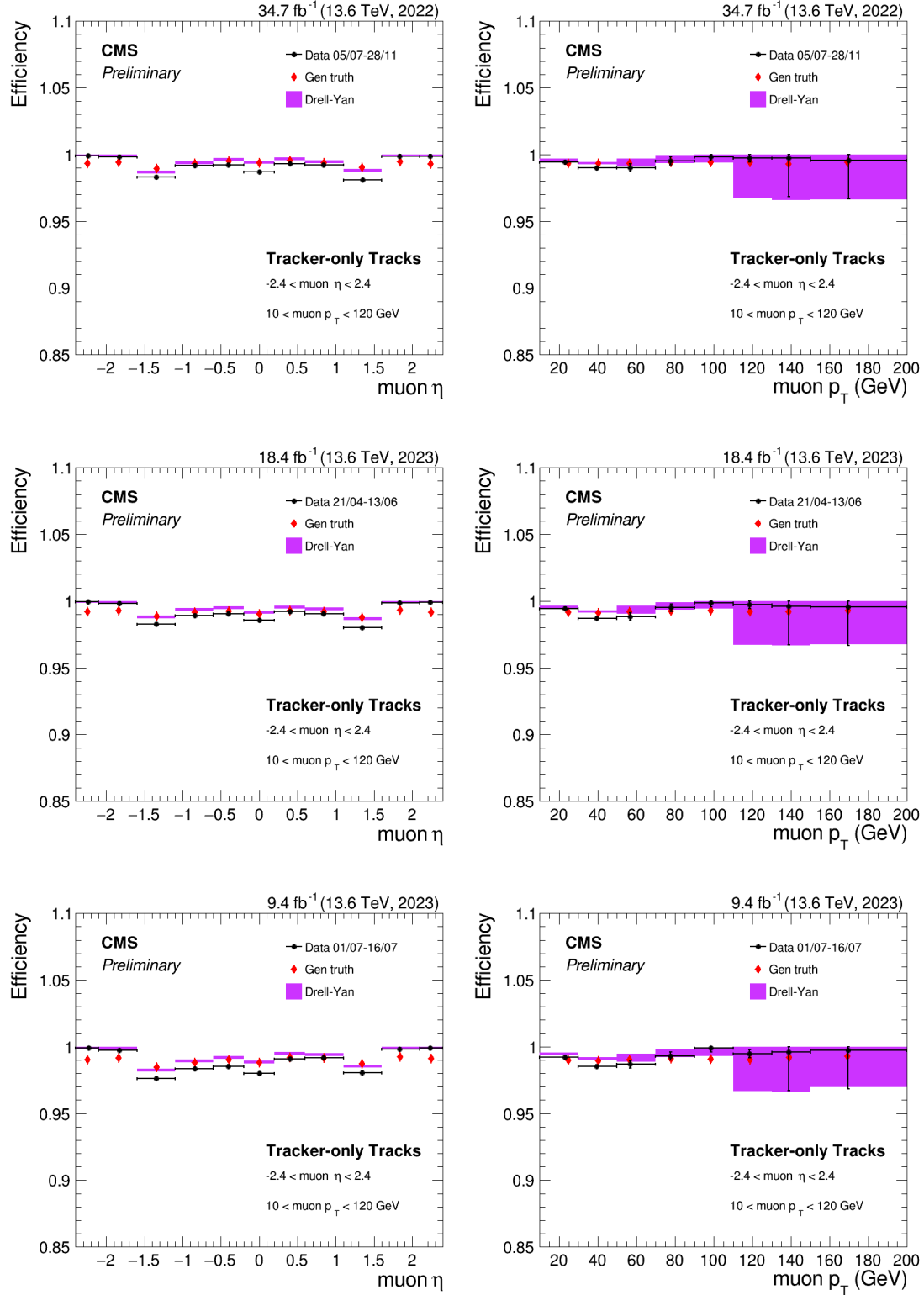


Table 46: Values of the systematic uncertainties on the signal yields taken from 2022 simulation, averaged over all wino-like electroweak production signal samples. The total listed on the last line is simply the sum in quadrature of each of the individual uncertainties.

source of uncertainty	run period	average value		
		$n_{\text{layers}} = 4$	$n_{\text{layers}} = 5$	$n_{\text{layers}} \geq 6$
pileup	2022	5.3%	4.9%	3.6%
	2022EE	3.6%	3.9%	3.4%
ISR	2022	6.8%	6.8%	6.7%
	2022EE	6.6%	6.7%	6.7%
trigger efficiency	2022	0.8%	0.4%	0.2%
	2022EE	0.5%	0.5%	0.2%
jet energy scale	2022	2.0%	2.5%	2.3%
	2022EE	0.8%	1.0%	1.9%
jet energy resolution	2022	1.6%	2.0%	2.0%
	2022EE	0.5%	0.6%	2.0%
$p_{\text{T}}^{\text{miss}}$	2022	0.6%	0.3%	0.4%
	2022EE	0.5%	0.5%	0.4%
E_{calo}	2022	2.3%	2.3%	2.3%
	2022EE	3.5%	3.5%	3.5%
missing inner hits	2022	0.3%	0.3%	0.9%
	2022EE	$\sim 0\%$	1.5%	0.7%
missing middle hits	2022	2.9%	2.8%	2.8%
	2022EE	3.8%	3.8%	3.8%
missing outer hits	2022	$\sim 0\%$	$\sim 0\%$	0.1%
	2022EE	$\sim 0\%$	$\sim 0\%$	0.2%
lepton veto scale factor	2022	0.02%	0.01%	0.10%
	2022EE	0.05%	0.02%	0.01%
track reconstruction efficiency	2022	0.7%	0.7%	0.7%
	2022EE	0.7%	0.7%	0.7%
total	2022	9.4%	9.3%	8.6%
	2022EE	9.2%	9.5%	9.6%

Table 47: Values of the systematic uncertainties on the signal yields taken from 2023 simulation, averaged over all wino-like electroweak production signal samples. The total listed on the last line is simply the sum in quadrature of each of the individual uncertainties.

source of uncertainty	run period	average value		
		$n_{\text{layers}} = 4$	$n_{\text{layers}} = 5$	$n_{\text{layers}} \geq 6$
pileup	2023	4.2%	5.2%	3.8%
	2023BPix	4.4%	3.7%	3.9%
ISR	2023	6.9%	7.0%	7.1%
	2023BPix	7.0%	7.1%	7.1%
trigger efficiency	2023	0.9%	0.8%	0.5%
	2023BPix	0.9%	0.8%	0.5%
jet energy scale	2023	4.8%	3.1%	3.1%
	2023BPix	0.8%	0.7%	1.4%
jet energy resolution	2023	4.6%	3.0%	2.8%
	2023BPix	0.4%	0.5%	1.3%
$p_{\text{T}}^{\text{miss}}$	2023	0.4%	0.4%	0.4%
	2023BPix	0.5%	0.5%	0.5%
E_{calo}	2023	18.8%	18.8%	18.8%
	2023BPix	20.7%	20.7%	20.7%
missing inner hits	2023	1.8%	0.2%	0.6%
	2023BPix	2.0%	4.2%	0.5%
missing middle hits	2023	5.9%	4.2%	4.2%
	2023BPix	4.2%	4.2%	4.2%
missing outer hits	2023	$\sim 0\%$	$\sim 0\%$	0.2%
	2023BPix	$\sim 0\%$	$\sim 0\%$	0.1%
lepton veto scale factor	2023	0.03%	0.03%	0.02%
	2023BPix	0.21%	0.03%	0.02%
track reconstruction efficiency	2023	0.7%	0.7%	0.7%
	2023BPix	2.0%	2.0%	2.0%
total	2023	21.5%	21.2%	20.9%
	2023BPix	22.4%	22.3%	22.4%

Table 48: Values of the systematic uncertainties on the signal yields taken from 2022 simulation, averaged over all higgsino-like electroweak production signal samples. The total listed on the last line is simply the sum in quadrature of each of the individual uncertainties.

source of uncertainty	run period	average value		
		$n_{\text{layers}} = 4$	$n_{\text{layers}} = 5$	$n_{\text{layers}} \geq 6$
pileup	2022	4.8%	4.5%	3.1%
	2022EE	5.0%	4.6%	4.0%
ISR	2022	6.8%	6.7%	6.7%
	2022EE	6.9%	6.7%	6.7%
trigger efficiency	2022	0.7%	0.5%	0.2%
	2022EE	0.8%	0.6%	0.2%
jet energy scale	2022	1.9%	2.4%	2.1%
	2022EE	2.2%	1.5%	2.5%
jet energy resolution	2022	1.7%	2.0%	1.4%
	2022EE	1.7%	1.2%	2.1%
$p_{\text{T}}^{\text{miss}}$	2022	0.5%	0.4%	0.5%
	2022EE	0.4%	0.5%	0.4%
E_{calo}	2022	2.3%	2.3%	2.3%
	2022EE	3.5%	3.5%	3.5%
missing inner hits	2022	0.3%	0.3%	0.9%
	2022EE	$\sim 0\%$	1.5%	0.7%
missing middle hits	2022	2.9%	2.8%	2.8%
	2022EE	3.8%	3.8%	3.8%
missing outer hits	2022	$\sim 0\%$	$\sim 0\%$	0.1%
	2022EE	$\sim 0\%$	$\sim 0\%$	0.2%
lepton veto scale factor	2022	0.01%	0.01%	0.10%
	2022EE	0.04%	0.02%	0.01%
track reconstruction efficiency	2022	0.7%	0.7%	0.7%
	2022EE	0.7%	0.7%	0.7%
total	2022	9.5%	9.5%	8.7%
	2022EE	10.4%	10.0%	10.0%

Table 49: Values of the systematic uncertainties on the signal yields taken from 2023 simulation, averaged over all higgsino-like electroweak production signal samples. The total listed on the last line is simply the sum in quadrature of each of the individual uncertainties.

source of uncertainty	run period	average value		
		$n_{\text{layers}} = 4$	$n_{\text{layers}} = 5$	$n_{\text{layers}} \geq 6$
pileup	2023	4.5%	4.1%	2.9%
	2023BPix	6.4%	3.7%	3.4%
ISR	2023	6.9%	7.0%	6.8%
	2023BPix	7.0%	7.0%	6.8%
trigger efficiency	2023	1.5%	0.9%	0.5%
	2023BPix	1.1%	0.9%	0.5%
jet energy scale	2023	1.6%	1.5%	3.1%
	2023BPix	1.3%	1.2%	3.0%
jet energy resolution	2023	1.2%	1.3%	3.1%
	2023BPix	1.1%	0.8%	2.4%
$p_{\text{T}}^{\text{miss}}$	2023	0.6%	0.3%	0.6%
	2023BPix	0.3%	0.8%	0.5%
E_{calo}	2023	18.8%	18.8%	18.8%
	2023BPix	20.7%	20.7%	20.7%
missing inner hits	2023	1.8%	0.2%	0.6%
	2023BPix	2.0%	4.2%	0.5%
missing middle hits	2023	5.9%	4.2%	4.2%
	2023BPix	4.2%	4.2%	4.2%
missing outer hits	2023	$\sim 0\%$	$\sim 0\%$	0.2%
	2023BPix	$\sim 0\%$	$\sim 0\%$	0.2%
lepton veto scale factor	2023	0.03%	0.03%	0.02%
	2023BPix	0.03%	0.04%	0.02%
track reconstruction efficiency	2023	0.7%	0.7%	0.7%
	2023BPix	2.0%	2.0%	2.0%
total	2023	21.6%	21.0%	21.1%
	2023BPix	23.4%	23.1%	22.9%

8 Results

The numbers of expected events from background sources are shown in Table 50. We proceed to set expected 95% CL upper limits on the production cross section of direct electroweak chargino production within the context of AMSB SUSY. Recall from Section 2.2 that this production occurs via the processes $pp \rightarrow \tilde{\chi}_1^\pm \tilde{\chi}_1^\mp$ and $pp \rightarrow \tilde{\chi}_1^0 \tilde{\chi}_1^\pm$ in a ratio of approximately 1 : 2. The upper limits on the total production cross section are calculated in a grid of chargino masses and lifetimes.

Table 50: Summary of numbers of events for the estimated backgrounds and the observed data. The uncertainties include those from statistical and systematic sources.

run period	n_{layers}	est. event count			observation
		leptons	fake tracks	total	
2022 CD	4	$0_{-0}^{+0.26} \pm 0$	$5.28 \pm 3.18 \pm 4.03$	$5.28_{-3.18}^{+3.19} \pm 4.03$	—
	5	$0_{-0}^{+0.10} \pm 0$	$0.59 \pm 0.39 \pm 0.49$	$0.59_{-0.39}^{+0.40} \pm 0.49$	—
	≥ 6	$1.01_{-0.24}^{+0.25} \pm 0.13$	$0.04_{-0.04}^{+0.05} \pm 0.05$	$1.05_{-0.24}^{+0.26} \pm 0.14$	—
2022 EFG	4	$0.42_{-0.30}^{+0.38} \pm 0.04$	$15.31 \pm 6.09 \pm 12.05$	$15.73 \pm 6.10 \pm 12.05$	—
	5	$0.3_{-0.22}^{+0.26} \pm 0.03$	$2.13 \pm 0.89 \pm 1.48$	$2.43_{-0.92}^{+0.93} \pm 1.48$	—
	≥ 6	$3.23_{-0.54}^{+0.54} \pm 0.38$	$0.08 \pm 0.07 \pm 0.04$	$3.31 \pm 0.55 \pm 0.38$	—
2023 C	4	$0.07_{-0.07}^{+0.12} \pm 0.01$	$8.21 \pm 4.81 \pm 14.96$	$8.28 \pm 4.81 \pm 14.96$	—
	5	$0.32_{-0.18}^{+0.19} \pm 0.04$	$2.02 \pm 1.20 \pm 3.69$	$2.34 \pm 1.21 \pm 3.69$	—
	≥ 6	$2.28_{-0.42}^{+0.42} \pm 0.26$	$0_{-0}^{+0.03} \pm 0$	$2.28 \pm 0.42 \pm 0.26$	—
2023 D	4	$0.07_{-0.07}^{+0.19} \pm 0.01$	$6.62 \pm 3.93 \pm 7.56$	$6.71_{-3.93}^{+3.94} \pm 7.56$	—
	5	$0_{-0}^{+0.14} \pm 0$	$1.70 \pm 1.03 \pm 2.02$	$1.7_{-1.03}^{+1.04} \pm 2.02$	—
	≥ 6	$0.27 \pm 0.16 \pm 0.02$	$0.07 \pm 0.06 \pm 0.16$	$0.34 \pm 0.17 \pm 0.16$	—

The limits are calculated with the modified frequentist method often referred to as CL_s [33, 34], using the LHC-type CL_s method [35]. This method uses a test statistic based on a profile likelihood ratio [36] and treats nuisance parameters in a frequentist context. Nuisance parameters for the systematic uncertainties in the integrated luminosity and in the signal selection efficiency are constrained with log-normal distributions. There are two types of nuisance parameters for the uncertainties in the background estimates, and they are specified separately for each of the four background contributions. Those that result from the limited size of the control samples are constrained with gamma distributions, while those that are associated with statistical uncertainties in multiplicative factors and the background systematic uncertainties, discussed in Section 7.1, are constrained with log-normal distributions.

The expected limits on the cross section are shown in Figure 45 as a function of chargino mass for several different chargino lifetimes. Figure 47 shows the expected and observed limits versus all masses considered. The intersection of the theory prediction and the upper limit on the cross section is used to set a constraint on the mass of the chargino, for a given chargino lifetime. This procedure is repeated for a large number of chargino lifetimes, in order to produce a two-dimensional constraint on the chargino mass and mean proper lifetime, which is shown in Figure 49 for a purely wino-like neutralino and Figure 46 for a purely higgsino-like neutralino. Charginos up to a mass of 777 GeV are excluded for a lifetime of 3 ns.

Combining the results of this analysis at the datacard level with the results of the previous Run II results for 2015-8 [2, 3] yields the expected limits on the cross section shown in Figures 51 and 52, improving the exclusion of chargino masses in the wino-like (higgsino-like) scenario up to 903 (762) GeV for a lifetime of 3 ns.

Figure 45: The expected 95% CL upper limits on the cross sections for direct chargino production as a function of chargino mass for various chargino lifetimes with a purely wino-like neutralino. These are calculated from the combined 2022 and 2023 results.

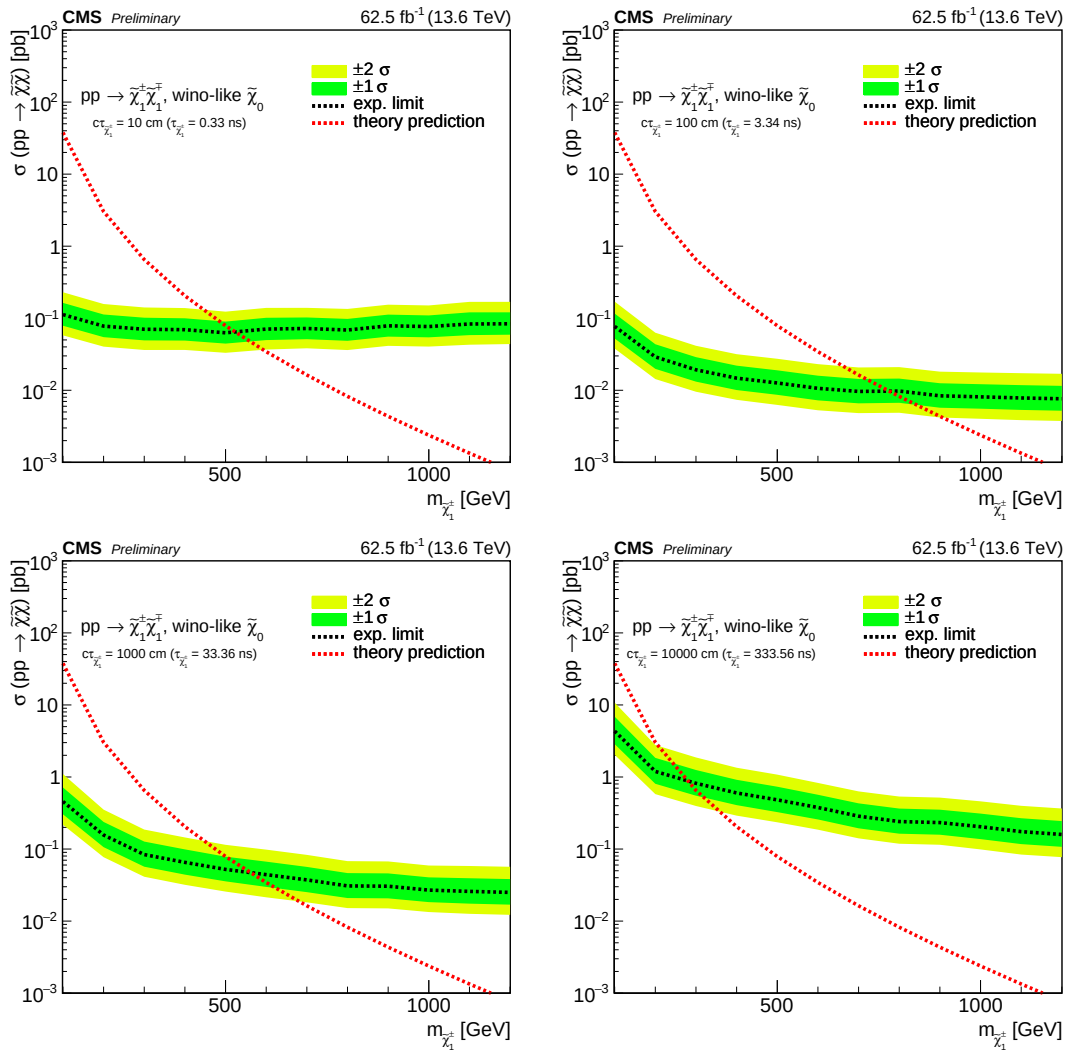


Figure 46: The expected 95% CL upper limits on the cross sections for direct chargino production as a function of chargino mass for various chargino lifetimes with a purely higgsino-like neutralino. These are calculated from the combined 2022 and 2023 results.

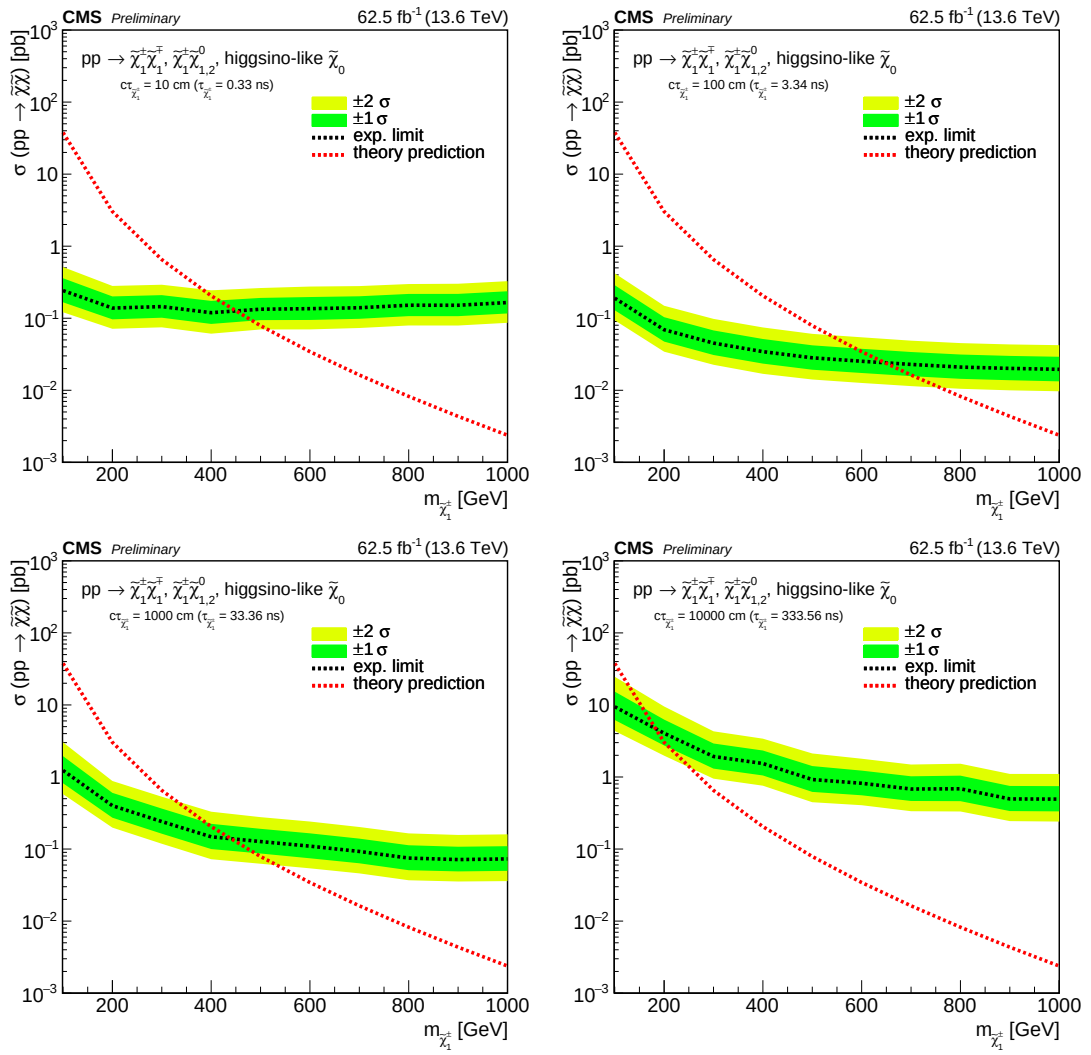


Figure 47: Upper limits at 95% CL on the cross section for direct chargino production as a function of chargino mass and lifetime for a purely wino-like neutralino. The plot shows the expected limits. These are calculated from the combined 2022 and 2023 results.

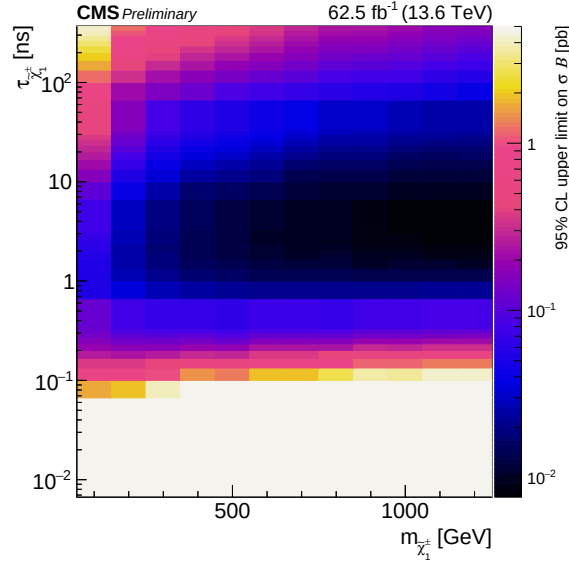


Figure 48: Upper limits at 95% CL on the cross section for direct chargino production as a function of chargino mass and lifetime with a purely higgsino-like neutralino. The plot shows the expected limits. These are calculated from the combined 2022 and 2023 results.

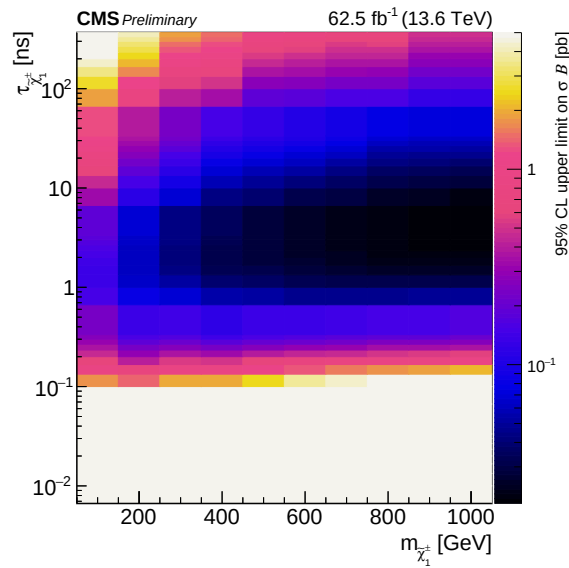


Figure 49: The expected constraints on chargino lifetime and mass for a purely wino-like neutralino. The region to the left of the curve is excluded at 95% CL. These are calculated from the combined 2022 and 2023 results.

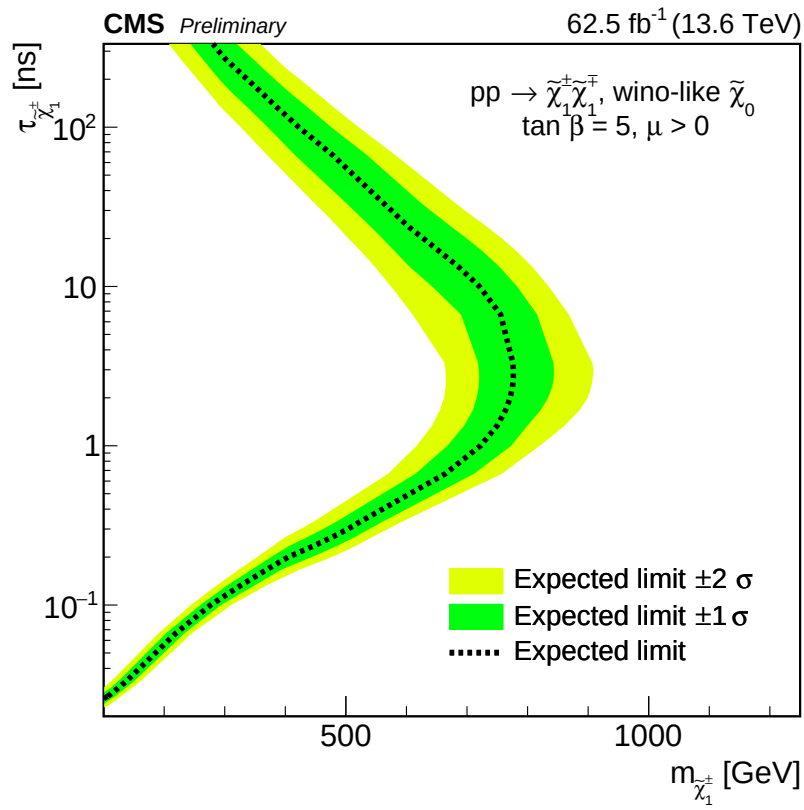


Figure 50: The expected constraints on chargino lifetime and mass for a purely higgsino-like neutralino. The region to the left of the curve is excluded at 95% CL. These are calculated from the combined 2022 and 2023 results.

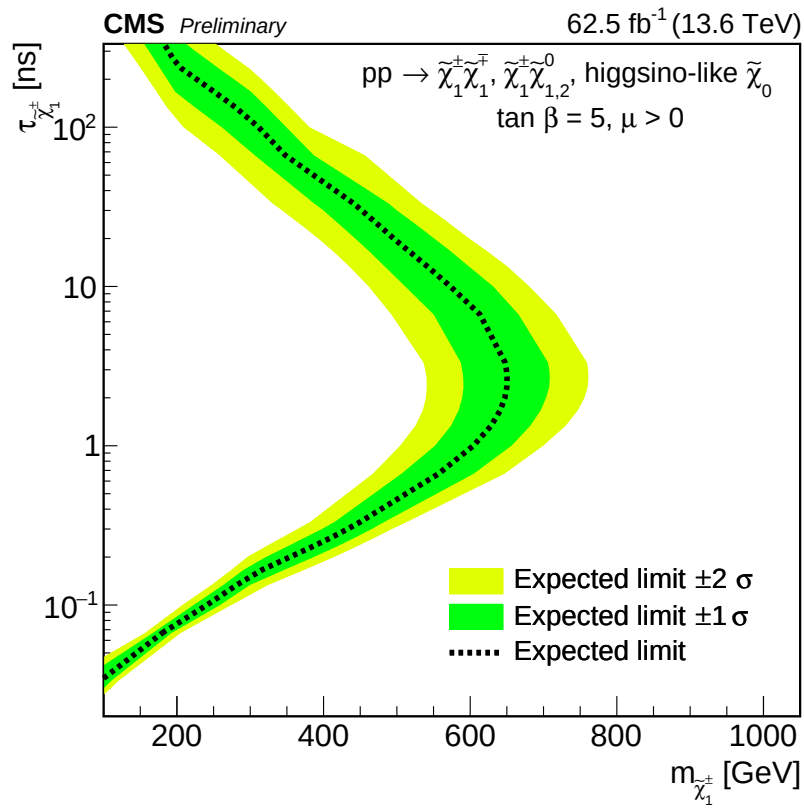


Figure 51: The expected constraints on chargino lifetime and mass for a purely wino-like neutralino, for the results in 2022 and 2023 combined with previously published results in 2015-8 for the full Run II+Run III results. The region to the left of the curve is excluded at 95% CL.

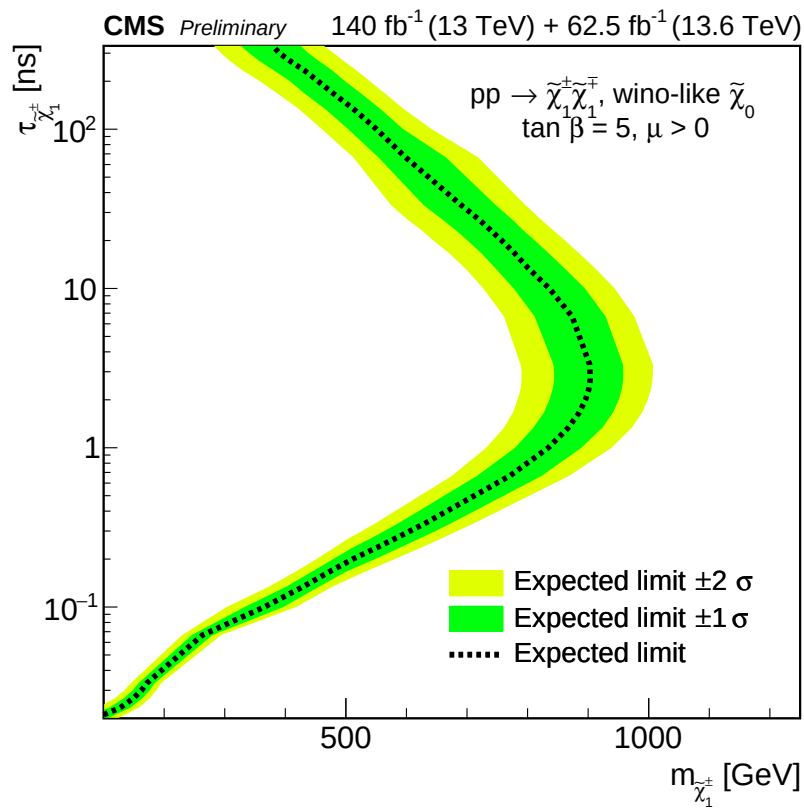
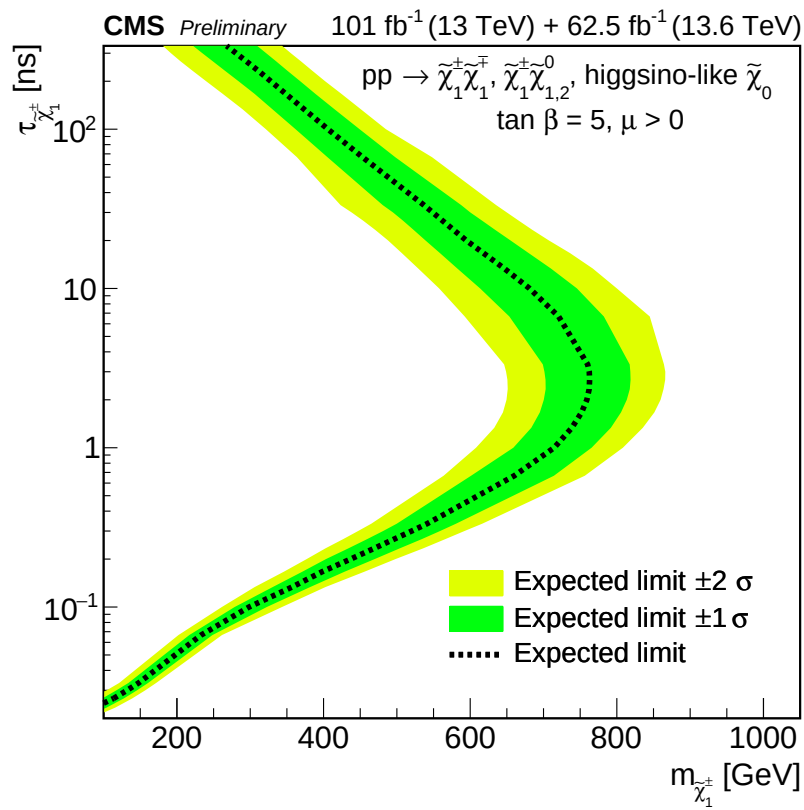


Figure 52: The expected constraints on chargino lifetime and mass for a purely higgsino-like neutralino, for the results in 2022 and 2023 combined with previously published results in 2015-8 for the full Run II+Run III results. The region to the left of the curve is excluded at 95% CL.



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A Signal simulation

For Run 3, the procedure to generate signal MC datasets using the AMSB model as benchmark was modified. As observed in AN-18-311 the largest source of systematic errors was the ISR corrections, where in 2017 it was 14.4-14.7% and in 2018 it was 12.0-12.2%. In those estimations two corrections were applied: one related to the mismodelling of weak bosons p_T when comparing MC with data; and the other related to the mismodelling of the ISR p_T when comparing Pythia8 (PY8) and MadGraph5 (MG5) $Z \rightarrow ll$ process, since Pythia8 was used to generate signal MC samples in Run 2, and it is known to not follow collision data very well²⁴, given that it only produces jets with extra jets coming from the ISR itself. MG5 has the possibility to include processes with extra hard jets and simulate real events more precisely.

Although not much could be done regarding the first correction, for the second, it was possible to change the event generation software to start using MadGraph5 and completely remove that correction. An extensive validation of signal events produced with MG5 and PY8 to cross-check for possible mistakes, and check for the distinction in the ISR p_T , was performed jointly with the CMS EXO MC&I and GEN groups²⁵.

The procedure to generate AMSB events with MadGraph5 is slightly different: instead of turning on one of the process options, as done in Pythia8, the process information is input in a process card with syntax given by $pp \rightarrow \chi\chi$. MG5 syntax allows to add extra processes to take into account the extra hard jets as, e.g., $pp \rightarrow \chi\chi j$. The processes of interest to be simulated in MadGraph5 are²⁶:

- $pp \rightarrow \chi_1^\pm \chi_1^0$
- $pp \rightarrow \chi_1^\pm \chi_1^{0j}$
- $pp \rightarrow \chi_1^\pm \chi_1^\pm$
- $pp \rightarrow \chi_1^\pm \chi_1^\pm j$

All of the run information, e.g., center of mass energy, protons PDF model, GEN-level cuts, and more, are input in the form of a run card. Given that some customizations were needed to account for the distinct chargino masses and lifetimes, a customize card was used to input the different mass/lifetime points into the events production. To match CMS's central events production, all cards and the events simulation details were saved in the format of MG5 gridpacks, that are compressed files containing all the information MG5 needs to produce events.

Using MG5 adds a few more checks to be done. Given its capability to generate processes with extra jets and to correctly merge these events using Pythia's interface with FastJet jet matching algorithms (the jet merging algorithm is applied during the hadronization step), the MG5 and PY8 jet merging scales need to be correctly determined. Following the procedure to correctly determine these values²⁷, the values of the matching scales in MG5 $x_{Q_{\text{cut}}}$ and in PY8 q_{cut} were obtained for each chargino mass that was generated by checking a grid of values. $x_{Q_{\text{cut}}}$ varied from 30 to 100 GeV in 10 GeV steps and from 200 to 1000 GeV in 100 GeV steps to test both low and high values, while q_{cut} was set as the default which is $\max(x_{Q_{\text{cut}}} + 10, 1.5 \times x_{Q_{\text{cut}}})$. Table 51 shows the values that were determined for each chargino mass.

²⁴It has been shown that the simulation of Pythia (without extra hard partons and the jet matching) doesn't match the inclusion of extra jets in the process. This can be seen in this presentation.

²⁵The last presentation on this topic was shown in the 14 May 2024 MC&I meeting.

²⁶These are the processes for a wino-like neutralino. The higgsino-like neutralino simulation would also include the same processes with χ_2^0 .

²⁷Which is described in this presentation. In summary, the distribution of the differential jet rate as a function of the partons p_T has to be smooth for the given choice of parameters.

Table 51: Values of $x_{Q_{\text{cut}}}$ and q_{cut} used for each range of chargino masses.

Chargino mass [GeV]	$x_{Q_{\text{cut}}}$ [GeV]	q_{cut} [GeV]
100 - 400	40.0	60.0
500 - 900	50.0	75.0
1000 - 1200	60.0	90.0

The first comparison with Pythia8 was to check if some relevant gen-level variables of the simulated charginos were following the expectations. In Figure 53, it is possible to see that most of the characteristics are matched in the comparison MG5/PY8, except for the p_T^{miss} and the p_T of the pair of generated charginos/chargino-neutralino calculated as the p_T of the resultant four-momentum sum of both generated particles, which are directly related to the ISR p_T and is expected.

This check was done for several chargino masses and lifetimes and all show the expected behavior. A comparison of the full disappearing tracks selection cutflow was also done to confirm the differences in Pythia8 vs MadGraph5 simulated events. As it can be seen in figure 54, the only relevant difference is in the trigger selection, $p_T^{\text{miss, no } \mu}$ and the p_T of the leading jet, which are all related to the ISR p_T .

Figure 53: Comparison of some variables of the chargino of the AMSB process simulated with Pythia8 which doesn't support the addition of extra hard partons (full red line), and MadGraph5 with the addition of one extra hard parton and performing the jet matching algorithm (dashed blue line). As expected, the difference of both methods is clear in the p_T^{miss} and the p_T of the summed four-momenta of the produced electroweakinos shown in the two bottom plots.

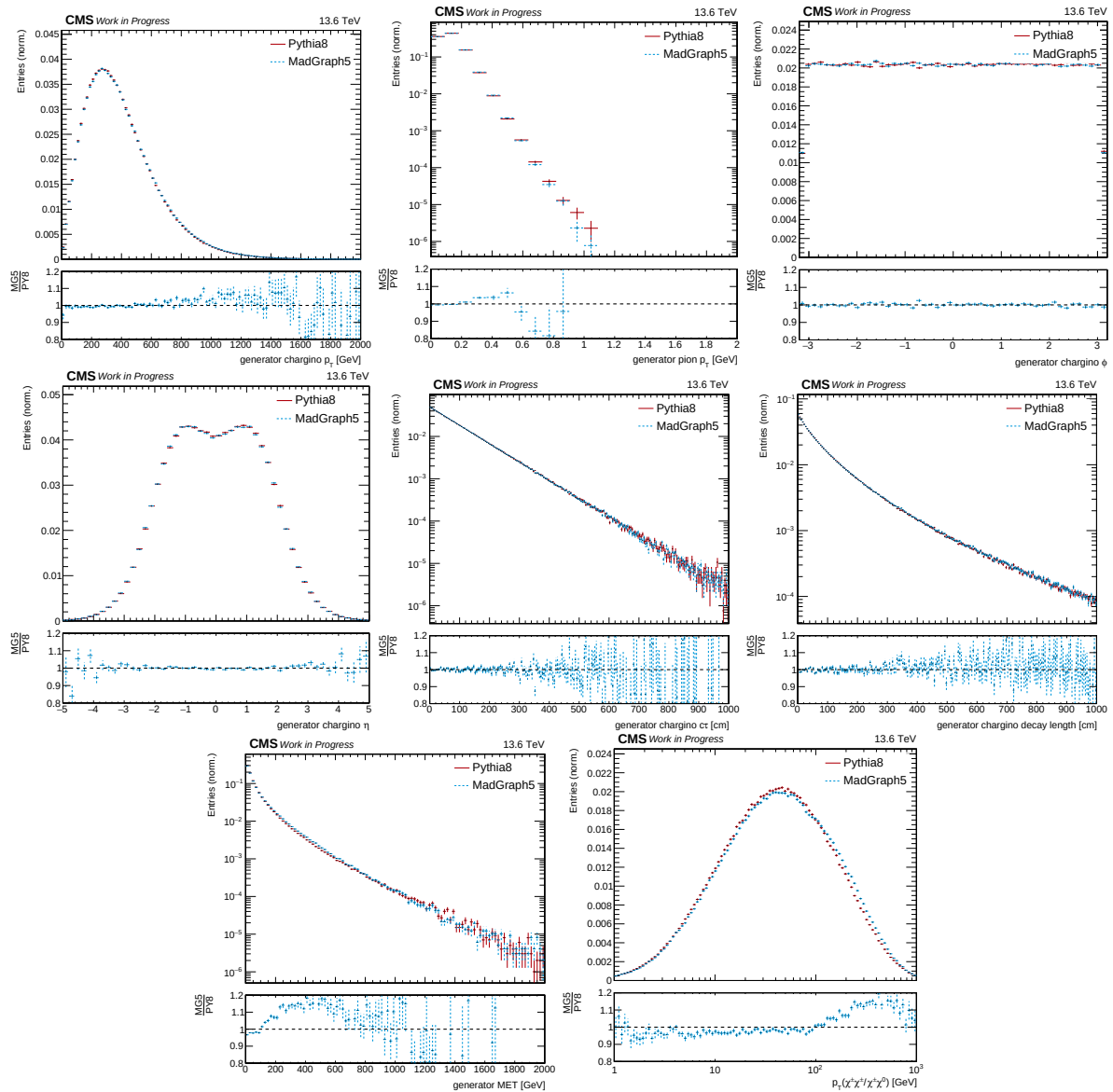
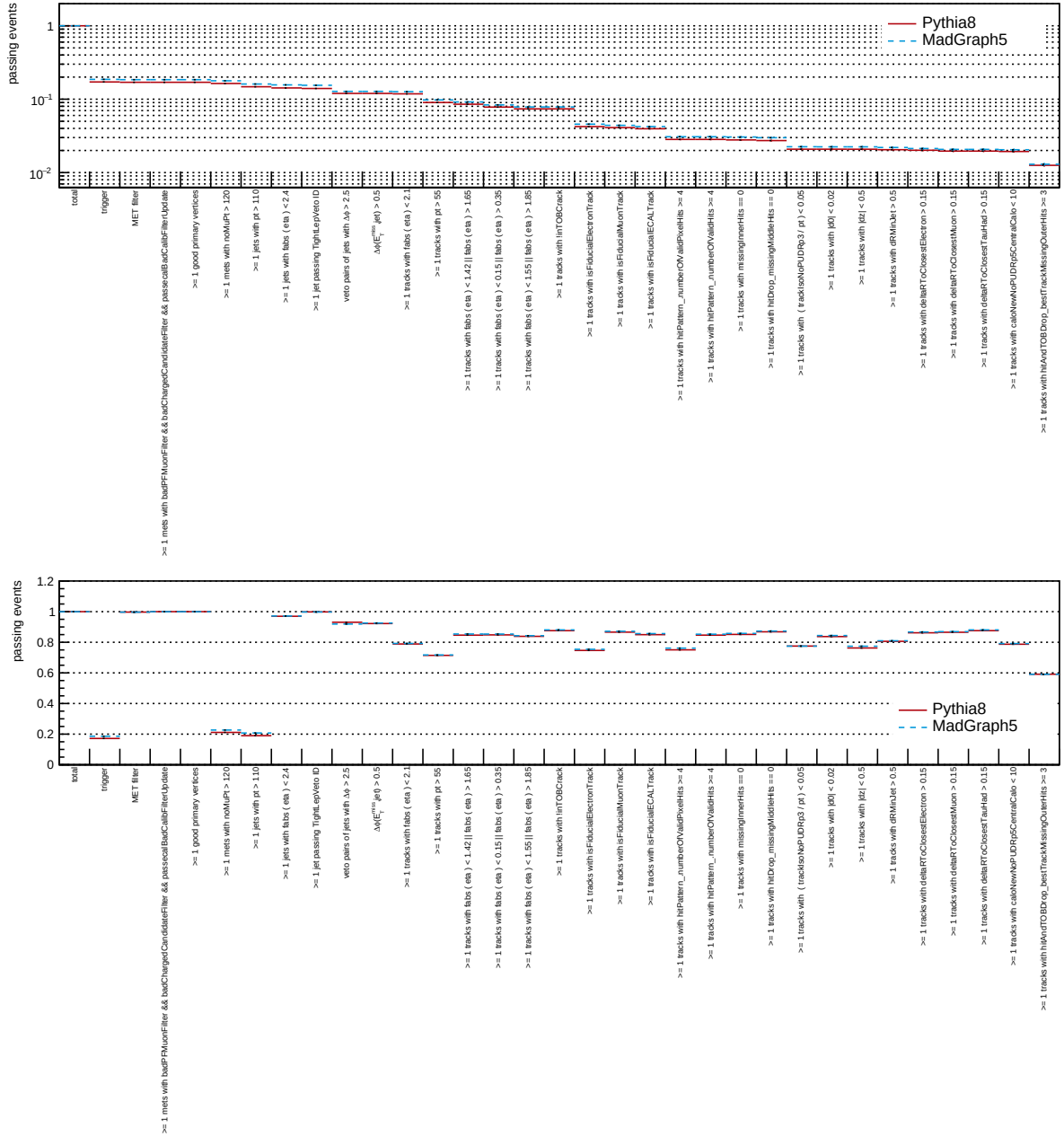


Figure 54: Top: comparison of the cutflow of the selections in the full disappearing tracks events selection. Bottom: comparison of the individual efficiency of each selection in the full disappearing selection. Pythia8 is shown with the full red line and MadGraph5 with the dashed blue line.



B Algorithms for finding missing hits corrections

B.1 Algorithm for finding missing middle hits correction

Given a two-dimensional histogram, `uncorrectedMC`, of the number of hits in the strip detector vs. the number of missing middle hits filled using simulated background events and a probability to drop a single hit, `hitDropProbability`, the following procedure, `getCorrectedMC`, returns the expected distribution of missing middle hits.

```

Sub getExtraMissingMiddleHitsPDF (nHits, hitDropProbability)
  Book 1D histogram pdf
  For iToy = 0 to 10000
    missingMiddleHits = 0
    countMissingMiddleHits = False

    For iHit = 0 to nHits
      x = Random number in [0, 1)
      hit = x > hitDropProbability

      If not hit and countMissingMiddleHits Then
        missingMiddleHits = missingMiddleHits + 1
      Else If hit Then
        countMissingMiddleHits = True
      End If
    Next iHit
    pdf.Fill (missingMiddleHits)
  Next iToy

  Normalize histogram pdf

  Return pdf
End Sub

Sub getCorrectedMC (uncorrectedMC, hitDropProbability)
  Book 1D histogram correctedMC
  For nHits = 1 to 20
    extraMissingMiddleHitsPDF = getExtraMissingMiddleHitsPDF (nHits, hitDropProbability)

    For originalMissingMiddleHits = 0 to 6
      bin = uncorrectedMC.FindBin (originalMissingMiddleHits, nHits)
      nTracks = uncorrectedMC.GetBinContent (bin)

      For extraMissingMiddleHits = 0 to 6
        bin = extraMissingMiddleHitsPDF.FindBin (extraMissingMiddleHits)
        extraMissingMiddleHitProbability = extraMissingMiddleHitsPDF.GetBinContent (bin)

        missingMiddleHits = originalMissingMiddleHits + extraMissingMiddleHits
        bin = correctedMC.FindBin (missingMiddleHits)
        currentNTracks = correctedMC.GetBinContent (bin)

        correctedMC.SetBinContent (bin, currentNTracks + nTracks * extraMissingMiddleHitProbability)
      Next extraMissingMiddleHits
    Next originalMissingMiddleHits
  Next nHits

  Return correctedMC
End Sub

```

B.2 Algorithm for finding missing outer hits corrections

Given a three-dimensional histogram, `uncorrectedMC`, of the number of hits in the strip detector vs. the number of hits in the TOB vs. the number of missing outer hits filled using simulated background events, a probability to drop a single hit, `hitDropProbability`, and a probability to drop every hit in the TOB, `tobDropProbability`, the following procedure, `getCorrectedMC`, returns the expected distribution of missing outer hits.

```

Sub getExtraMissingOuterHitsPDF (nHits, hitDropProbability)
  Book 1D histogram pdf
  For iToy = 0 to nToys
    missingOuterHits = 0
    For iHit = 0 to nHits
      x = Random number in [0, 1)
      hit = x > hitDropProbability

      If not hit Then
        missingOuterHits = missingOuterHits + 1
      Else
        Break
      End If
    Next iHit
    pdf.Fill (missingOuterHits)
  Next iToy

  Normalize histogram pdf

  Return pdf
End Sub

Sub getCorrectedMC (uncorrectedMC, hitDropProbability, tobDropProbability, preOrPostTOB)
  Book 1D histogram correctedMC
  For nHits = 1 to 20
    extraMissingOuterHitsPDF = getExtraMissingOuterHitsPDF (nHits, hitDropProbability)
    For originalMissingOuterHits = 1 to 16
      For tobHits = 1 to 16
        bin = uncorrectedMC.FindBin (originalMissingOuterHits, tobHits, nHits)
        nTracks = uncorrectedMC.GetBinContent (bin)

        For extraMissingOuterHits = 0 to 16
          bin = extraMissingOuterHitsPDF.FindBin (extraMissingOuterHits)
          extraMissingOuterHitsProbability = extraMissingOuterHitsPDF.GetBinContent (bin)

          If preOrPostTOB = "PRE_TOB" Then
            missingOuterHits = originalMissingOuterHits + extraMissingOuterHits
            missingOuterHitsWithTOBDrop = originalMissingOuterHits + tobHits
          Else If preOrPostTOB = "POST_TOB" Then
            missingOuterHits = originalMissingOuterHits
            missingOuterHitsWithTOBDrop = originalMissingOuterHits + extraMissingOuterHits + tobHits
          End If

          bin = correctedMC.FindBin (missingOuterHits)
          currentNTracks = correctedMC.GetBinContent (bin)
          correctedMC.SetBinContent (bin, currentNTracks + nTracks \
            * extraMissingOuterHitsProbability * (1 - tobDropProbability))

          bin = correctedMC.FindBin (missingOuterHitsWithTOBDrop)
          currentNTracks = correctedMC.GetBinContent (bin)
          correctedMC.SetBinContent (bin, currentNTracks + nTracks \
            * extraMissingOuterHitsProbability * tobDropProbability)

        Next extraMissingOuterHits
      Next tobHits
    Next originalMissingOuterHits
  Next nHits

  Return correctedMC
End Sub

```

C Trigger efficiencies of inclusive p_T^{miss} paths

The efficiency of the OR of all signal paths is shown in Figures 5 and 35 above. The efficiencies of the `HLT_MET105_IsoTrk50_v*` path path and each of its legs are shown below in Figure 55, and the efficiencies of each individual p_T^{miss} path are shown in Figures 56–59.

Figure 55: Efficiencies of the `HLT_MET105_IsoTrk50_v*` path (top left), its p_T^{miss} leg (top right) and track leg (bottom) of as a function of $p_T^{\text{miss, no } \mu}$ and track p_T respectively. The 2022 data set is shown in black, the 2023 data set is shown in red and the MC data set is shown in blue.

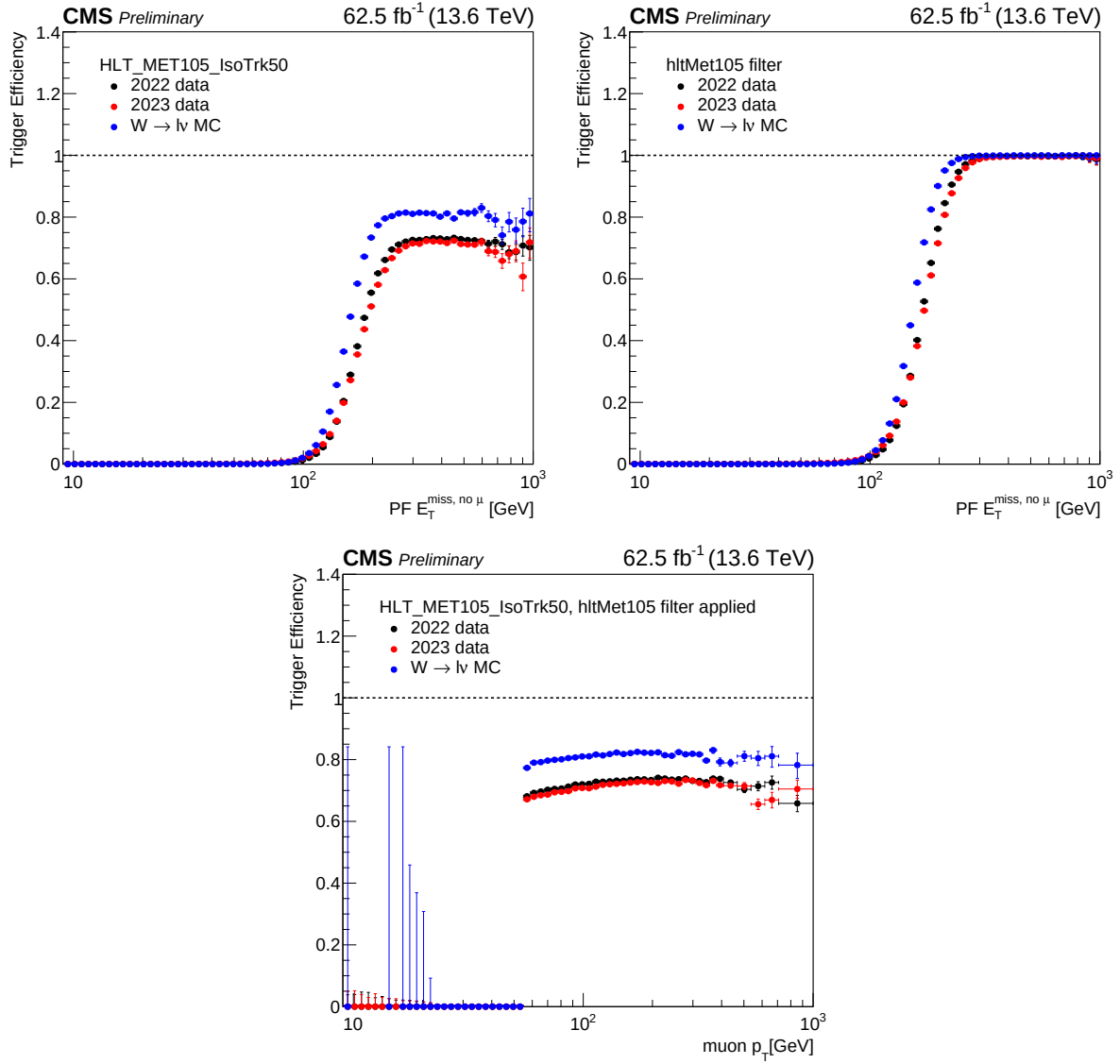


Figure 56: Efficiencies of `HLT_MET120_IsoTrk50` and `HLT_PFMET105_IsoTrk50` paths used throughout both 2022 and 2023. The name of each path is included in the legend of each sub-figure. The 2022 data set is shown in black, the 2023 data set is shown in red and the MC data set is shown in blue.

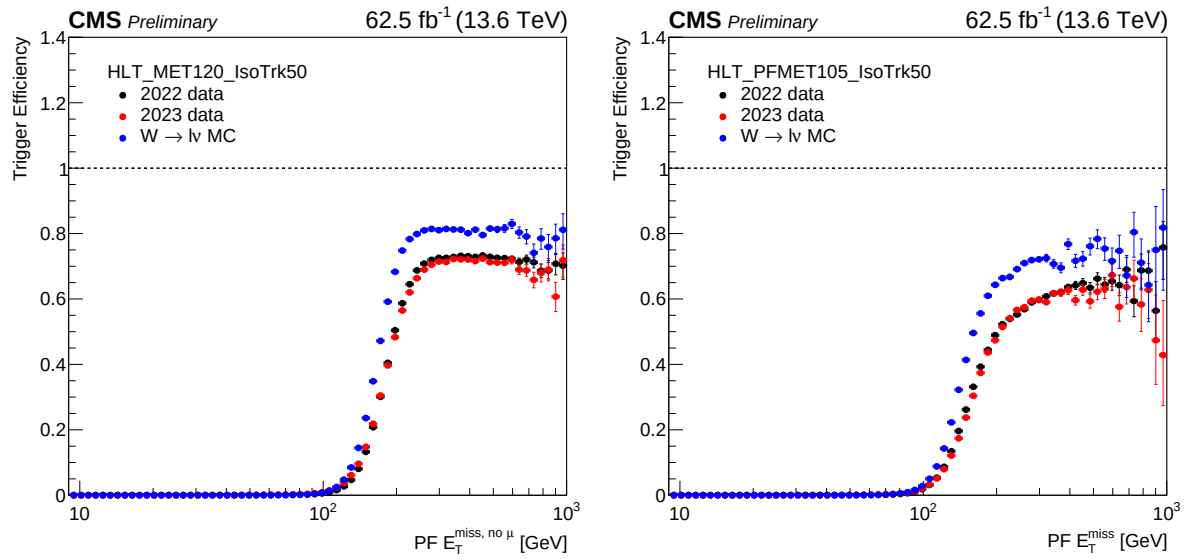


Figure 57: Efficiencies of `HLT_PFMET*_PFMHT*_IDTight*` paths used throughout both 2022 and 2023. The name of each path is included in the legend of each sub-figure. The 2022 data set is shown in black, the 2023 data set is shown in red and the MC data set is shown in blue.

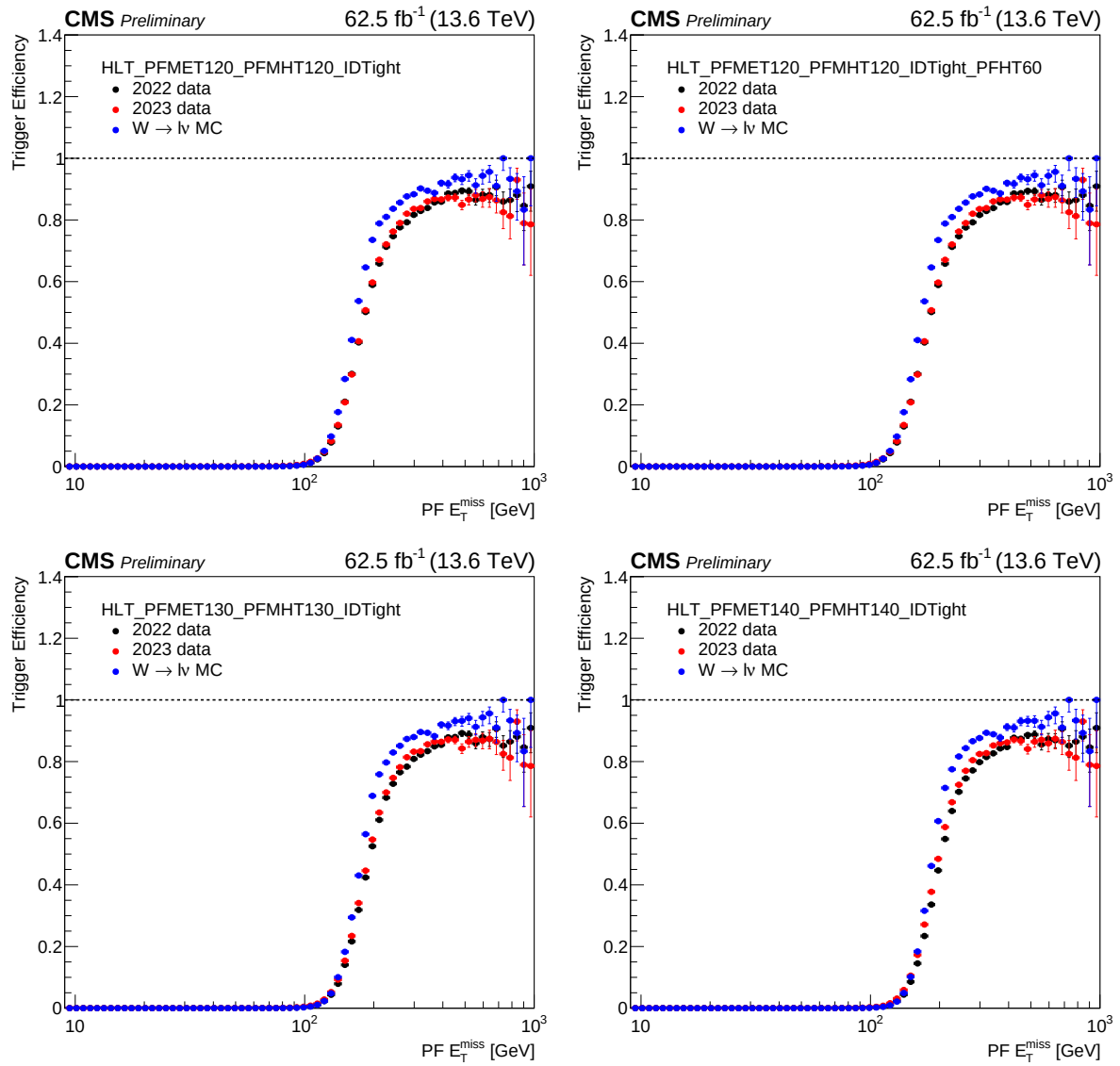


Figure 58: Efficiencies of 110 and 120 GeV $\text{HLT_PFMETNoMu*PFMHTNoMu*IDTight*}$ paths used throughout both 2022 and 2023. The name of each path is included in the legend of each sub-figure. The 2022 data set is shown in black, the 2023 data set is shown in red and the MC data set is shown in blue.

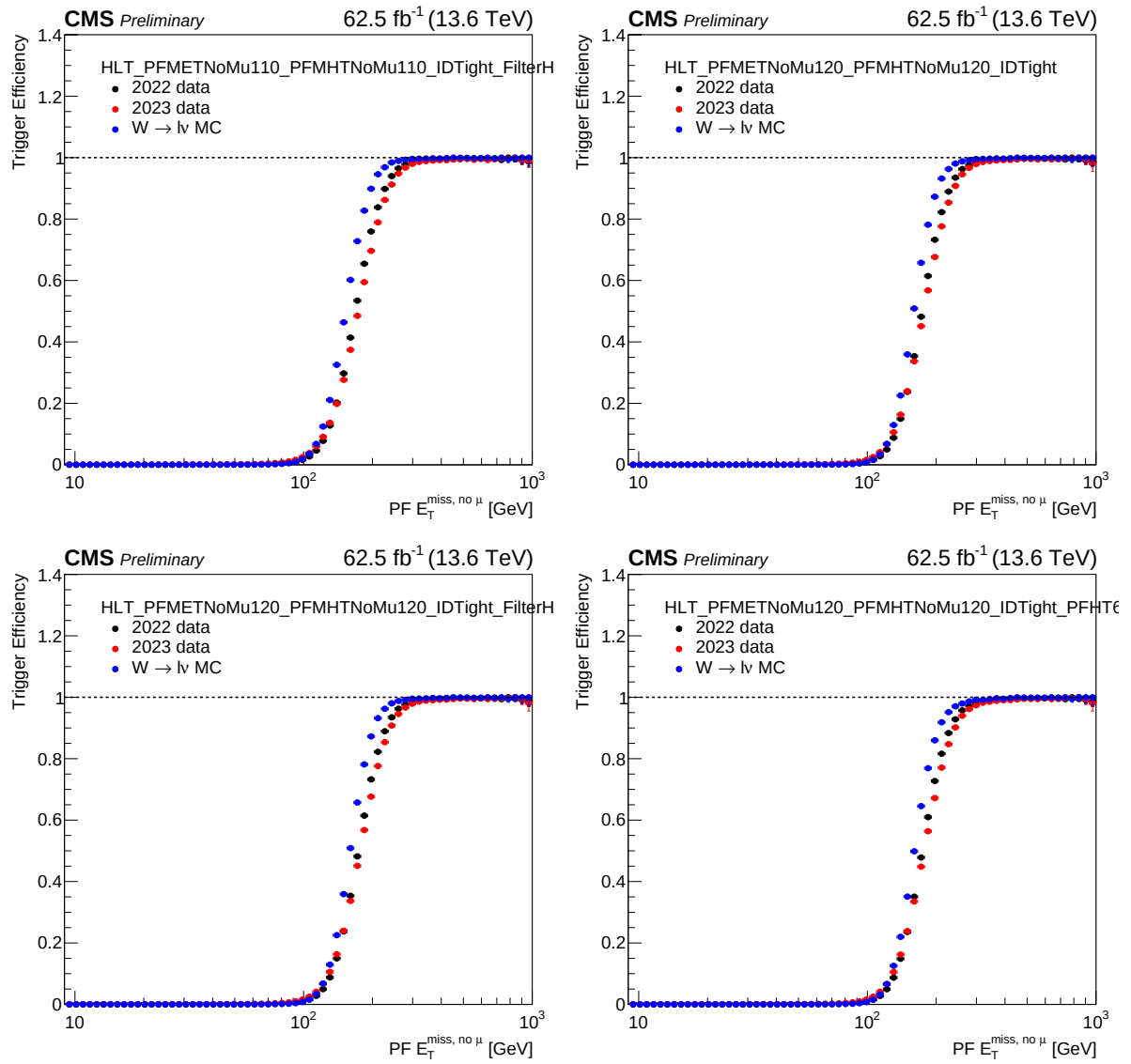
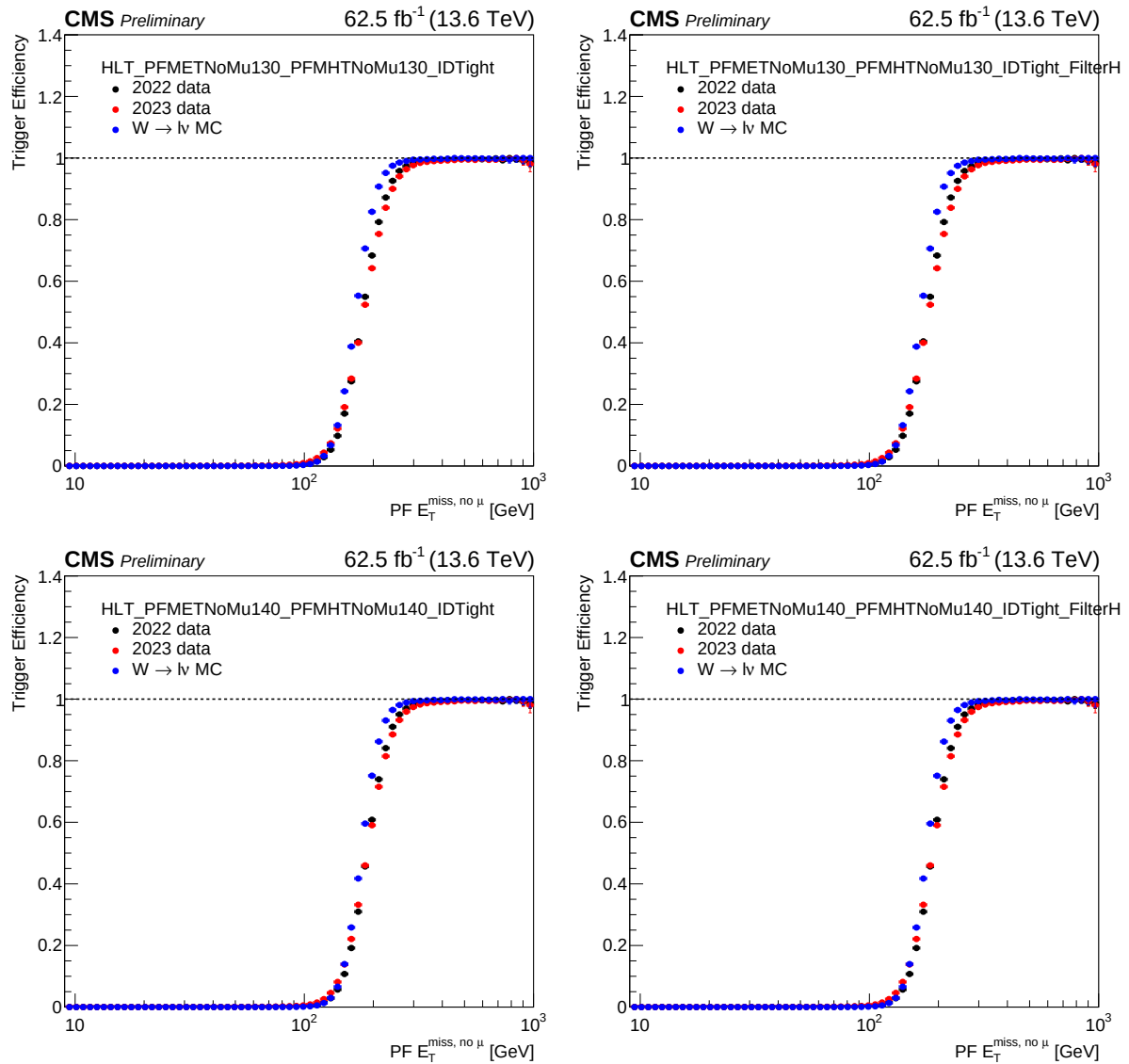


Figure 59: Efficiencies of 130 and 140 GeV `HLT_PFMETNoMu*_PFMHTNoMu*_IDTight*` paths used throughout both 2022 and 2023. The name of each path is included in the legend of each sub-figure. The 2022 data set is shown in black, the 2023 data set is shown in red and the MC data set is shown in blue.



D Additional Fake Track Estimate Details

D.1 Projection of d_{xy} sideband to signal region

Concerning the second assumption discussed in Section 5.2.1, this section details the projection of the d_{xy} from the sideband to the signal region by use of the transfer factor ζ . Table 52 compares the projected number of signal region tracks to the observed numbers of signal region tracks.

As a further cross-check for the $n_{\text{layers}} = 5, \geq 6$ categories, one could consider if the underlying d_{xy} distributions are actually flat instead of shaped as $n_{\text{layers}} = 4$. In this scenario the transfer factor would be a constant $\zeta = 0.02 \text{ cm}/0.45 \text{ cm} = 0.044$ with zero uncertainty, compared to the gaussian-shaped values of 0.10 ($Z \rightarrow \mu\mu$) and 0.11 ($Z \rightarrow ee$) in 2022 CD, 0.09 ($Z \rightarrow \mu\mu$) and 0.10 ($Z \rightarrow ee$) in 2022 EFG, and 0.07 ($Z \rightarrow \mu\mu$) and 0.07 ($Z \rightarrow ee$) in 2023 C and 0.09 ($Z \rightarrow \mu\mu$) and 0.09 ($Z \rightarrow ee$) in 2023 D. Taking this roughly one-third reduction in the projections in Table 52, this alternate hypothesis would result in a poorer agreement with the observed number of signal region tracks in these control regions, especially as the fit uncertainty would no longer be relevant.

Table 52: Projections of the number of sideband tracks (N_{sb}) to the signal region by the application of the transfer factor ζ ($N_{\text{sb}} \cdot \zeta$). The listed uncertainties are from first the statistical uncertainty in N_{sb} , then secondly the fit uncertainty in ζ . These projections are in reasonable agreement with the observed signal region tracks (N_{sr}).

run period	n_{layers}	N_{sb}		$N_{\text{sb}} \cdot \zeta$		N_{sr}	
		$Z \rightarrow \mu\mu$	$Z \rightarrow ee$	$Z \rightarrow \mu\mu$	$Z \rightarrow ee$	$Z \rightarrow \mu\mu$	$Z \rightarrow ee$
2022 CD	4	125	73	$12.2 \pm 1.1 \pm 7.3$	$8.4 \pm 1.0 \pm 12.3$	14	6
	5	14	5	$1.4 \pm 0.4 \pm 0.8$	$0.6 \pm 0.3 \pm 0.9$	5	2
	≥ 6	1	1	$0.1 \pm 0.1 \pm 0.1$	$0.1 \pm 0.1 \pm 0.2$	1	1
2022 EFG	4	373	271	$33.2 \pm 1.7 \pm 13.1$	$28.2 \pm 1.7 \pm 11.2$	46	30
	5	52	35	$4.6 \pm 0.6 \pm 1.8$	$3.6 \pm 0.6 \pm 1.4$	9	5
	≥ 6	2	1	$0.2 \pm 0.1 \pm 0.1$	$0.1 \pm 0.1 \pm 0.04$	1	1
2023 C	4	276	180	$20.5 \pm 1.2 \pm 13.7$	$12.8 \pm 1.0 \pm 12.1$	36	27
	5	68	49	$5.0 \pm 0.6 \pm 3.4$	$3.5 \pm 0.5 \pm 3.3$	3	6
	≥ 6	0	1	$0.0 \pm 0.0 \pm 0.0$	$0.1 \pm 0.1 \pm 0.1$	0	0
2023 D	4	195	102	$17.3 \pm 1.2 \pm 9.9$	$8.9 \pm 0.9 \pm 4.5$	34	17
	5	50	16	$4.4 \pm 0.6 \pm 2.5$	$1.4 \pm 0.3 \pm 0.7$	12	1
	≥ 6	2	3	$0.2 \pm 0.1 \pm 0.1$	$0.3 \pm 0.2 \pm 0.1$	1	0

D.2 Fake Track Estimate Probabilities

This section provides the full details of each $|d_{xy}|$ sideband used in the check of the third assumption, described in Section 5.2.1. The transfer factors ζ are shown in Table 53, the $P_{\text{fake}}^{\text{raw}}$ is shown in Table 54, and the full version of Table 34 with all sidebands is shown in Table 55.

E Background Closure Test (2022G)

A background closure test was performed on data from 2022 G by using an inverted selection on our signal region. Our signal region requires that the $\Delta\phi(\text{leading jet, MET}) > 0.5$ which we invert to define a closure test region. As seen in figure 12 this region is free of signal.

Table 53: Comparison of the transfer factors in 2022 EFG, ζ , derived from the integration of the $|d_{xy}|$ fit in each sideband in the $n_{\text{layers}} = 4$ category, defined in Eq. 15.

$ d_{xy} $ (cm)	ζ	
	$Z \rightarrow \mu\mu$	$Z \rightarrow ee$
(0.05, 0.10)	0.43	0.43
(0.10, 0.15)	0.48	0.50
(0.15, 0.20)	0.57	0.62
(0.20, 0.25)	0.71	0.81
(0.25, 0.30)	0.91	1.10
(0.30, 0.35)	1.2	1.52
(0.35, 0.40)	1.6	2.06
(0.40, 0.45)	2.0	2.63
(0.45, 0.50)	2.5	3.13

Table 54: Comparison of the raw fake probability in 2022 EFG, $P_{\text{fake}}^{\text{raw}}$, in the $Z \rightarrow \mu\mu$ and $Z \rightarrow ee$ control regions for all nine independent definitions of the sideband regions in the 2017 data. The product of these values with the transfer factor ζ gives the P_{fake} for each sideband.

n_{layers}	$ d_{xy} $ (cm)	$P_{\text{fake}}^{\text{raw}}$	
		$Z \rightarrow \mu\mu$	$Z \rightarrow ee$
4	(0.05, 0.10)	$(4.7 \pm 0.7) \times 10^{-6}$	$(1.0 \pm 0.1) \times 10^{-5}$
	(0.10, 0.15)	$(6.0 \pm 0.8) \times 10^{-6}$	$(8.6 \pm 1.3) \times 10^{-6}$
	(0.15, 0.20)	$(7.1 \pm 0.8) \times 10^{-6}$	$(10.8 \pm 1.4) \times 10^{-6}$
	(0.20, 0.25)	$(6.0 \pm 0.8) \times 10^{-6}$	$(6.6 \pm 1.1) \times 10^{-6}$
	(0.25, 0.30)	$(4.0 \pm 0.6) \times 10^{-6}$	$(4.6 \pm 0.9) \times 10^{-6}$
	(0.30, 0.35)	$(2.6 \pm 0.5) \times 10^{-6}$	$(2.9 \pm 0.7) \times 10^{-6}$
	(0.35, 0.40)	$(2.2 \pm 0.5) \times 10^{-6}$	$(1.5 \pm 0.5) \times 10^{-6}$
	(0.40, 0.45)	$(1.2 \pm 0.3) \times 10^{-6}$	$(2.9 \pm 0.7) \times 10^{-6}$
	(0.45, 0.50)	$(1.7 \pm 0.4) \times 10^{-6}$	$(1.6 \pm 0.5) \times 10^{-6}$
5	(0.05, 0.10)	$(1.2 \pm 0.3) \times 10^{-6}$	$(2.9 \pm 0.7) \times 10^{-6}$
	(0.10, 0.15)	$(8.6 \pm 2.9) \times 10^{-7}$	$(5.4 \pm 3.2) \times 10^{-7}$
	(0.15, 0.20)	$(8.6 \pm 2.9) \times 10^{-7}$	$(1.8 \pm 0.6) \times 10^{-7}$
	(0.20, 0.25)	$(1.9 \pm 1.3) \times 10^{-7}$	$(1.8 \pm 1.8) \times 10^{-7}$
	(0.25, 0.30)	$(4.8 \pm 2.1) \times 10^{-7}$	$(3.7 \pm 2.6) \times 10^{-7}$
	(0.30, 0.35)	$(5.7 \pm 2.3) \times 10^{-7}$	$(3.7 \pm 2.6) \times 10^{-7}$
	(0.35, 0.40)	$(3.8 \pm 1.9) \times 10^{-7}$	$(1.8 \pm 1.8) \times 10^{-7}$
	(0.40, 0.45)	$(9.5 \pm 9.5) \times 10^{-8}$	$0_{-0}^{+2.1} \times 10^{-7}$
	(0.45, 0.50)	$(2.9 \pm 1.6) \times 10^{-7}$	$0_{-0}^{+2.1} \times 10^{-7}$
≥ 6	(0.05, 0.10)	$(1.9 \pm 1.3) \times 10^{-7}$	$0_{-0}^{+2.1} \times 10^{-7}$
	(0.10, 0.15)	$(0.0_{-0.0}^{+1.1}) \times 10^{-7}$	$(1.8 \pm 1.8) \times 10^{-7}$
	(0.15, 0.20)	$(0.0_{-0.0}^{+1.1}) \times 10^{-7}$	$0_{-0}^{+2.1} \times 10^{-7}$
	(0.20, 0.25)	$(0.0_{-0.0}^{+1.1}) \times 10^{-7}$	$0_{-0}^{+2.1} \times 10^{-7}$
	(0.25, 0.30)	$(0.0_{-0.0}^{+1.1}) \times 10^{-7}$	$0_{-0}^{+2.1} \times 10^{-7}$
	(0.30, 0.35)	$(0.0_{-0.0}^{+1.1}) \times 10^{-7}$	$0_{-0}^{+2.1} \times 10^{-7}$
	(0.35, 0.40)	$(0.0_{-0.0}^{+1.1}) \times 10^{-7}$	$0_{-0}^{+2.1} \times 10^{-7}$
	(0.40, 0.45)	$(0.0_{-0.0}^{+1.1}) \times 10^{-7}$	$0_{-0}^{+2.1} \times 10^{-7}$
	(0.45, 0.50)	$(0.0_{-0.0}^{+1.1}) \times 10^{-7}$	$0_{-0}^{+2.1} \times 10^{-7}$

Table 55: Comparison of the fake track event probabilities, P_{fake} , and fake track background estimates obtained from the $Z \rightarrow \mu\mu$ and $Z \rightarrow ee$ control regions, for all nine independent definitions of the sideband regions in the 2017 data.

n_{layers}	$ d_{xy} $ (cm)	P_{fake}		$N_{\text{est}}^{\text{fake}}$	
		$Z \rightarrow \mu\mu$	$Z \rightarrow ee$	$Z \rightarrow \mu\mu$	$Z \rightarrow ee$
4	(0.05, 0.10)	$(2.0 \pm 0.6) \times 10^{-6}$	$(4.4 \pm 1.4) \times 10^{-6}$	9.7 ± 2.7	21.2 ± 6.6
	(0.10, 0.15)	$(2.9 \pm 0.8) \times 10^{-6}$	$(4.3 \pm 1.5) \times 10^{-6}$	14.0 ± 4.1	21.0 ± 7.3
	(0.15, 0.20)	$(4.1 \pm 1.3) \times 10^{-6}$	$(6.7 \pm 2.6) \times 10^{-6}$	19.7 ± 6.5	32.4 ± 12.7
	(0.20, 0.25)	$(4.2 \pm 1.6) \times 10^{-6}$	$(5.3 \pm 2.6) \times 10^{-6}$	20.6 ± 7.9	25.8 ± 12.4
	(0.25, 0.30)	$(3.6 \pm 1.7) \times 10^{-6}$	$(5.0 \pm 2.9) \times 10^{-6}$	17.7 ± 8.2	24.4 ± 14.2
	(0.30, 0.35)	$(3.1 \pm 1.7) \times 10^{-6}$	$(4.4 \pm 3.1) \times 10^{-6}$	15.0 ± 8.3	21.6 ± 14.9
	(0.35, 0.40)	$(3.5 \pm 2.2) \times 10^{-6}$	$(3.0 \pm 2.4) \times 10^{-6}$	16.8 ± 10.5	14.6 ± 11.7
	(0.40, 0.45)	$(2.5 \pm 1.7) \times 10^{-6}$	$(7.7 \pm 6.0) \times 10^{-6}$	12.2 ± 8.5	37.4 ± 29.0
	(0.45, 0.50)	$(4.2 \pm 2.9) \times 10^{-6}$	$(5.2 \pm 4.0) \times 10^{-6}$	20.5 ± 14.2	25.0 ± 19.4
4	mean	$(3.3 \pm 0.6) \times 10^{-6}$	$(5.1 \pm 1.1) \times 10^{-6}$	16.2 ± 2.8	24.8 ± 5.2
5	(0.05, 0.10)	$(5.3 \pm 1.9) \times 10^{-7}$	$(1.3 \pm 0.5) \times 10^{-6}$	2.6 ± 0.9	6.2 ± 2.3
	(0.10, 0.15)	$(4.1 \pm 1.8) \times 10^{-7}$	$(2.8 \pm 1.8) \times 10^{-7}$	2.0 ± 0.9	1.3 ± 0.9
	(0.15, 0.20)	$(4.9 \pm 2.2) \times 10^{-7}$	$(1.1 \pm 0.6) \times 10^{-6}$	2.4 ± 1.1	5.5 ± 2.7
	(0.20, 0.25)	$(1.3 \pm 1.1) \times 10^{-7}$	$(1.5^{+1.6}_{-1.5}) \times 10^{-7}$	0.7 ± 0.5	$0.7^{+0.8}_{-0.7}$
	(0.25, 0.30)	$(4.3 \pm 2.7) \times 10^{-7}$	$(4.0 \pm 3.6) \times 10^{-7}$	2.1 ± 1.3	2.0 ± 1.7
	(0.30, 0.35)	$(6.9 \pm 4.5) \times 10^{-7}$	$(5.6 \pm 5.3) \times 10^{-7}$	3.3 ± 2.2	2.7 ± 2.6
	(0.35, 0.40)	$(6.0 \pm 4.7) \times 10^{-7}$	$(3.8^{+4.6}_{-3.8}) \times 10^{-7}$	2.9 ± 2.3	$1.8^{+2.2}_{-1.8}$
	(0.40, 0.45)	$(1.9^{+2.3}_{-1.9}) \times 10^{-7}$	$(0.0^{+5.5}_{-0.0}) \times 10^{-7}$	$0.9^{+1.1}_{-0.9}$	$0.0^{+2.7}_{-0.0}$
	(0.45, 0.50)	$(7.0 \pm 6.1) \times 10^{-7}$	$0^{+6.5}_{-0} \times 10^{-7}$	3.4 ± 3.0	$0^{+3.2}_{-0}$
5	mean	$(4.6 \pm 1.1) \times 10^{-7}$	$(4.6 \pm 1.2) \times 10^{-7}$	2.2 ± 0.5	2.2 ± 0.6
≥ 6	(0.05, 0.10)	$(8.1 \pm 6.1) \times 10^{-8}$	$(0.0^{+9.1}_{-0.0}) \times 10^{-8}$	0.4 ± 0.3	$0.0^{+0.4}_{-0.0}$
	(0.10, 0.15)	$(0.0^{+5.2}_{-0.0}) \times 10^{-8}$	$9.2^{+9.6}_{-9.2} \times 10^{-8}$	$0.0^{+0.3}_{-0.0}$	$0.4^{+0.5}_{-0.4}$
	(0.15, 0.20)	$(0.0^{+6.2}_{-0.0}) \times 10^{-8}$	$(0.0^{+1.3}_{-0.0}) \times 10^{-7}$	$0.0^{+0.3}_{-0.0}$	$0.0^{+0.6}_{-0.0}$
	(0.20, 0.25)	$(0.0^{+7.7}_{-0.0}) \times 10^{-8}$	$(0.0^{+2.1}_{-0.0}) \times 10^{-8}$	$0.0^{+0.4}_{-0.0}$	$0.0^{+0.8}_{-0.0}$
	(0.25, 0.30)	$(0.0^{+9.9}_{-0.0}) \times 10^{-8}$	$(0.0^{+2.3}_{-0.0}) \times 10^{-7}$	$0.0^{+0.5}_{-0.0}$	$0.0^{+1.1}_{-0.0}$
	(0.30, 0.35)	$(0.0^{+1.3}_{-0.0}) \times 10^{-7}$	$(0.0^{+3.2}_{-0.0}) \times 10^{-7}$	$0.0^{+0.6}_{-0.0}$	$0.0^{+1.5}_{-0.0}$
	(0.35, 0.40)	$(0.0^{+1.7}_{-0.0}) \times 10^{-7}$	$(0.0^{+4.3}_{-0.0}) \times 10^{-7}$	$0.0^{+0.8}_{-0.0}$	$0.0^{+2.1}_{-0.0}$
	(0.40, 0.45)	$(0.0^{+2.2}_{-0.0}) \times 10^{-7}$	$(0.0^{+5.5}_{-0.0}) \times 10^{-7}$	$0.0^{+1.1}_{-0.0}$	$0.0^{+2.7}_{-0.0}$
	(0.45, 0.50)	$(0.0^{+2.7}_{-0.0}) \times 10^{-7}$	$(0.0^{+6.5}_{-0.0}) \times 10^{-7}$	$0.0^{+1.3}_{-0.0}$	$0.0^{+3.2}_{-0.0}$
≥ 6	mean	$(9.0 \pm 9.0) \times 10^{-9}$	$(1.0 \pm 1.0) \times 10^{-8}$	0.0 ± 0.0	$0.0^{+0.1}_{-0.0}$

For each signal region, zero events were observed in our closure test signal region, consistent with the estimated backgrounds. The small background estimates arise from the limited number of events passing selection in the background control regions. In certain lepton background/signal regions, no events were available to compute the probability of non-reconstructed leptons. These regions are assigned an estimate of zero with large upper bounds. The upper limits are sizable because the probability of non-reconstructed leptons includes cases where $\frac{0}{0}$ occurs. To account for this, we conservatively set $P = 1$ as a “worst-case” scenario, using this value to calculate the upper bounds.

	n_{layers}		
	4	5	≥ 6
fake	0.052 ± 0.032	$0.004^{+0.006}_{-0.004}$	$(0.4^{+0.5}_{-0.4}) \times 10^{-3}$
electron	$(0.0^{+4.9}_{-0.0}) \times 10^{-6}$	$(0.0^{+6.7}_{-0.0}) \times 10^{-7}$	$(3.1 \pm 0.9) \times 10^{-5}$
muon	$(0.0^{+2.8}_{-0.0}) \times 10^{-6}$	$(0.0^{+6.0}_{-0.0}) \times 10^{-8}$	$(0.0^{+1.0}_{-0.0}) \times 10^{-6}$
tau	$(0.0^{+9.5}_{-0.0}) \times 10^{-5}$	$(0.0^{+3.8}_{-0.0}) \times 10^{-5}$	$(0.0^{+1.6}_{-0.0}) \times 10^{-4}$
total background	0.052 ± 0.032	$0.004^{+0.006}_{-0.004}$	$(0.4^{+0.7}_{-0.4}) \times 10^{-3}$
observed	1	0	0
(obs-bkgd)/ σ	29.63	1.0	1.0

Table 56: Results of closure test performed on 2022CD data. An inverted requirement on leading jet/MET phi was used to create a signal free region ($\Delta\Phi(MET, leadingjet) < 0.5$). All observations in the data were 0 and the estimates were consistent with that observation.

	n_{layers}		
	4	5	≥ 6
fake	0.124 ± 0.049	0.017 ± 0.007	$(0.7 \pm 0.5) \times 10^{-4}$
electron	$(1.6^{+3.3}_{-1.6}) \times 10^{-5}$	$(2.1^{+2.5}_{-2.1}) \times 10^{-6}$	$(6.8 \pm 1.4) \times 10^{-6}$
muon	$(6.3^{+8.1}_{-6.3}) \times 10^{-6}$	$(0.0^{+8.7}_{-0.0}) \times 10^{-7}$	$(5.5 \pm 4.0) \times 10^{-6}$
tau	$(1.4^{+9.2}_{-1.4}) \times 10^{-4}$	$(5.7^{+5.5}_{-4.8}) \times 10^{-4}$	$(6.4^{+6.1}_{-4.9}) \times 10^{-4}$
total background	0.124 ± 0.049	0.017 ± 0.007	$(7.1^{+6.1}_{-4.9}) \times 10^{-4}$
observed	0	0	0
(obs-bkgd)/ σ	2.53	2.43	1.45

Table 57: Results of closure test performed on 2022EFG data. An inverted requirement on leading jet/MET phi was used to create a signal free region ($\Delta\Phi(MET, leadingjet) < 0.5$). All observations in the data were 0 and the estimates were consistent with that observation.

	n_{layers}		
	4	5	≥ 6
fake	0.089 ± 0.060	0.022 ± 0.015	$(0.0^{+3.7}_{-0.0}) \times 10^{-4}$
electron	$(0.0^{+4.6}_{-0.0}) \times 10^{-7}$	$(1.7 \pm 1.4) \times 10^{-5}$	$(6.2 \pm 1.4) \times 10^{-5}$
muon	$(4.0^{+6.4}_{-4.0}) \times 10^{-6}$	$(1.1^{+2.4}_{-1.1}) \times 10^{-6}$	$(5.5^{+5.6}_{-5.5}) \times 10^{-6}$
tau	$(0.0^{+1.1}_{-0.0}) \times 10^{-4}$	$(0.0^{+8.7}_{-0.0}) \times 10^{-5}$	$(1.1^{+1.7}_{-1.1}) \times 10^{-4}$
total background	0.089 ± 0.060	0.022 ± 0.015	$(1.7^{+4.1}_{-1.1}) \times 10^{-4}$
observed	0	0	0
(obs-bkgd)/ σ	1.48	1.47	1.55

Table 58: Results of closure test performed on 2023Cdata. An inverted requirement on leading jet/MET phi was used to create a signal free region ($\Delta\Phi(MET, leadingjet) < 0.5$). All observations in the data were 0 and the estimates were consistent with that observation.

	n_{layers}		
	4	5	≥ 6
fake	0.103 ± 0.059	0.026 ± 0.016	0.001 ± 0.001
electron	$(0.0^{+1.3}_{-0.0}) \times 10^{-6}$	$(0.0^{+1.1}_{-0.0}) \times 10^{-5}$	$(6.5 \pm 4.9) \times 10^{-6}$
muon	$(0.0^{+7.7}_{-0.0}) \times 10^{-8}$	$(0.0^{+1.1}_{-0.0}) \times 10^{-6}$	$(7.8 \pm 5.8) \times 10^{-6}$
tau	$(3.3^{+8.4}_{-3.3}) \times 10^{-4}$	$(0.0^{+1.5}_{-0.0}) \times 10^{-4}$	$(0.0^{+1.5}_{-0.0}) \times 10^{-4}$
total background	0.103 ± 0.059	0.026 ± 0.007	0.001 ± 0.001
observed	0	0	0
(obs-bkgd)/ σ	1.75	3.71	1.0

Table 59: Results of closure test performed on 2023D data. An inverted requirement on leading jet/MET phi was used to create a signal free region ($\Delta\Phi(MET, leadingjet) < 0.5$). All observations in the data were 0 and the estimates were consistent with that observation.

F Comparing old cutflow

F.1 Old cutflow

We've upgraded to a new code base and want to test that the cutflows in the new code agrees with the older version and if there is a disagreement we need to understand the cause, regardless of which code base is correct.

We're first looking at the Muon TnP selection. In the old code the single muon HTL filter is part of the cutflow table, but in the new code this was done in the skim so the cutflow is starting with this filter already in place. To do a more meaningful comparison I've rescaled the old cutflow such that it calculates individual cut efficiencies starting after the HLT filter has been applied.

Table 60: Selections used in the tag-and-probe study of electron reconstruction inefficiency. The requirements are cumulative, with only the objects passing a given criterion being considered in subsequent criteria. The selected electron is the tag, while the selected track is the probe. With the exception of the p_T and tag-and-probe pair requirements, the probe track selection is the disappearing track selection with the electron veto (defined in Table 21) removed.

quantity	object	selection
	event passes SingleElectron/EGamma triggers	
≥ 1	electrons	$p_T > 35 \text{ GeV}$ ($> 32 \text{ GeV}$ in 2018)
≥ 1	electrons	$ \eta < 2.1$
≥ 1	electrons	passing tight electron ID
	exactly one passing electron chosen randomly	
≥ 1	tracks	$p_T > 30 \text{ GeV}$
≥ 1	tracks	$ \eta < 2.1$
≥ 1	tracks	$ \eta < 0.15 \text{ OR } \eta > 0.35$
≥ 1	tracks	$ \eta < 1.42 \text{ OR } \eta > 1.65$
≥ 1	tracks	$ \eta < 1.55 \text{ OR } \eta > 1.85$
≥ 1	tracks	$\min \Delta R_{\text{track, noisy/dead ECAL ch.}} > 0.05$
≥ 1	tracks	$ d_z > 0.5 \text{ cm OR } \lambda > 10^{-3}$
≥ 1	tracks	$\eta < 0 \text{ OR } \eta > 1.42 \text{ OR } \phi < 2.7$
≥ 1	tracks	number of pixel hits ≥ 4
≥ 1	tracks	missing inner hits = 0
≥ 1	tracks	missing middle hits = 0
≥ 1	tracks	rel. PF-based iso. < 0.05
≥ 1	tracks	$ d_{xy} < 0.02 \text{ cm}$
≥ 1	tracks	$ d_z < 0.5 \text{ cm}$
≥ 1	track-jet pairs	$\Delta R_{\text{track, jet}} > 0.5$
≥ 1	track-electron pairs	$M_{\text{track, electron}} > 10 \text{ GeV}$
≥ 1	tracks	$\min \Delta R_{\text{track, muon}} > 0.15$
≥ 1	tracks	$\min \Delta R_{\text{track, had. tau}} > 0.15$
	exactly one passing track chosen randomly	
= 1	track-electron pairs	$ M_{\text{track, electron}} - M_Z < 10 \text{ GeV}$
= 1	track-electron pairs	$q_{\text{track}} \cdot q_{\text{electron}} < 0$
≥ 1	track	$n_{\text{layers}} \geq 4$ (three signal region bins)

Muon background estimation cutflow using new code:

Now looking at the components used to calculate Pveto for 2023D:

Table 61: Muon tag-and-probe cutflow for the 2023C dataset.

Cut	Events	ϵ_{prev}	ϵ_{total}
Event passes SingleMuon triggers	217572089	1.0000	1.0000
≥ 1 muon with $p_T > 26$ GeV	193195922	0.8880	0.8880
≥ 1 muon with $ \eta < 2.1$	174272109	0.9020	0.8010
≥ 1 muon passing tight muon ID	165021709	0.9469	0.7585
≥ 1 muon with $M_T(p_T^{\text{miss}}, \mu) < 40$ GeV	107536164	0.6516	0.4943
Exactly one passing muon chosen randomly	107536164	1.0000	0.4943
≥ 1 track with $p_T > 30$ GeV	89305972	0.8305	0.4105
≥ 1 track with $ \eta < 2.1$	87362008	0.9782	0.4015
≥ 1 track with $ \eta < 0.15$ or $ \eta > 0.35$	80760822	0.9244	0.3712
≥ 1 track with $ \eta < 1.42$ or $ \eta > 1.65$	72350188	0.8959	0.3325
≥ 1 track with $ \eta < 1.55$ or $ \eta > 1.85$	66245788	0.9156	0.3045
≥ 1 track with $ d_z > 0.5$ cm or $ \lambda > 10^{-3}$	66206140	0.9994	0.3043
≥ 1 track with ≥ 4 pixel hits	46567039	0.7034	0.2140
≥ 1 track with missing inner hits = 0	46275726	0.9937	0.2127
≥ 1 track with missing middle hits = 0	42564013	0.9198	0.1956
≥ 1 track with relative PF-based isolation < 0.05	38163324	0.8966	0.1754
≥ 1 track with $ d_{xy} < 0.02$ cm	37679554	0.9873	0.1732
≥ 1 track with $ d_z < 0.5$ cm	37660467	0.9995	0.1731
≥ 1 track–jet pair with $\Delta R(\text{track}, \text{jet}) > 0.5$	37619123	0.9989	0.1729
≥ 1 track–muon pair with $M_{\text{track}, \mu} > 10$ GeV	3886523	0.1033	0.0179
≥ 1 track with $\min \Delta R(\text{track}, e) > 0.15$	3857279	0.9925	0.0177
≥ 1 track with $E_{\text{calo}} < 10$ GeV	3521875	0.9130	0.0162
Exactly one passing track chosen randomly	3521875	1.0000	0.0162
≥ 1 track–muon pair with $ M_{\text{track}, \mu} - M_Z < 10$ GeV	3039925	0.8632	0.0140
≥ 1 track–muon pair with $q_{\text{track}} q_{\mu} < 0$	3039744	0.9999	0.0140
≥ 1 track with $N_{\text{layers}} \geq 4$	3039744	1.0000	0.0140

Table 62: Muon tag-and-probe cutflow for the 2023D dataset.

Cut	Events	ϵ_{prev}	ϵ_{total}
Event passes SingleMuon triggers	119708673	1.0000	1.0000
≥ 1 muon with $p_T > 26$ GeV	106112076	0.8864	0.8864
≥ 1 muon with $ \eta < 2.1$	95384460	0.8989	0.7968
≥ 1 muon passing tight muon ID	90559043	0.9494	0.7565
≥ 1 muon with $M_T(p_T^{\text{miss}}, \mu) < 40$ GeV	57765710	0.6379	0.4826
Exactly one passing muon chosen randomly	57765710	1.0000	0.4826
≥ 1 track with $p_T > 30$ GeV	47764681	0.8269	0.3990
≥ 1 track with $ \eta < 2.1$	46694952	0.9776	0.3901
≥ 1 track with $ \eta < 0.15$ or $ \eta > 0.35$	43226884	0.9257	0.3611
≥ 1 track with $ \eta < 1.42$ or $ \eta > 1.65$	38711383	0.8955	0.3234
≥ 1 track with $ \eta < 1.55$ or $ \eta > 1.85$	35323350	0.9125	0.2951
≥ 1 track with $ d_z > 0.5$ cm or $ \lambda > 10^{-3}$	35302519	0.9994	0.2949
≥ 1 track with ≥ 4 pixel hits	24083001	0.6822	0.2012
≥ 1 track with missing inner hits = 0	23953298	0.9946	0.2001
≥ 1 track with missing middle hits = 0	21969765	0.9172	0.1835
≥ 1 track with relative PF-based isolation < 0.05	19626974	0.8934	0.1640
≥ 1 track with $ d_{xy} < 0.02$ cm	19336109	0.9852	0.1615
≥ 1 track with $ d_z < 0.5$ cm	19323793	0.9994	0.1614
≥ 1 track–jet pair with $\Delta R(\text{track}, \text{jet}) > 0.5$	19291255	0.9983	0.1612
≥ 1 track–muon pair with $M_{\text{track}, \mu} > 10$ GeV	2065957	0.1071	0.0173
≥ 1 track with $\min \Delta R(\text{track}, e) > 0.15$	2050720	0.9926	0.0171
≥ 1 track with $E_{\text{calo}} < 10$ GeV	1882106	0.9178	0.0157
Exactly one passing track chosen randomly	1882106	1.0000	0.0157
≥ 1 track–muon pair with $ M_{\text{track}, \mu} - M_Z < 10$ GeV	1631582	0.8669	0.0136
≥ 1 track–muon pair with $q_{\text{track}} q_{\mu} < 0$	1631474	0.9999	0.0136
≥ 1 track with $N_{\text{layers}} \geq 4$	1631474	1.0000	0.0136

Table 63: Muon tag-and-probe cutflow for the 2024 dataset.

Cut	Events	ϵ_{prev}	ϵ_{total}
Event passes SingleMuon triggers	1435442329	1.0000	1.0000
≥ 1 muon with $p_T > 26$ GeV	1249464454	0.8704	0.8704
≥ 1 muon with $ \eta < 2.1$	1123199727	0.8989	0.7825
≥ 1 muon passing tight muon ID	1050410888	0.9352	0.7318
≥ 1 muon with $M_T(p_T^{\text{miss}}, \mu) < 40$ GeV	679343948	0.6467	0.4733
Exactly one passing muon chosen randomly	679343948	1.0000	0.4733
≥ 1 track with $p_T > 30$ GeV	561353083	0.8263	0.3911
≥ 1 track with $ \eta < 2.1$	547686409	0.9757	0.3815
≥ 1 track with $ \eta < 0.15$ or $ \eta > 0.35$	506469529	0.9247	0.3528
≥ 1 track with $ \eta < 1.42$ or $ \eta > 1.65$	454903375	0.8982	0.3169
≥ 1 track with $ \eta < 1.55$ or $ \eta > 1.85$	416931539	0.9165	0.2905
≥ 1 track with $ d_z > 0.5$ cm or $ \lambda > 10^{-3}$	405990666	0.9738	0.2828
≥ 1 track with ≥ 4 pixel hits	262338703	0.6462	0.1828
≥ 1 track with missing inner hits = 0	260402734	0.9926	0.1814
≥ 1 track with missing middle hits = 0	233725637	0.8976	0.1628
≥ 1 track with relative PF-based isolation < 0.05	202132629	0.8648	0.1408
≥ 1 track with $ d_{xy} < 0.02$ cm	197941666	0.9793	0.1379
≥ 1 track with $ d_z < 0.5$ cm	197769638	0.9991	0.1378
≥ 1 track-jet pair with $\Delta R(\text{track}, \text{jet}) > 0.5$	192810010	0.9749	0.1343
≥ 1 track-muon pair with $M_{\text{track}, \mu} > 10$ GeV	19431352	0.1008	0.0135
≥ 1 track with $\min \Delta R(\text{track}, e) > 0.15$	19261693	0.9913	0.0134
≥ 1 track with $E_{\text{calo}} < 10$ GeV	16915183	0.8782	0.0118
Exactly one passing track chosen randomly	16915183	1.0000	0.0118
≥ 1 track-muon pair with $ M_{\text{track}, \mu} - M_Z < 10$ GeV	14479580	0.8560	0.0101
≥ 1 track-muon pair with $q_{\text{track}} q_{\mu} < 0$	14478232	0.9999	0.0101
≥ 1 track with $N_{\text{layers}} \geq 4$	14478232	1.0000	0.0101

Table 64: Muon tag-and-probe cutflow for the 2025 dataset.

Cut	Events	ϵ_{prev}	ϵ_{total}
Event passes SingleMuon triggers	1676034238	1.0000	1.0000
≥ 1 muon with $p_T > 26$ GeV	1419992647	0.8472	0.8472
≥ 1 muon with $ \eta < 2.1$	1284657844	0.9047	0.7665
≥ 1 muon passing tight muon ID	1186228241	0.9234	0.7078
≥ 1 muon with $M_T(p_T^{\text{miss}}, \mu) < 40$ GeV	775839888	0.6540	0.4629
Exactly one passing muon chosen randomly	775839888	1.0000	0.4629
≥ 1 track with $p_T > 30$ GeV	624261528	0.8046	0.3725
≥ 1 track with $ \eta < 2.1$	610327529	0.9777	0.3641
≥ 1 track with $ \eta < 0.15$ or $ \eta > 0.35$	564718204	0.9253	0.3369
≥ 1 track with $ \eta < 1.42$ or $ \eta > 1.65$	507299779	0.8983	0.3027
≥ 1 track with $ \eta < 1.55$ or $ \eta > 1.85$	465465094	0.9175	0.2777
≥ 1 track with $ d_z > 0.5$ cm or $ \lambda > 10^{-3}$	465193295	0.9994	0.2776
≥ 1 track with ≥ 4 pixel hits	246253812	0.5294	0.1469
≥ 1 track with missing inner hits = 0	244177344	0.9916	0.1457
≥ 1 track with missing middle hits = 0	217171982	0.8894	0.1296
≥ 1 track with relative PF-based isolation < 0.05	173815984	0.8004	0.1037
≥ 1 track with $ d_{xy} < 0.02$ cm	167051259	0.9611	0.0997
≥ 1 track with $ d_z < 0.5$ cm	166773974	0.9983	0.0995
≥ 1 track-jet pair with $\Delta R(\text{track}, \text{jet}) > 0.5$	166383685	0.9977	0.0993
≥ 1 track-muon pair with $M_{\text{track}, \mu} > 10$ GeV	17006221	0.1022	0.0101
≥ 1 track with $\min \Delta R(\text{track}, e) > 0.15$	16825722	0.9894	0.0100
≥ 1 track with $E_{\text{calo}} < 10$ GeV	15200794	0.9034	0.0091
Exactly one passing track chosen randomly	15200794	1.0000	0.0091
≥ 1 track-muon pair with $ M_{\text{track}, \mu} - M_Z < 10$ GeV	13201312	0.8685	0.0079
≥ 1 track-muon pair with $q_{\text{track}} q_{\mu} < 0$	13199350	0.9999	0.0079
≥ 1 track with $N_{\text{layers}} \geq 4$	13199350	1.0000	0.0079

run period	flavor	n_{layers}	$N_{T\&P}$	$N_{T\&P}^{\text{veto}}$	$N_{SS,T\&P}$	$N_{SS,T\&P}^{\text{veto}}$	P_{veto}
2023C	muon	4	121	75	64	60	$0.2632^{+0.21}_{-0.21}$
		5	61	19	17	14	$0.1136^{+0.13}_{-0.13}$
		≥ 6	2365445	3	90	2	$4.228e-07^{+9.5e-07}_{-9.5e-07}$
2023D	muon	4	74	43	48	48	$0^{+0.37}_{-0}$
		5	30	9	7	7	$0.08696^{+0.18}_{-0.18}$
		≥ 6	1063304	4	40	0	$3.762e-06^{+1.9e-06}_{-1.9e-06}$
2024	muon	4	1152	562	611	561	$0.001848^{+0.062}_{-0.062}$
		5	668	143	172	142	$0.002016^{+0.034}_{-0.034}$
		≥ 6	14440388	20	640	13	$4.848e-07^{+4e-07}_{-4e-07}$
2025	muon	4	1343	837	917	877	$0^{+0.097}_{-0}$
		5	924	249	302	258	$0^{+0.036}_{-0}$
		≥ 6	13254159	30	792	17	$9.809e-07^{+5.2e-07}_{-5.2e-07}$